Dissertation Type: research

DEPARTMENT OF COMPUTER SCIENCE

Towards Ethical Moral Machines

Torty Sivill

A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Engineering in the Faculty of Engineering.

Tuesday 30th October, 2018

Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of MEng in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

Torty Sivill, Tuesday 30th October, 2018

Contents

1 Introduction **1**

2 The Fantasy of Intelligent Machines **3**
2.1 What is Artificial Intelligence? . . . 3
2.2 History of Artificial Intelligence . . . 3
2.3 Motivation for Ethical Artificial Intelligence . . . 4

3 The Philosophical Foundations of Artificial Intelligence **6**
3.1 What are Ethics and Morals? . . . 6
3.2 Approaches to Building Ethical Machines . . . 6
3.3 Understanding Human Moral Behaviour . . . 7
3.4 A Hybrid Approach . . . 8

4 Machine Learning in Artificial Intelligence **9**
4.1 Machine Learning Models . . . 9
4.2 Learning Under Uncertainty . . . 10
4.3 Bayesian Reasoning . . . 12
4.4 Bayesian Hierarchical Models . . . 14
4.5 Motivations for Approximations . . . 16
4.6 Markov Chain Monte Carlo Algorithms . . . 18

5 Related Work **22**
5.1 Top Down Related Work . . . 22
5.2 Bottom Up Related Work . . . 22
5.3 Moral Theory as presented by Kleiman-Weiner et al. . . . 23
5.4 Representing Moral Theory through Utility Calculus . . . 24
5.5 Inferring Moral Theories . . . 25

6 Data **27**
6.1 Big Data Explosion . . . 27
6.2 Problems with Big Data . . . 27
6.3 MIT Moral Machine Dataset . . . 28
6.4 Autonomous Vehicle Dataset: Wachter et al. . . . 29

7 Model Design and Implementation **31**
7.1 Introduction to Kim et al. Computational Model . . . 31
7.2 Decomposing Character Vectors to Abstract Feature Space . . . 31
7.3 Composing Utility Functions . . . 32
7.4 Hierarchical Model . . . 32
7.5 LKJ Covariance Matrix . . . 33
7.6 Implementation of Model in PyMC3 . . . 34
7.7 Evaluation of Model . . . 35
7.8 Results . . . 36

8 Questioning Prior Assumptions **40**
8.1 Importance of Prior Assumptions . . . 40
8.2 The Strength of Non-informative Priors . . . 41

9 Questioning Utility Functions **44**
9.1 Moral Principles . . . 44
9.2 Do Humans Use Utility Functions to Make Decisions? . . . 45
9.3 Incorporation of Deontological Statements Into Model . . . 45

10 Challenging the Hierarchical Model **47**
10.1 Decomposing a Population into Sub-Groups . . . 47

11 Aggregating Preferences **49**
11.1 Ethics without Rules . . . 49
11.2 Voting Based Ethical Decision Making . . . 49

12 Cross Domain Model **52**
12.1 Transferring Model to a New Domain . . . 52

13 Critical Evaluation and Future Work **54**
13.1 Oversimplification of Decision Making . . . 55
13.2 Towards a Hybrid Model . . . 56
13.3 Limits of Data . . . 56
13.4 Bias in Aggregation . . . 56
13.5 Algorithmic Fairness . . . 57
13.6 Transparent Artificial Intelligence . . . 57

A Results from "Learning a commonsense moral theory" **65**

B Results from "A Computational Model of Commonsense Moral Decision Making" **67**

List of Figures

4.1 Graphical example of hierarchical and non-hierarchical models [79] 15
4.2 Transition graph for the Markov chain example with $\mathcal{X} = x_1, x_2, x_3$ [9] 18
4.3 Example trajectory achieved over one iteration of the NUTS algorithm. The shape outlined in blue denotes a contour of the target distribution. The black hollow circles are the values of θ given by the leapfrog algorithm. The black solid circle is the initial value of θ. The red solid circles are values of θ that are discounted as they fall below the slice variable μ. The black circles with crosses through them must be excluded to ensure detailed balance. The blue arrow corresponds to the trajectory of the leftmost to the rightmost value of the subtree in the 3rd peak of the overall tree satisfying detailed balance. The magenta arrow is the normalised trajectory of the final position in the tree. The doubling process stops here as the angle made by both arrows signifies a U-turn. [63] 21

5.1 Kleiman-Weiner et al.'s proposed hierarchical probabilistic model for inferring latent moral theories from sparse behavior [72] 26

6.1 Example scenario from the MIT Moral Machine Platform [1] 29
6.2 Example scenario from the German Autonomous Vehicle Study [46] 30

7.1 Binary matrix, A, that decomposes the features from the character space to features from the abstract principle space [71] 32
7.2 Histogram showing Gelman Rubin statistic for all model parameters under a centred parametrization 36
7.3 Histogram showing Gelman Rubin statistic for all model parameters under a non-centred parametrization 36
7.4 Energies explored by Hamiltonian Markov Chain under the NUTS algorithm for a centred parametrization. 36
7.5 Energies explored by Hamiltonian Markov Chain under the NUTS algorithm for a non-centred parametrization. 36
7.6 Example of a) A Centred Parametrization b) A Non-Centred Parametrization 37
7.7 Inferred posterior distributions for group norm model parameters. Top Left: self preservation, Top Right: elderly, Middle Left: infancy, Middle Right: middle aged, Bottom Left: kneeling down, Bottom Right: pedestrian 37
7.8 Pointplot showing the differences in predictive accuracies between three different models. The Hierarchical model is the implemented model detailed above, Benchmark 1 shows results from the flat model, Benchmark 2 shows results from a model that assumes all individuals have equal weights over all moral principles. The Y axis accuracy shows predictive accuracy over individual decisions. The X axis represents the number of respondents in sample. The vertical bars show variance from cross validation. 39
7.9 Pointplot showing the differences in predictive accuracies between point estimates of individual moral weights. The MCMC Estimate shows point estimates achieved by taking the mean of inferred posterior distributions obtained from the MCMC, the MAP estimate shows accuracy achieved through point estimates obtained through maximum a posteriori. 39

8.1 Box plot showing results of experimentation with the parametrisation of the LKJ covariance prior. 41
8.2 Box plot showing predictive accuracy of the hierarchical model, comparing modeling the standard deviation hyperprior as a narrow normal distribution and a half Cauchy distribution. 42

8.3 Box plot showing predictive accuracy of Hierarchical model when modeled with different hyperprior distributions . . . 43

9.1 Box plot showing predictive accuracy of the hierarchical model with priors specified by Kim et al. [71] when compared to the model initialised with informative, deontological priors. . . . 46

11.1 Boxplot comparing model accuracy of the aggregation via the summarised method of Noothigattu et al. with group norm aggregation. Accuracy here refers to the number of individual decisions predicted by each aggregated model that are equivalent to applying the Borda count to individual preferences. . . . 50
11.2 Boxplot comparing the out of sample predictive accuracy of model under different aggregation methods when using different methods of aggregation to generate universal moral weightings. . . . 51

12.1 An example moral dilemmma posed to respondents in the Hayakawa et al. study [61] . . 53
12.2 Pointplot showing the relationship between model predictive accuracy and the number of abstract moral features used in parametrisation on the Hayakawa et al. dataset [61] . . . 53

A.1 Kleiman-Weiner et al. maximum a posteriori (MAP) estimate of direct reciprocity relations, indirect reciprocity relations and individual weights from a learner observing behavior of the society shown in Figure A.2 [72] . . . 65
A.2 A population of 20 agents used throughout Kleiman-Weiner et al. work showing a) their relationship to other agents b) the weights they place over abstract moral principles and c) the weighting each agent places on all other agents which is a weighted sum over abstract features. [72] . . . 66

B.1 Kim et al. reported individual predictive accuracy of their proposed hierarchical model when compared to three benchmarks. Benchmark 2 is the equivalent of Benchmark 1 used in this dissertation's results [72]. . . . 67

List of Tables

10.1 Table showing the difference in inferred moral norms between gender groups 48
10.2 Table showing the difference in inferred moral norms between age groups 48

List of Algorithms

4.1 Pseudocode for the Metropolis Hastings Algorithm [9] 19
4.2 Pseudocode for the Hamiltonian Monte Carlo Algorithm [63] 20

Executive Summary

Early this century academics called for developers to take responsibility for the impacts of their innovation [27], so why has advancement in the area of ethical artificial intelligence has been so slow? Some academics blame the enormity and complexity of the challenge but most blame an under-appreciation of the issue. This dissertation discusses ethical artificial intelligence, and raises important questions that will hopefully ignite progress in this field. This dissertation's central focus is on the challenge of automating ethical decision making. Some critics argue that machines will never be moral beings [32], yet systems that have to make ethical decisions are being used today. This dissertation is not proposing an answer to these critics but highlighting issues that must be addressed if we, as a society, are to trust in these systems. Furthermore, as Moor suggests in his instructive work, the biggest challenge facing ethical machines is the ambiguity surrounding the definition of ethics [90]. This dissertation extends recent studies that use models of human moral behaviour to help define ethical behaviour of autonomous systems. The primary research hypothesis is:

> *Is it possible to computationally model human moral decision making and can these models of morality help to achieve the overall goal of automating ethical decision making?*

Answering these questions begins with examining the state of the art in computational models for learning human morality which we evaluate on an original dataset to question its generality. We then propose the use of Monte Carlo algorithms such that point estimates are generated more robustly than previous studies. We then dissect and question the assumptions of human morality, made by state of the art, proposing adjustments to the model. Finally, we question the use of aggregation methods to combine individual moral preferences into an overall model for ethical decision making and extend the model beyond the trolley problem. The focus of this dissertation is to raise a plethora of questions and issues, surrounding current perspectives in the field of ethical artificial intelligence, that must be addressed. However, the main achievements of this dissertation can be summarised below:

- Researched and replicated state of the art in computational models of ethics using the PyMC3 framework and Markov Chain Monte Carlo algorithms. This involved the use of convergence heuristics to ensure model was specified most effectively.
- Challenged the assumptions made by state of the art to propose a parametrisation founded in mathematical theory. This adaptation to the state of the art resulted in a 2.5% increase in predictive model accuracy.
- Challenged the assumption made by state of the art to propose a novel parametrisation based in philosophical theory. This adaptation raised important questions surrounding utility and moral theory and found evidence of a correlation between two central ethical perspectives .
- Challenged the state of the art by extending idea of population sub division to achieve 63.6% accuracy when using learned individual morals to predict gender. Although most importantly, raised interesting questions surrounding the impact of unfounded specific assumptions.
- Implemented state of the art in computational social choice to question whether models of morality could be used to automate ethical decision making. Implemented an aggregated model that achieves 78% predictive accuracy over moral decisions.
- First, to our knowledge, to test a model of moral decision making over multiple domains. Implemented model achieves an accuracy of 68% over more diverse, complex moral dilemmas.

Supporting Technologies

- PyMC3, an open source, probabilistic programming framework has been used to implement all computational models presented in the dissertation.
- The "German Autonomous Vehicle" dataset used throughout this dissertation was sourced from the study, "Human decisions in moral dilemmas are largely described by Utilitarianism: virtual car driving study provides guidelines for ADVs" [46]. I was granted access to this dataset upon direct request to M.Wachter and F.Blind.
- The "Foreign Language" dataset used in Chapter 12 of this dissertation was sourced from the study, "Thinking More or Feeling Less? Explaining the Foreign-Language Effect on Moral Judgment" [61]. It was downloaded directly from `https://osf.io/rtfkj/`.

Notation and Acronyms

MCMC	:	Markov Chain Monte Carlo
AI	:	Artificial Intelligence
AGI	:	Artificial General Intelligence
cdf	:	Cumulative Density Function
pdf	:	Probability Density Function
MAP	:	Maximum a Posteriori
MLE	:	Maximum Likelihood Estimation
NUTS	:	No-U-Turn Sampler
drecip	:	direct reciprocity relationship
irecip	:	indirect reciprocity relationship
GFT	:	Google Flu Trends
CDC	:	Centre for Diseases Control and Prevention
AAAI	:	Association for the Advancement of Artificial Intelligence
ACM	:	Association for Computing Machinery
psrf	:	Potential Scale Reduction Factor
K-S Statistic	:	Kolmogorov-Smirnov Statistic
TM	:	Thurstone-Mosteller (process)
BNP	:	Bayesian Non-parametric (model)
IBP	:	Indian Buffet Process

Acknowledgements

Firstly, I would like to thank Dr Carl Henrik Ek for his excellent guidance, feedback and enthusiasm throughout this project. Secondly, I would like to thank M.Wachter and F.Blind for their support and allowing me access to their dataset.

Chapter 1

Introduction

As computers become increasingly prevalent, the subject of designing intelligent systems that function responsibly is increasingly important [90]. This dissertation attempts to break down the challenge of designing ethical artificial agents, considering first the challenge of computationally modeling human moral behaviour and second, questioning how these models of morality could be used to automate ethical decision making. Ethical artificial intelligence is a great challenge requiring input from multiple disciplines and whose resolution remains up for debate [90]. The focus of this project therefore, is not to propose a solution but to highlight issues and raise questions that must be addressed in order to drive future progress in this field.

We begin by studying the history of artificial intelligence, establishing it as a field that has historically pushed the limits of computation. This delivers us to the present, where artificial intelligence has exploded into every area of society. We explore how past mistakes and recent events have sparked an ethical artificial intelligence movement, calling for society, industry and public alike to take responsibility for innovation. We focus on autonomous ethical decision making as an area requiring immediate attention and express that, despite the use of ethical agents across society today, we remain far from the delivery of ethical agents we can trust. Further, a 2017 IPSOS Mori survey, conducted on the behalf of the Royal Society [102], found that only 22% of the public feel that the benefits of military robots, that can think for themselves, outweigh the risks. We present a critical discussion that gives insight into the difficulties of designing ethical machines and foregrounds the arguments presented in this dissertation.

We continue to explore the challenge of engineering ethical machines by looking at historic philosophical arguments underpinning ethics and morality and how these perspectives have shaped state of the art in this field. We use central philosophical and psychological perspectives to propose a framework for engineering ethical machines that emphasizes the importance of learning human morality in formulating ethical rules. The challenge of learning a model of morality brings us to examine the role machine learning plays in modern day artificial intelligence, concentrating on the need for probabilistic models in intelligent systems. We establish Bayesian theory as an approach to reasoning under uncertainty and discuss the challenges for Bayesian techniques. We discuss the benefits of Markov Chain Monte Carlo, a groundbreaking computational technique that addresses some of these challenges.

We proceed by presenting previous study in the area of automating ethical decision making, navigating the journey from top-down rule based to bottom-up evolutionary approaches. This dissertation absorbs the challenges faced by these studies to motivate our approach of learning a model of morality prior to considering the formulation of an ethical machine. We then introduce the recent work of Kleiman-Weiner et al. [72] as the state of the art in learning a commonsense moral theory, forming the basis for our implementation. Kleiman-Weiner et al.'s main contributions can be summarised as 1. Using utility calculus to encode moral theories 2. A Bayesian hierarchical model to infer individual moral beliefs. Kleiman-Weiner et al. acknowledge the limitations of their model when tested on simulated data. We thus evaluate the effectiveness of the theoretical model on real world data whilst considering some common pitfalls of big data led applications. We then discuss the Moral Machine dataset, the first of its kind to crowd-source morality on a large scale [116] and question the ethicality of gamifiying this form of data collection. We therefore present an alternative moral behaviour dataset, collected by a German autonomous vehicle study, conducted by Faulhaber et al. [46], used in this dissertation's model implementation.

We present the design and implementation of our model of morality. We introduce the work of Kim et al.[71] who develop the theory of Kleiman-Weiner et al. into a computational model evaluated on the Moral Machine dataset. We adapt the method of Kim et al. to build a model around the Faulhaber dataset and evaluate the model using Markov Chain Monte Carlo convergence heuristics. Our results show that our implemented hierarchical model achieves 82% predictive accuracy over our dataset. However, the discussion's emphasis is not on the quantified success of the model but on the questions and issues our implementation has raised. We therefore challenge assumptions made by Kim et al. and highlight how these affect results. Continuing our line of questioning, we consider utility calculus as a mechanism to capture human morality. We explore the philosophical debate surrounding the use of moral principles in decision making and demonstrate how experimental results could be used to ground ethical theory. Further, we discuss the link between deontology and utilitarianism, showing, through our implementation, how deontological statements can be used as a prior for the utilitarian model to achieve greater predictive accuracy. We continue to challenge the model of Kim et al. by considering the hierarchical structure of the model and ethicality of decomposing populations into sub cultures. Motivated by this, we expand the idea of decomposition to include groups differentiated by gender and age, to ascertain whether models of morality like Kim et al.'s could be used to gauge the moral compass of arbitrary groups.

Returning our attention to automating ethical decision making, we discuss the feasibility of using a model of human moral behaviour within these intelligent systems. This brings us to presenting the theory of Noothigattu et al. [98], who use computational social choice to aggregate individual models of morality into an overall ethical decision maker. We use our model to argue the successes and ethical problems associated with the method proposed by Noothigattu et al. Finally, we test the model of Kim et al. on moral dilemmas that extend the trolley problem to consider more complex cases of morality and find that the future of ethical machines is heavily dependent on mathematics.

To conclude, we present the limitations of our methodology, with specific, detailed recommendations for future work alongside concluding arguments summarising the questions raised throughout the dissertation.

The high-level objectives of this project are:

1. Debate and test the feasibility of building a computational model of human moral decision making.
2. Question the effects of prior assumptions on hierarchical models.
3. Discover whether utility functions are the optimum mechanism for modeling moral theories.
4. Challenge the idea of population decomposition to see if moral preferences can be inferred over arbitrary groups.
5. Explore the possibility of automating ethical decision making using individual models of morality.
6. Test a model across multiple moral dilemma domains to question the possibility of an universal ethical machine.

Chapter 2

The Fantasy of Intelligent Machines

2.1 What is Artificial Intelligence?

Artificial intelligence is a concept that dominates technological conversations today. Despite its popularity as a buzzword however, numerous references in popular culture and poorly conceived political campaigns have muddied the significance of its underlying meaning. So what actually is artificial intelligence? In his paper, "What is artificial Intelligence" [47], James Fetzer emphasizes the difficulty of this question, concluding that there is a significant difference between "natural intelligence" and "artificial intelligence". However, other modern definitions blur the boundaries between the two. The English Oxford Dictionary, for example, defines artificial intelligence as "The theory and development of computer systems able to perform tasks normally requiring human intelligence" [41]. In his seminal book "Artificial Intelligence: A Modern Approach" [110], Russel Norvig makes a further claim that artificial intelligence definitions can be divided into those that address human-like intelligence and those that deal with an idealised, rational intelligence. This argument clearly shows the ambiguity of artificial intelligence. However, it is obvious that most definitions rely on an implied understanding of intelligence.

The Turing Test, as proposed by Alan Turing in 1950, presents an operational definition of intelligence. To pass the Turing Test, a machine must be perceived as human. Despite the Turing Test's prominence today, many scientists have disregarded it in the search to understand the cognitive mechanisms underpinning intelligence rather than imitating it. Contrastingly, the field of logic was born from the "right thinking" syllogisms of Aristotle as an attempt to encapsulate the idea of rational intelligence. Logisist artificial intelligence aims to expand this way of thinking to build intelligent machines. However, the difficulty of this approach lies in the stark difference between theory and practice. John McCarthy, a computer scientist often referred to as the "father of artificial intelligence" [8], encapsulates both rational and human-like arguments through his definition of intelligence [87]. McCarthy stresses that, whilst there is currently no way of defining intelligence outside the boundaries of human intelligence, artificial intelligence is not limited to simulating human intelligence.

Artificial intelligence can be traced back as far as 200 BC, where in Egypt, evidence alludes to the existence of mechanical gates that would open and close automatically [81]. Additionally, the Greek hero Homer imagined the impossible with "mechanical tripods" waiting upon the gods [27]. For centuries, scientists have been questioning the idea of intelligence and testing the limits of computation. This rhetoric can be seen most clearly today in modern artificial intelligence through pioneering applications such as predicting heart attacks, discovering planets, mind-reading and, of course, driving cars.

2.2 History of Artificial Intelligence

Many consider the birthplace of modern day artificial intelligence to have taken place in 1956 at the Dartmouth workshop. Four innovative mathematicians and computer scientists, Shannon, McCarthy, Rochester and Minsky, gathered to organise the Dartford Symposium, which invited academics from fields such as psychology, philosophy, mathematics and computer science, to solve a common goal: simulate human intelligence. The academics present conjectured that **every** intelligent human function could

be precisely defined and thus simulated by a machine [88]. Although this objective ultimately failed, the Symposium brought together a collective who would later become experts in the field of artificial intelligence [110].

The excitement generated by the Dartmouth workshop inspired a decade of exciting breakthroughs in the field of artificial intelligence. In his influential PhD thesis "Semantic information Processing" [89], Marvin Minsky summarises the progress of the preceding decade as primarily finding heuristics to control trial and error search, and secondly, discovering different methods of learning. One of the main successes of this era was the General Problem Solver as presented by Newell and Simon [95]. The General Problem Solver could solve a limited set of problems in a human like order, therefore being (probably) the first embodiment of the human-like definition of artificial intelligence aforementioned. Beyond 1968, the field of artificial intelligence encountered a period of difficulty as the complexity of problems continued to rise. During this time, artificial intelligence suffered a knowledge drought as most early programs contained no knowledge of the problems they were trying to solve, only succeeding through syntactic manipulation. A second problem that arose was the intractability of many mathematical problems that artificial intelligence was trying to solve.

Following this challenging period, artificial intelligence became popular during the 1980s, which saw a revolution in the way artificial intelligence was conducted. Preceding this period, the field of artificial intelligence had used symbolic computation to distance itself from the rest of computer science. However, the late 1980s sparked a change that saw artificial intelligence embracing scientific method. This revolution also saw the fusion of data and artificial intelligence as those in the field began to question whether machine learning could be used to solve artificial intelligence challenges through neural networks and data mining. The advent of the Internet brought with it repositories of real world data that provided knowledge bases for many modern artificial intelligence programs. During this era the excitement and possibilities drove innovation forward with disregard for the quality of methodologies or social implications [127]. This attitude is a disaster from an ethical perspective and is beginning to be dramatically brought to light today. This dissertation begins to explore some of the ethical issues surrounding artificial intelligence that are necessary to address if we are to maintain control of this exciting and possibility rich field.

2.3 Motivation for Ethical Artificial Intelligence

Artificial intelligence today is extremely popular. We are living in the age of the "knowledge revolution" [94] where knowledge, often in the form of data, is becoming a new currency, demonstrated perfectly by the recent Cambridge Analytica scandal. Today, artificial intelligence applications are mechanisms for extracting and utilising knowledge to perform increasingly more complicated tasks. It is for this reason that governments and industries across the world are vying to be world leaders in this field. Dave Coplin, Envisioning Officer at Microsoft UK, even proclaimed artificial intelligence to be "the most important technology that anybody on the planet is working on today" [113].

Most modern day artificial intelligence applications fall into the domain of "narrow artificial intelligence" [103], the field concerned with creating programs that show expertise in one specific domain, such as chess or weather forecasting. However, Artificial General Intelligence (AGI) is a field gathering momentum within the artificial intelligence community. AGI is concerned with developing intelligent systems that can solve multiple, complicated problems completely autonomously, controlling its own feelings, emotions and judgments. We have seen how this goal was unattainable for those at the Dartmouth conference half a century ago. History would seem to be repeating itself if it were not for renewed confidence and a different approach. In fact many modern computer scientists view AGI as merely an engineering problem [103].

The concept of AGI has opened up discussion on many classical philosophical arguments such as, can a machine have consciousness? What does it mean to be human? Can a machine be a moral agent? [32]. Alongside philosophical debate however, we are now seeing artificial intelligence applied to contexts that have dramatic social implications. We are seeing the introduction of artificial intelligence in potential life or death situations, the criminal judiciary system and positions of care for those most vulnerable in society.

As artificial intelligence becomes intrinsically linked with human lives there are calls from academia to start taking responsibility for our own innovations: "With our successes in AI, however, come increased

responsibility to consider the societal implications of technological success and educate decision makers and the general public so they can plan for them" [27]. This motivation has paved the way for the emerging field of ethical artificial intelligence, which the government cites as critical to the success of technological innovation in the United Kingdom [86].

There are many emerging ethical issues surrounding artificial intelligence today. These include accountability, transparency, and bias of systems. However, this project is primarily concerned with the use of intelligent systems that are being used to make ethical decisions. One such venture is the domain of autonomous vehicles where the significance of this ethical problem was highlighted recently in Arizona, where a driverless car ran down a pedestrian [78]. This tragedy highlights the necessity to explore and question how these autonomous vehicles make difficult decisions such that on their inevitable occurrence we can be assured that an intelligent system acts in the most ethical way.

The challenge of designing intelligent systems that are capable of making ethical decisions is somewhat captured by the philosophical argument over the three different types of knowledge: explicit, implicit and tacit. Explicit knowledge is the type of knowledge that can be extracted from an individual over suitable enquiry [44]. Implicit knowledge, on the other hand, is made up of "what we know but aren't aware of" [85]. In contrast to both of these forms of knowledge, tacit knowledge refers to knowledge that is unable to be articulated and "tied to the senses" [97]. Ethics is a form of tacit knowledge possessed by humans such that we are unable to explain why we think, or act, a certain way other than an intuition that guides our decisions over ethical dilemmas. Furthermore, much literature has been dedicated to the problem of extracting and understanding tacit knowledge from human behaviour [97] [124]. However, creating definitions over tacit knowledge is extremely difficult: how do we explain what we do not understand? Furthermore, this seemingly unsolvable paradox significantly affects the field of engineering ethical machines, how are we to bridge the semantic gap between human ethics and codifying ethical principles?

The area of ethical artificial intelligence is unavoidably extremely challenging. In their cautionary paper, "The Dark Side of Ethical Robots" [121], Winfield et al. discuss the risks of mass deployment of autonomous ethical machines including the possibility of corruption and question the existence of such intelligent systems, suggesting that the potential risks far outweigh the benefits. However, we live in a reality where these sorts of systems are and will continue to be deployed. This chapter has highlighted how challenging the prospect of designing ethical artificial intelligence is going to be, especially with those such as Winfield questioning our current technological capability to even achieve any semblance of automated ethicality. This project therefore attempts to expand on Winfield et al.'s arguments, challenge existing literature and raise new questions that will help to progress this field.

Chapter 3

The Philosophical Foundations of Artificial Intelligence

The field of machine ethics moves the wider field of ethical artificial intelligence away from questioning the uses of artificial intelligence to "monitoring the behaviour of autonomous systems"[5]. This chapter approaches the challenge of designing ethical machines and highlights philosophical arguments underpinning this problem.

There are multiple approaches to designing ethical machines. Allen et al. summarise these approaches as being either top-down or bottom-up. Top-down approaches use ethical theories as a set of rules that determine actions. Bottom-up approaches are instead concerned with machines that are not restricted by a specific ethical theory but learn appropriate behaviour from a given environment. We shall now see that top-down and bottom-up approaches to ethical machines are synonymous with the definitions of ethics and morality.

3.1 What are Ethics and Morals?

> *"A man is moral if he conforms to the established practices and customs of the group in which he is. He is ethical if he voluntarily obligates himself to live in the light of an ideal good."*
>
> Paul Weiss, The Journal of Philosophy [126]

Much of the literature considers morals and ethics as two interchangeable terms. However, in this project, it is important to discern the differences between the two. In his definition, Paul Weiss captures the idea that ethics refers to external rules that govern our sense of right and wrong whereas morals concern individual preferences governing actions [126]. Within the context of ethical artificial intelligence, it is critical to correctly navigate the distinction between the two concepts. Do we want ethical or moral machines?

3.2 Approaches to Building Ethical Machines

Is it possible to ingrain a set of ethics within a machine? James. H. Moor of Dartmouth College argues that there are a variety of ways in which an ethical machine can be achieved [90]. Firstly, Moor argues that an agent could contain software that is implicitly ethical as opposed to containing code that outlines explicitly ethical rules. An example of an implicitly ethical agent is pharmacy software that monitors drug interactions. This software has a duty of care to only prescribe drugs that are to deliver more harm than good [90]. Secondly, Moor introduces the idea of explicit ethical agents as those machines that "represent ethics explicitly and then operate effectively on the basis of this knowledge" [90]. There are few real world examples of explicit ethical agents as the field is plagued by questions that a millennia of study in classical ethical philosophy have still not resolved, primarily a debate surrounding the optimum ethical theory. In their contribution to "Technology in Society" [62], Herschel et al. discuss classical ethical theories and bridge the gap between technology and classical philosophy. An ethical theory contains a set of rules that enables "the individual to make persuasive, logical, and reasoned arguments" [62]. Some

ethical theories are discussed in brief below:

Kantianism is an ethical theory that is concerned not with what we do but with what we ought to do. The philosopher Kant identified rules as the critical component underlying actions and it is according to these rules that the ethicality of an action is judged. Unlike Kantianism, the ethical perspective of utilitarianism judges the right or wrong of an action based on its outcome. Utilitarianism associates a cost with every action and the right choice is therefore the action that maximises utility for an associated population.

If we are to engineer intelligent systems that are explicit ethical agents, which ethical theory should be used to govern the actions of the agent? Central figures to the field of machine ethics, Anderson and Anderson attempted to create an explicit ethical agent, "Jeremy", that employed the use of utilitarian ethical theory [7]. Jeremy would attempt to optimise the net quantity of pleasure from a given action. However, critics of utilitarianism argue that justice is ignored to the end of doing what is good but not what is right [7]. It is clear from this argument that there is still further research required before an universal ethical theory can exist within a machine [90].

Moor concludes by briefly presenting "full ethical agents" [90] as machines that are capable of making explicit ethical judgments whilst being able to "reasonably justify them". It is clear that we are a long way from achieving this capability in intelligent systems. In fact, full ethical agents seem congruous to the field of artificial general intelligence presenting similar philosophical arguments with some academics, such as Moor, questioning the feasibility of full ethical agents.

Whilst we have seen that there exist frameworks in creating top-down ethical autonomous systems, we must continue to question whether this is the right approach. Many ethicists, including Weiss, argue that the basis for ethics must be within morals and not the other way round. This argument enforces the importance of morals in engineering ethical machines and provokes a new line of questioning into the origins and development of morality.

3.3 Understanding Human Moral Behaviour

The "moral brain" [37] is an interdisciplinary perspective on the question of morality. Pretot et al. [105] begin by exploring the origins of human morality. They recognise some moral behaviours in animals, such as empathy in rodents, thus drawing the conclusion that morality is a behaviour that has evolved within human DNA. However, the argument that humans are born innately moral is fiercely revoked by those who believe humans are born as complete blank slates. For instance, Hamlin states that although the literature considers innate morality as a tenable hypothesis [60], there is little further study into its resolution. However, he continues to argue that new evidence suggests that humans are born with sprouts of morality that later shape our moral compass [60]. Despite both arguments, Hamlin argues that there is no doubt that both socialization and cognitive construction massively influence moral development. Moral development, contrastingly, is an aspect of moral psychology that is not so heartedly disputed. There is a vast amount of evidence to suggest that the moral development process begins in early childhood. It is clear from moral theory, therefore, that moral behaviour is something that is (nearly) entirely learned. If infants are able to develop morality, would it therefore be possible for a machine to do the same?

This is the question answered by the bottom-up approach to engineering ethical machines. Machine learning techniques can be applied to educate an intelligent system to recognise moral behavior. Associated learning, for example is a process that mirrors the way in which children develop moral behaviour through reward and punishment [3]. If we are to apply bottom-up techniques to engineer ethical machines, then we require a knowledge base that encapsulates human moral behaviour.

In the book, "The Social Psychology of Morality", Ditto and Liu claim that moral behaviour "becomes most interesting when moral principles come into conflict" [42]. Moral psychologists have therefore become fascinated with the concept of moral dilemmas where no moral choice is made without undesirable consequences. Moral dilemmas perfectly capture the moral preferences of the individual [42] and could be the most effective way of encoding human morality in a form that could be understood by a machine. The most famous form of moral dilemmas is the trolley problem which asks variants of the question: is it morally permissible to pull a lever that sacrifices the life of one person to save the lives of five others?

3.4 A Hybrid Approach

This chapter has shown that engineering an ethical machine hinges on capturing a model of ethics or morality. The question still remains as to whether a bottom-up approach that learns human moral behaviour or a top-down approach that stipulates an ethical theory, is the optimum solution. Philosopher Joseph Katz argues that perhaps it is preferential to focus on ethics devoid of morality as morals often suffer from the concept of superego where superego denotes a psychoanalytic theory developed by Sigmund Freud referring to one's idealised self image [70]. Contrastingly, Anja Steinbauer argues that in matters of morality we cannot allow ourselves to be lead by authority [118]. In the context of engineering ethical machines however, many academics believe that a hybrid approach is the best approach. Learning human moral behaviour is directed towards the expansion of choices and flexibility in behaviour whereas top down ethical constraints redirect a model towards acting for the good of a society.

If we are to create machines that are capable of making ethical decisions we must, therefore, first learn models of human behaviour and then combine these models with legislative or ethical theory that guarantees moral machines always make an ethical decision; in other words, we must endeavour to engineer ethical moral machines. This project, therefore, looks first at modeling human moral decision making over moral dilemmas by implementing, questioning and suggesting improvements to current state of the art. The project then considers how the above could be transformed into a model that would allow an autonomous system to make ethical decisions, and indeed whether a solution to this challenge is even possible.

Chapter 4

Machine Learning in Artificial Intelligence

The previous chapter introduces the idea of engineering ethical machines by learning a model of human moral behaviour. Since the 1980s, learning has become a core philosophy of artificial intelligence. In their contribution to the "Journal of Engineering and Computer Science" [119], Talwar et al. discuss the motivations of machine learning within artificial intelligence. Talwar et al. recognise that deep understanding in intelligent machines would allow them to make optimised decisions in the most efficient way. They note the transition from "heavy machines with explicit programming" to the introduction of algorithms that teach machines to "detect different patterns and to adapt to new circumstances" [119], ultimately leading to intelligent systems being able to make decisions independently. Talwar et al. identify that "designing the algorithm and using it in most appropriate way is the real challenge for the developers and scientists" [119]. This challenge has led to the expansion of the field of machine learning. Carbonell et al., authors of "Machine Learning Volume I" [29], describe machine learning as "the study and computer modelling of learning processes in their many manifestations" [29].

From a technical perspective, machine learning involves teaching computers to optimise some criterion based on a knowledge base consisting of data or past experience. This process involves defining a model of parameters. Learning, then optimises those parameters over a given knowledge base. Both the algorithms used for learning and inference need to be highly efficient thus rendering the construction of effective models highly complicated, relying heavily on computer science theory and mathematics.

4.1 Machine Learning Models

4.1.1 Parametric vs. Non Parametric Models

In his book, "Machine Learning: A Probabilistic Perspective" [108], Murphy expresses that when defining a model to describe a given dataset, it is imperative to consider whether a parametric model or non-parametric model is most suitable. A parametric model specifies a fixed number of parameters, whereas the non-parametric model has a flexible number of parameters that grows with the size of the knowledge base. Parametric models are often more efficient yet require stronger assumptions to be made about the underlying data. Non parametric models give much more flexibility but are often intractable. Further, non parametric models often suffer from what is known as the "curse of dimensionality" [108]. When the feature space describing a set of data points grows in dimensionality, the space becomes sparser and sparser and therefore more likely to be correctly separated by a hyperplane. However, when the higher dimension classification is projected back to the low dimensional space represented by the observable data, the classification fails [108].

This section has explored some motivations for machine learning techniques to be used in the creation of intelligent systems and considered some of the challenges in designing models that can be learned efficiently by machine learning algorithms. Much of this project is about experimenting with and questioning different models within the context of learning human morality. However, it is important at this point to introduce the "No free lunch theorem" [129], coined by David Wolpert in 1997 as a way of explaining

that there is no optimal all-round model. A model that works well in one domain may completely fail in the next.

4.2 Learning Under Uncertainty

Murphy claims that the best way to solve machine learning problems is to use the tools provided by probability theory [108]. His reasoning follows that probability theory can be applied to any context involving uncertainty and machine learning is a field riddled with uncertainty: what is the best prediction about the future given some past data? How should I model my data? What is likely to happen next, given some data? Probability theory thus provides the framework for the quantification and manipulation of uncertainty. Statistical inference is the process that links learning with uncertainty using data analysis to deduce properties of an underlying probability distribution.

4.2.1 Probability Theory Background

Many consider Andrei Nikolaevich Kolmogorov the founding father of probability theory [99]. His monograph, "Foundations of the The Theory of Probability" [73], is said to have transformed probability theory from a "collection of calculations to a mathematical theory" [99]. Within his seminal paper, Kolmogorov introduces his axioms of probability [73], which still form the basis for modern day probability theory. Kolmogorov's main contributions can be summarised below.

Kolmogorov introduces the idea of a *probability space*: $(\Omega, \mathcal{F}, P)$, as a tuple of three elements where Ω represents the set of all possible events, $\mathcal{F}$ is a subset of Ω and P is a measure on the space $(\Omega, \mathcal{F})$, such that

$$P(\Omega) = 1.$$

This rule is also known as the second axiom of probability [99]. The first axiom of probability can be described as the following: for any event $A \in \mathcal{F}$, $P(A)$ denotes the belief in the event A being true. Probability theory requires that

$$0 \leq P(A) \leq 1,$$

also be true. Kolmogorov's third axiom is also known as the axiom of additivity. For $A, B \in \mathcal{F}$, if A and B have no element in common then

$$P(A + B) = P(A) + P(B)[73].$$

Kolmogorov also introduces the concept of random variables. A random variable is a mapping $X : \Omega \to M$, such that for any real number a, the set $\omega \in \Omega : X(\omega) < a$ is a member of the subset $\mathcal{F}$ [99]. Kolmogorov was also the first to provide a mathematical definition for conditional probabilities. Given events $A, B \in \mathcal{F}$ and $P(A) > 0$,

$$P(B|A) = \frac{P(AB)}{P(A)}[73].$$

From this theorem, Kolmogorov continues to define joint probabilities as

$$P(AB) = P(A)P(B|A),$$

from here, Kolmogorov induces the multiplication theorem as

$$P(A_1, A_2, A_3, A_4...A_n) = P(A_1)P(A_2|A_1)...P(A_n|A_{n-1}, A_{n-2}..., A_1).$$

These contributions represent just a proportion of Kolmogorov's paper. However, these rules and definitions, alongside some extensions and consequences, remain as the central foundations of probability theory.

4.2.2 Discrete Distributions

Here, we present the mathematical background, taken from Murphy [108], behind some parametric distributions defined on discrete state spaces that are used later in the dissertation.

Binomial Distribution

The binomial distribution limits events to having only one of two possible outcomes. Let p be the probability of one event and $1-p$ be the probability of the other event. If the trial is repeated n times, then we are interested in the probability of getting k occurrences of the first event. This probability is defined as $p(k|n,p)$. Under the assumption that all the trials are independent, the probability of a sequence with r occurrences of the first event is $p^k(1-p)^{n-k}$. However, this probability represents just one of the possible sequences containing r occurrences of the first event. The total number of unique sequences of trials is the number of different ways of selecting r from n which is shown by the following equation,

$$^{n}C_k = \frac{n!}{k!(n-k)!} \quad .$$

Therefore, the probability of getting k occurrences of the first event is

$$p(k|n,p) = \frac{n!}{k!(n-k)!} \quad p^k(1-p)^{n-k}.$$

The Bernoulli distribution is a special case of the Binomial distribution where $n = 1$. The Bernoulli distribution is therefore given as

$$f(k|p) = \begin{cases} p & \text{if } k = 1 \\ 1-p & \text{if } k = 0. \end{cases}$$

4.2.3 Continuous Distributions

So far, when talking about random variables we have assumed they have represented discrete values. We now extend the idea of random variables to cover uncertain continuous quantities. If X is an uncertain continuous quantity, the probability that X lies in the range,

$$W : (a \leq X \leq b),$$

where the events A and B, defined as $A = (X \leq a), B = (X \leq b)$, can be calculated as

$$p(W) = p(B) - p(A).$$

The cumulative distribution function (cdf) defines a mapping $F(q) = p(X \leq q)$. Using the cdf notation we can now redefine $P(W)$ as

$$p(a \leq X \leq b) = F(b) - F(a).$$

The probability density function (pdf) is defined as

$$f(x) = \frac{d}{dx}F(x).$$

Using the pdf we can now compute the probability of a continuous variable being in some finite range as

$$P(a \leq X \leq b) = \int_a^b f(x).$$

The Normal Distribution

The normal distribution, sometimes referred to as the Gaussian distribution is the most commonly used distribution in machine learning [108]. Its pdf is given by

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2},$$

where μ is the mean and σ^2 is the variance of the distribution. The precision of a distribution refers to the narrowness of the distribution which is controlled by the variance. A lower variance term describes a more precise distribution [108].

4.2.4 Joint Probability Distributions

A joint probability distribution has the form $p(x_1, x_2, ...x_n)$ and models the relationships between a set of n variables.

Covariance and Correlation

The covariance between two random variables measures the degree to which the two variables are related [108]. If $\mathbf{x}$ is a d dimensional random vector, its covariance matrix is therefore defined as

$$cov\big[\boldsymbol{x}\big] = \mathbb{E}\Big[(\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])(\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])^{\intercal}\Big].$$

Covariances can range between zero and infinity which can cause problems when working with them. For this reason covariances can be normalised into correlations. The Pearson correlation coefficient can be defined as

$$corr\big[X, Y\big] = \frac{cov\big[X, Y\big]}{\sqrt[2]{var\big[X\big]var\big[Y\big]}}.$$

Correlations can lie between the interval $[-1, 1]$. A correlation of one between two variables X, Y signifies that X and Y are linearly dependent. If X and Y are independent then their covariance is zero and thus uncorrelated. However, it is important to note that uncorrelated does not imply independent.

Multivariate Normal Distribution

Like the aforementioned univariate normal distribution, the multivariate normal distribution is the most common joint probability density function in machine learning for continuous variables. The pdf for the multivariate normal distribution in D dimensions is given as:

$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{|\boldsymbol{\Sigma}|^{\frac{1}{2}}(2\pi)^{\frac{D}{2}}} e^{\Big[-\frac{1}{2}((\boldsymbol{x}-\boldsymbol{\mu})^{\intercal}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\Big]},$$

where $\boldsymbol{\mu}$ is $\mathbb{E}\big[\boldsymbol{x}\big]$ and $\boldsymbol{\Sigma}$ is the $D \times D$ covariance matrix.

4.3 Bayesian Reasoning

We have previously discussed motivations behind intelligent systems being needed to cope with uncertainty and therefore being capable of reasoning probabilistically. This section will therefore outline the technical background behind Bayesian reasoning.

4.3.1 Bayesian Vs. Frequentist

Within probability theory, there are two central arguments, Bayesian and Frequentist. Frequentists believe probability to describe long run sequences of events. For example, if the probability of drawing the ace of spades from a pack of cards is $\frac{1}{52}$ then a Frequentist perspective on this probability would be that if we turned over a whole deck of cards, one of these would be the ace of spades. Although logical and useful, the Frequentist approach often does not work well in situations such as predicting a one off event or inference from data. It is therefore often much more useful to think of probabilities as a measure of uncertainty. In his book, "Probability Theory: The Logic of Science" [68], Jaynes states that the Bayesian perspective of probability is concerned with information rather than repeated trials. Within the machine learning domain, the Frequentist perspective of probability theory does not make sense, but the Bayesian approach does [68].

4.3.2 Bayes' Theorem

Bayesian philosophy was originally derived from an interpretation of Bayes' Theorem. Bayes' theorem can be obtained from rearranging Kolmogorov's [73] definitions of joint and conditional probability aforementioned:

$$p(h|e) = \frac{p(e|h)p(h)}{p(e)}. \tag{4.1}$$

This definition asserts that the probability of a hypothesis h given some evidence e, also known as the posterior, can be calculated by its likelihood $p(e|h)$ times its probability prior to any evidence $p(h)$, normalized by $p(e)$. The normalization factor ensures that probabilities of all hypotheses sum to 1 [74]. In their paper, "Bayesian Hierarchical Models" [109], Rouder et al. home in on the main idea underpinning Bayesian probability, that "beliefs may be updated rationally in light of data" [109]. Although Bayes' theorem is an "irrefutable manipulation of probability theory" [109], the further claim that this is the proper way of adjusting hypothesis beliefs is far more controversial and has led to the philosophical debate, "Conditionalization" [74]. Luckily, however, the practical applications of using Bayes' theorem for inference does not depend on the resolution of the Conditionalization argument [74].

4.3.3 Bayesian Inference

Statistical inference refers to the deduction of an underlying posterior distribution using data analysis. Bayesian inference is a form of statistical inference that allows model parameters to be random variables and prior information to be included in calculation. However, the requirement of prior information can also be a disadvantage of Bayesian inference. Other drawbacks include the intractability of mathematical calculations which make solutions impossible to find. Often, approximations to mathematically challenging inference problem can be found using techniques like the Monte Carlo algorithm [106].

Bayes' theorem is fundamental in achieving Bayesian inference. If M is a hypothesis or model and D is underlying data then Bayes theorem can be rewritten as

$$P(M|D) = \frac{P(D|M)P(M)}{P(D)}. \tag{4.2}$$

To infer $P(M|D)$ Bayes' theorem requires the likelihood: $P(D|M)$, the probability of observing the data given that the model is true. Bayes theorem also requires the unconditional probability that the model is true, the prior $P(M)$. Lastly, Bayes' theorem requires the unconditional probability of the data $P(D)$. Given these three probabilities the posterior can be calculated.

4.3.4 Linear Bayesian Models

In Bayesian paradigms, useful models contain unknown parameters that must be inferred from data. We now use examples, taken from Rouder et al. [109], to explain the computational problems associated with Bayesian reasoning.

Example 1: Consider an example where a coin is flipped repeatedly. The number of heads may be modeled as $Y \sim Binomial(p, n)$, where n is the number of trials and p is a free parameter corresponding to the probability of getting a head and should be inferred. As we are considering Bayesian models, the likelihood should be considered alongside a prior belief. Using Bayes' theorem to redefine the problem, we get

$$p(p|Y) = \frac{Pr(Y|p)p(p)}{Pr(Y)}, \tag{4.3}$$

where $p(p|Y)$ is the posterior distribution showing beliefs about the parameter conditional on the observed data. $p(p)$ is the prior distribution of beliefs about the probability of getting heads. An example distribution over the prior could be

$$p(p) = \frac{1}{2}. \tag{4.4}$$

The term $Pr(Y|p)$ is the likelihood and is determined by the model. For the Binomial model, the likelihood is,

$$Pr(Y|p) = ({}^{n}C_{Y})(p^{Y}(1-p)^{n-Y}). \tag{4.5}$$

The last term $Pr(Y)$ can be reduced by marginalising Y over all the possible values of p such that

$$Pr(Y) = \int_0^1 Pr(Y|p)p(p)dp, \tag{4.6}$$

by the total law of probability. Whilst it is not always necessary to calculate $Pr(Y)$ to compute posterior beliefs, the difficulty of computing integrals has caused the area of Bayesian computation massive challenges throughout the years.

Bayes' rule may be extended to describe models with multiple parameters.
Example 2: Consider a new example where a group of students take a class test. The random variable $\boldsymbol{Y}$ that represents a vector of test marks may be modeled by

$$\boldsymbol{Y} \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2), \tag{4.7}$$

where μ and σ^2, are free parameters of the model and should be inferred. This model requires a statement about the joint posterior and joint prior beliefs of μ and σ^2. An application of Bayes' theorem thus gives

$$p(\mu, \sigma^2|\boldsymbol{Y}) \propto Pr(\boldsymbol{Y}|\mu, \sigma^2)p(\mu, \sigma^2). \tag{4.8}$$

As this model only has two dimensions, computing the joint posteriors and priors is easy. However, as models grow in dimensionality, computing these probabilities becomes increasingly difficult. As a solution to this problem, often marginal probabilities are used. The marginal probability for a given parameter is calculated by integrating the uncertainty in all other parameters.

The marginal posteriors for μ and σ are therefore given by

$$f(\mu|\boldsymbol{Y}) = \int_{\sigma_2} f(\boldsymbol{Y}|\mu, \sigma^2)d\sigma^2, \tag{4.9}$$

and

$$f(\sigma^2|\boldsymbol{Y}) = \int_{\mu} f(\boldsymbol{Y}|\mu, \sigma^2)d\mu. \tag{4.10}$$

Although useful, these marginal probabilities become intractable as dimensionality increases. To express each marginal all other parameters must be integrated out and the resulting integrals could be extremely high dimensionalf. This problem of high dimensional integration was a "major pragmatic barrier" [109] for the adoption of Bayesian methods until the 1980s, which saw the arrival of efficient computational techniques [109].

4.4 Bayesian Hierarchical Models

One of the main advantages of adopting a Bayesian approach in modeling is the ease in which a linear Bayesian model can be transformed into a hierarchical Bayesian model, capable of accounting for variation in real world settings. [109]. A Bayesian hierarchical model can be simply described as a statistical model defined over multiple levels which capture different levels of observational data.

In the influential work "Data analysis using regression and multilevel/hierarchical models" [50], Gelman et al. highlight the rich motivation for hierarchical models. The first of these incentives is "learning about treatment effects that vary" [50]. One of the most interesting goals of data analysis is learning how y changes as a result of x. Hierarchical models allow x to vary within a population or within subgroups of that population. Gelman et al. expand on this idea by highlighting the benefits of taking into account group differences when making predictions, which is allowed by hierarchical models. Secondly, Gelman et al. discuss how hierarchical models can be useful in estimating group levels parameter with limited data. Finally, Gelman et al. discuss the efficiency hierarchical models afford in inference when dealing with inherently structured data.

The difference between hierarchical and non-hierarchical models can be summarised by Figure 4.1 taken from Levy [79], which shows how observations $\mathbf{Y}$ belong to a number of clusters. The outcomes of the model are therefore determined by both the parameters θ, that are shared amongst clusters, and b, shared amongst observations within a cluster. There is also a second probability distribution, Σ_b, over the cluster

specific parameters b. Parameters that form the prior distribution are called hyperparameters, where the distributions over the hyperparameters are termed hyperpriors [79].

Taken from Jose Bernado and Adrian Smiths instructive work, "Bayesian Theory" [18], we now discuss the framework for some typical hierarchical models. Equations 4.11, 4.12 and 4.13 show a specification of a simple Bayesian hierarchical model.

$$p(x|\theta) = p(x_1, ...x_k) = \prod_{i=1}^{k} p(x_i|\theta_i) \tag{4.11}$$

$$p(\theta|\phi) = p(\theta_1, ...\theta_k|\phi) = \prod_{i=1}^{k} p(\theta_i|\phi \tag{4.12}$$

$$p(\phi). \tag{4.13}$$

The vector $x_1, ...x_k$ contains data points from k individual, but linked, sources for example, students within a class. The first level of the hierarchical model, Equation 4.11, therefore specifies the likelihood of observing each of the k data points given the parameter vector $\theta_1\theta_k$. Equations 4.12 and 4.13 show the second and third levels of the model that define the prior distribution over theta. The hyperparameter, here ϕ, usually relays some information about the group of the k individuals, i.e. mean and covariance [18]. Applying Bayes' theorem to the model and marginalising out ϕ sees the same posterior inference calculations we have seen in the linear models:

$$p(\theta_i|x) = \int p(\theta_i|\phi, x)p(\phi|x)d\phi, \tag{4.14}$$

$$p(\theta_i|\phi, x) \propto p(x|\theta_i,)p(\theta_i|\phi), \tag{4.15}$$

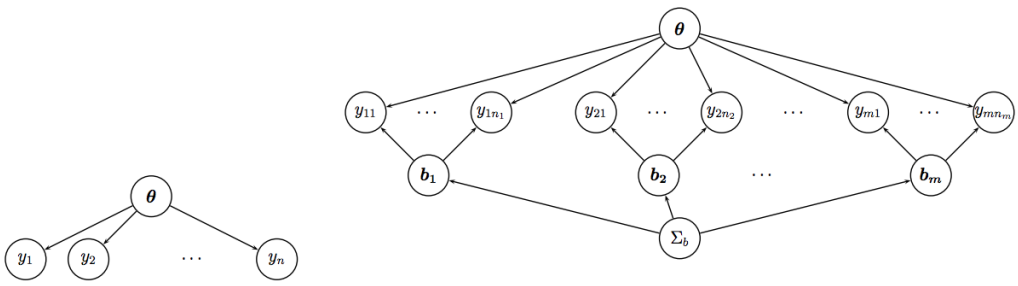
$$p(\phi|x) \propto p(x|\phi)p(\phi), \tag{4.16}$$

and

$$p(x|\phi) = \int p(x|\theta)p(\theta\phi)d\theta. \tag{4.17}$$

Although this is a simplistic example of a hierarchical model, there is no limit to the number, or complexity, of levels. Therefore, the evaluation of the appropriate integrals in posterior cases may not be trivial and are actually nearly always intractable [18]. However, by using computational techniques, these models can often be implemented in a "fully Bayesian way". [18].

Figure 4.1: Graphical example of hierarchical and non-hierarchical models [79]



(a) A non-hierarchical model (b) A simple hierarchical model, in which observations are grouped into m clusters

4.5 Motivations for Approximations

The previous two sections describe some of the problems Bayesian inference encounters in practice. In their famous paper, "Bayesian Computation via the Gibbs Sampler and Related Markov Chain Monte Carlo Methods" [115], Smith et al. state that barring a select few "stylised" Bayesian problems, the integrals defined for inference require analytical or numerical approximation. Andrieu et al. identify "normalization" and "marginalisation" as the two biggest difficulties integrals cause for Bayesian inference due to their intractability over high dimensions [9]. Since analysis is no longer an option, an alternative is to estimate the integral using numerical methods.

4.5.1 Expectations

Andrieu et al. however, are also able to further motivate approximation techniques beyond those seen in the previous section [9]. One particularly prevalent motivator highlighted by Andrieu et al. is computing expectations, whereby the goal of analysis is to compute

$$\mathbb{E}_{p(x|y)}(f(x)) = \int_{\mathcal{X}} f(x)p(x|y)dx. \tag{4.18}$$

This idea is also expressed by Markov Chain Monte Carlo computational tool, STAN, founder, Michael Betancourt in his paper "A Conceptual Introduction to Hamiltonian Monte Carlo" [19]. Betancourt begins by describing the problem with computing expectations analytically. Similarly to computing normalization factors in Bayesian inference, computing expectation integrals for most target distributions is intractable.

Betancourt continues by considering the "good statistical questions" [19] concerning posterior distributions. Under the assumption that all important information is contained in the posterior, Betancourt asks, "how can we extract this information?" [19], and summarises two approaches. Firstly, optimization techniques, including maximum likelihood and maximum a posteriori which find the mode point estimate within a posterior distribution. Secondly, expectations which integrate over the entire posterior space to find a global summary.

Betancourt argues that "the only approach that makes sense is expectations" [19]. He continues by considering the geometry of high dimensional space, directly comparing volume and density over these spaces. Betancourt asserts that density is highly dependent on parametrization and therefore, manipulations of the posterior that use density and ignore volume, such as optimisations, are making unhealthy assumptions. Further, optimization techniques consider the mode of posterior distributions where density is high. The volume however, is largest out in the tails of the posterior distribution with smaller density values [19]. Moreover, this disparity grows exponentially with the dimension of parameter space. Taking expectations integrates over all of the target space; the tails are taken into account and may make significant contributions to the resultant expectation calculation. Betancourt summarises that balancing density with volume is thus an important and critical task in computing expectations.

In response to this balancing challenge, Betancourt proposes the idea of the "typical set", the neighbourhood of solutions that lie between the density heavy mode and the volume heavy tail ends of the posterior space. Betancourt then concludes that expectation approximations can be achieved by identifying the "typical set" and argues that the Markov Chain Monte Carlo algorithm is a tool that does this.

4.5.2 The Origins of Monte Carlo

In 1946, while recovering from being ill, polish scientist Stanisaw Marcin Ulam was playing solitaire. Probably bored out of his mind, Ulam started to play with probabilities, contemplating the chances of one particular solitaire, the Canfield, would come out successfully after laying out 52 cards. At first, Ulam attempted exhaustive combinatorial calculations, he then swapped to the more practical approach of trial and error; laying out solitaires at random and counting the number of successes and failures. The idea of using a statistical sample to approximate a hard mathematical, combinatorial problem is the core philosophy behind the Monte Carlo simulation [9]. Joining forces with John Neumann who understood the potential of using this theory to use computers to solve problems in mathematical physics, both scientists managed to come up with multiple Monte Carlo algorithms over the years. Today, Monte Carlo

algorithms have contributed massively to a variety of fields including economics and computer science, with the Metropolis algorithm being said to be amongst the top ten influential algorithms of the 20th century [15].

4.5.3 The Monte Carlo Idea

Following the above discussion we can summarise the aim of the Monte Carlo method as either or both of the following points, taken from the research conducted by D Mackay [84].

- Randomly select independent and identically distributed samples

$$\{x^{(i)}\}_{i=1}^{N}, \tag{4.19}$$

  from a target density $P(x)$ defined over a high dimensional space $\mathcal{X}$. This space may be a posterior distribution or the combinatorial space of solitaire solutions.

- To estimate expectations of functions under this target density. For example, the function $\phi(x)$ can be evaluated as

$$\Phi = \langle \phi(x) \rangle = \int d^N x P(x)\phi(x). \tag{4.20}$$

Mackay et al. conjecture that by solving the first problem, we can use the generated samples to approximate a solution for the second problem as

$$\hat{\Phi} \equiv \frac{1}{N}\sum_{i=1} \phi(x^i). \tag{4.21}$$

As the samples are generated from $P(x)$, the expectation $\hat{\Phi} = \Phi$. Additionally, the inverse proportionality between the variance of $\hat{\Phi}$ and the number of samples N is one of the key components of the Monte Carlo algorithm. This property means that the accuracy of the Monte Carlo algorithm in approximating a solution to expectations is independent from the dimensionality of the space sampled [84].

We have shown how the problem described in 4.20 can be reduced to drawing random samples from a probability space $P(X)$. However, now we shall explore why sampling from this space is hard. Mackay et al. first assume that the density from which samples are taken $P(X)$ can be calculated as

$$P(x) = \frac{P^*(x)}{Z}, \tag{4.22}$$

where Z is the normalizing factor discussed in length in Chapter 4.3 as intractable. The fact that we cannot solve the integrals of the normalization factor means that taking samples from $P(x)$ is difficult. However, Mackay et al. argue that even if it were possible to solve for Z, sampling from $P(x)$ would still be challenging due to the high dimensionality of the space. There are, in fact only a handful of high dimensional spaces that are easy to sample from [84].

Having established that drawing samples from $P(x)$ is extremely difficult, even if $P^*(x)$ is easy to compute, there is clear motivation to study Monte Carlo methods such as Markov Chain Monte Carlo which propose solutions to the sampling challenge.

4.5.4 Markov Chain Monte Carlo

Markov Chain Monte Carlo (MCMC) is a technique for approximating samples whilst exploring the state space $p(x)$ using a Markov chain mechanism. Its use is motivated by the fact that preceding sampling algorithms such as Importance Sampling fail to obtain proposal distributions that are easy to sample from whilst giving a good approximation [9]. The Markov chain is designed such that the algorithm spends most of its time in the important parts of the state space, the "typical set", as outlined by Betancourt [19]. Moreover, the algorithm is particularly designed so that the random samples mimic as closely as possible the samples from the target distribution $P^*(x)$.

A Markov chain is a stochastic process, an indexed set of random variables where the value of a particular variable in the set is dependent on the random variable in the preceding position. It is natural to introduce Markov chains on finite spaces, where samples x^i can take a discrete number N of values such that $x^i \in \{x_1, ...x_n\}$. Further, a Markov chain must satisfy

$$p(x_i|x_{i-1}, ...x_1) = T(x_i|x_{i-1}), \tag{4.23}$$

which specifies that the future state depends on the current state but not past states. For a Markov chain to be homogeneous, $T = T(x_i|x_{i-1})$ remains invariant for all values of i [9]. Figure 4.2 shows the transition graph for an example Markov chain with $N = 3$, taken from Andrieu et al. [9]. The transition matrix for this graph can therefore be derived as,

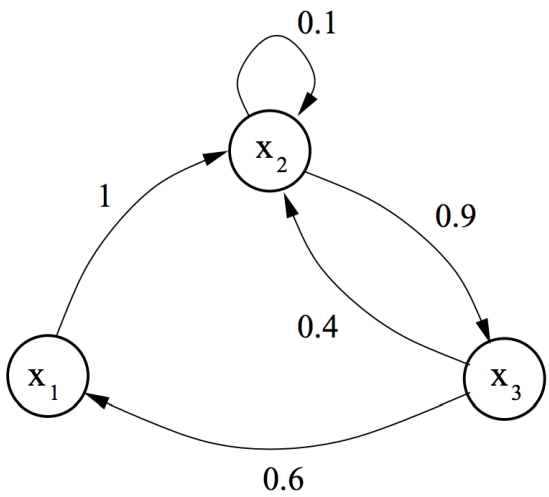
$$T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0.1 & 0.9 \\ 0.6 & 0.4 & 0 \end{bmatrix}. \tag{4.24}$$

If the initial state of the vector is $\mu(x^1) = (0.5, 0.2, 0.3)$, it follows that $\mu(x^1)T = (0.18, 0.64, 0.18)$ and after several iterations the Markov chain converges at $(0.2, 0.4, 0.4)$. The fact that Markov chains converge at a stable point plays an integral role in Markov Chain Monte Carlo. Further, Andrieu et al. [9] state that the Markov chain will always converge to the invariant distribution $p(x)$ as long as T is a stochastic transition matrix that is both irreducible and aperiodic (doesn't get trapped in cycles). A further condition that is not required but desired ensures that $p(x)$ is the desired invariant distribution. A Markov chain that satisfies this condition abides by the following reversibility condition, "detailed balance",

$$p(x^{(i)})T(x^{(i)}|x^{(i-1)} = p(x^{(i-1)})T(x^{(i)}|x^{(i-1)}. \tag{4.25}$$

Markov Chain Monte Carlo samplers are invariant and aperiodic Markov chains that have the target distribution as the invariant distribution. It is critical when designing these samplers that detailed balance is achieved and that they converge as quickly as possible [9].

Figure 4.2: Transition graph for the Markov chain example with $\mathcal{X} = x_1, x_2, x_3$ [9]



4.6 Markov Chain Monte Carlo Algorithms

4.6.1 The Metropolis Hastings Algorithm

The Metropolis Hastings algorithm is the most widely used of all Markov Chain Monte Carlo samplers. It is also the simplest and most flexible [9]. The algorithm works by generating candidate state transitions from a supplemental distribution.

1. Initialise θ^0
2. **for** $i = 0$ **upto** $N - 1$ **do**
   - Sample $\mu \sim Uniform_{[0,1]}$
   - Sample $\theta' \sim q(\theta'|\theta^i)$
   - **if** $\mu < A(\theta^i, \theta') = min\{1, \frac{p(\theta')q(\theta^i|\theta')}{p(\theta^i)q(\theta'|\theta^i)}\}$ **then**
     - $\theta^{i+1} = \theta'$
   - **end**
   - **else**
     - $\theta^{i+1} = \theta^i$
   - **end**

end

Algorithm 4.1: Pseudocode for the Metropolis Hastings Algorithm [9]

To exemplify the Metropolis Hastings sampler, consider the algorithm for a single parameter θ. The algorithm uses the supplemental distribution, the "proposal distribution" to produce candidate variables $q_t(\theta'|\theta^i)$, where θ^i represents the current value of θ. The generated variable θ' is a possible value for θ^i at the next step. The algorithm also calculates the probability of transitioning back to θ^i from θ' as $q_t(\theta^i|\theta')$ [9]. The above probabilities are then combined to determine a value termed the "acceptance ratio":

$$a(\theta', \theta^i) = \frac{q_t(\theta'|\theta^i)p(\theta')}{q_t(\theta^i|\theta')p(\theta^i)}. \tag{4.26}$$

The value of θ^i at the next step, θ^{i+1} is then determined by the following equation

$$\theta^{i+1} = \begin{cases} \theta' & \text{with prob.} \quad \min(a(\theta', \theta^i), 1) \\ \theta^i & \text{with prob.} \quad 1 - \min(a(\theta', \theta^i), 1) \end{cases}$$

.

The pseudocode for the algorithm can be seen in Algorithm 4.1 [9]. The algorithm is simple, however, there are significant challenges in designing the proposal distribution $p(\theta'|\theta)$. Andrieu et al. [9] prove that the Metropolis Hastings algorithm satisfies the detailed balanced condition and under certain conditions, is aperiodic and irreducible which means that the algorithm's samples mimic those from the target distribution.

4.6.2 The Hamiltonian Monte Carlo Algorithm

Whilst the Metropolis Hastings algorithm is flexible and easy to implement, as it is a random walk sampler it may not be statistically efficient for many models [9]. To address this, the Hamiltonian or Hybrid Monte Carlo algorithm was proposed. It prevents random walk behaviour by simulating a physical system governed by Hamiltonian dynamics.

The Hamiltonian Monte Carlo approach adds an extra momentum variable, r, to accompany the model variable, θ [63]. For the Hamiltonian Monte Carlo algorithm, new samples are generated by simulating a physical stream where particles move about a high dimensional space subject to kinetic and potential energy changes. Particles within the space are characterized by a position vector $\theta \in \mathbb{R}^D$ and velocity vector $r \in \mathbb{R}^D$ [93]. The Hamiltonian is therefore defined as

$$H(\theta, r) = E(\theta) + K(r) = E(\theta) + \frac{1}{2}\sum_i ri^2, \tag{4.27}$$

where E denotes potential energy and K refers to kinetic energy. The position vector and velocity vector are modified throughout the simulation to ensure the Hamiltonian remains constant. The differential equations that maintain a constant Hamiltonian are shown below:

$$\frac{d\theta_i}{dt} = \frac{\delta H}{\delta r_i} = ri, \tag{4.28}$$

$$\frac{dr_i}{dt} = \frac{\delta H}{\delta \theta_i} = -\frac{\delta E}{\delta \theta_i}. \tag{4.29}$$

1. Given $\theta^0, \epsilon, L, \mathcal{L}, M$:
2. **for** $m = 1$ **upto** M **do**
   - Sample $r^0 \sim \mathcal{N}(0, I)$
   - Set $\theta^m \leftarrow \theta^{m-1}, \theta' \leftarrow \theta^{m-1}, r' \leftarrow r^0$
   for $i = 1$ **upto** L **do**
   Set $\theta', r' \leftarrow Leapfrog(\theta', r', \epsilon)$
   end
   With probability $\alpha = \min\left\{1, \frac{\exp \mathcal{L}(\theta') - \frac{1}{2} r' r'}{\exp \mathcal{L}(\theta^{m-1}) - \frac{1}{2} r^0 r^0}\right\}$, set $\theta^m \leftarrow \theta', r^m \rightarrow -r'$

end

Function Leapfrog (θ, r, ϵ)
-Set $r' \leftarrow r + (\epsilon/2)\Delta_\theta \mathcal{L}(\theta)$
Set $\theta' \leftarrow \theta \epsilon r'$
Set $r' \leftarrow r' + (\frac{\epsilon}{2})\Delta_\theta \mathcal{L}(\theta')$
return $\theta' r'$

Algorithm 4.2: Pseudocode for the Hamiltonian Monte Carlo Algorithm [63]

Neal [93] shows how the above dynamics idea could be used as transition operators of a Markov Chain that leave $p(\theta, r)$ invariant. Unfortunately, however, Hamiltonian dynamics cannot be simulated in its entirety due to the problem of time discretization. The Leapfrog algorithm was therefore proposed as an implementation of Hamiltonian dynamics that preserves volume and detailed balance [93]. The Leapfrog algorithm operates over the following 3 steps:

$$r_i(t + \frac{\epsilon}{2}) = r(t) - \frac{\epsilon}{2}\frac{\delta}{\delta\theta_i}E(\theta(t)), \tag{4.30}$$

$$\theta_i(t + \epsilon) = \theta_i(t) + \epsilon r_i(t + \frac{\epsilon}{2}), \tag{4.31}$$

$$r_i(t + \epsilon) = r(t + \frac{\epsilon}{2}) - \frac{\epsilon}{2}\frac{\delta}{\delta\theta_i}E(\theta(t + \epsilon)). \tag{4.32}$$

In practice, using discretized timesteps, ϵ , does not preserve the Hamiltonian, $H(\theta, r)$, exactly and will actually introduce bias into the system. To reverse these effects exactly, Hamiltonian Monte Carlo adds a Metropolis accept/reject stage after n leapfrog steps. The new state, $X' = (\theta', r')$, is accepted with probability $P_{acc}(X, X')$ defined as

$$P_{acc}(X, X') = min\Big(1, \frac{\exp^{-H(\theta', r')}}{\exp^{-H(\theta, r)}}\Big). \tag{4.33}$$

Algorithm 4.2 shows the pseudocode for drawing M samples with the Hamiltonian Monte Carlo algorithm, taken from Hoffman et al. [63]. For each sample, m, the momentum parameter, r, is first sampled from a normal distribution. The Leapfrog algorithm is then applied L times to the position and momentum variables to generate proposed values, θ', r', where $\mathcal{L}(\theta)$ refers to the logarithm of θ. The proposed values are then accepted or rejected according to the Metropolis acceptance ratio.

4.6.3 The No U-Turn Sampler

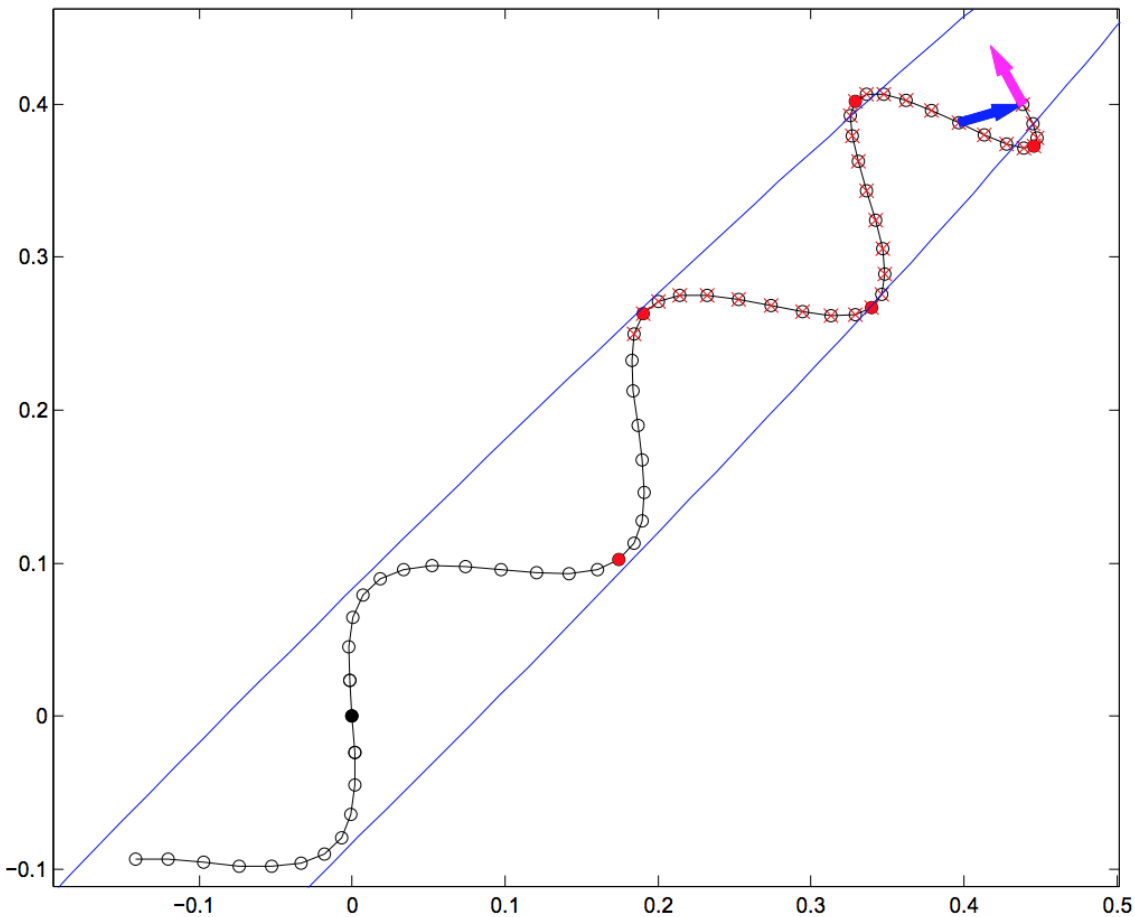
One of the biggest disadvantages of the Hamiltonian Monte Carlo algorithm is the extensive tuning required to make it sample efficiently. In particular, the algorithm is extremely sensitive to the parameters, Leapfrog step size and Leapfrog number of steps, that require specification by the user [63]. If either of these are parametrised badly this has severe consequences on the efficiency of the algorithm.

The No-U Turn Sampling (NUTS) algorithm, introduced by Hoffman and Gelman in 2014 [63], automatically tunes the step size and number of steps without user intervention. Particularly problematic is the need to specify the parameter for the number of steps. If the step size is too small then the algorithm displays undesirable random walk properties. If however, the step size is too large, the algorithm will end up going back on itself and wasting computation [63]. The No-U Turn Sampling algorithm eliminates the need to specify the number of leap frog steps whilst retaining the ability to prevent random walk behaviour by making a criterion that prevents the simulation running for longer than necessary,

i.e. when the distance between θ and θ' is no longer increased. Hoffman and Gelman propose a criterion based on the dot product of the current momentum, r, and a position vector that captures the difference between the current θ' and initial position θ. This quantity is proportional to the progress the algorithm makes away from θ. We therefore want the algorithm to stop when this value drops below zero.

Unfortunately, Hoffman et al. identify that this algorithm does not satisfy time reversibility. They therefore propose a modified algorithm that satisfies this condition by running the algorithm both forward and backwards in time. Hoffman et al. start by introducing a slice variable, μ, which aids implementation of the algorithm. After sampling μ the algorithm uses the Leapfrog algorithm to build a binary tree using doubling: first moving forwards or backwards one step then forwards or backwards two steps then forwards or backwards four steps. This doubling process ensures that the binary tree is balanced. The process halts when the trajectory from furthest left node to the furthest right node of **any** balanced subtree doubles back on itself. Figure 4.2 shows an example of a NUTS trajectory taken from Hoffman et al. [63].

Figure 4.3: Example trajectory achieved over one iteration of the NUTS algorithm. The shape outlined in blue denotes a contour of the target distribution. The black hollow circles are the values of θ given by the leapfrog algorithm. The black solid circle is the initial value of θ. The red solid circles are values of θ that are discounted as they fall below the slice variable μ. The black circles with crosses through them must be excluded to ensure detailed balance. The blue arrow corresponds to the trajectory of the leftmost to the rightmost value of the subtree in the 3rd peak of the overall tree satisfying detailed balance. The magenta arrow is the normalised trajectory of the final position in the tree. The doubling process stops here as the angle made by both arrows signifies a U-turn. [63]



Chapter 5

Related Work

Chapter 3 discusses the varying approaches to engineering machines that are capable of making autonomous ethical decisions. Chapter 2 also introduces the concept of bottom-up and top-down approaches to ethical machines, summarising their similarities to ethical and moralistic approaches. In their innovative work [125], Wallace and Allen summarise all approaches to ethical machines as either top-down or bottom-up. We shall now review some related work across both perspectives and introduce the work that has greatly influenced this project.

5.1 Top Down Related Work

Within the context of engineering, a top-down approach refers to the process of iteratively reducing a problem into its sub tasks until one can be solved directly. Problems solved by a top-down approach are those that are able to be formally specified in their entirety. Charisi et al. [30] identify that the most frequent form of top-down approach being used to implement ethical behaviour is system governance via a set of ethical theory determined rules. This approach is adopted by Arkin et al. in their introduction of an "ethical governor" [11] which is a component of the ethical architecture for a military autonomous system. The ethical governor's purpose is to conduct an evaluation of the "ethical appropriateness" [11] of an action prior to its completion by a robot. The governor ensures a response is ethical by ensuring its non-membership of a pre-defined set of possible unethical outcomes. The "ethical appropriateness" of a response is determined by both utilitarian and deontological ethical theory.

The question of which ethical theory should determine autonomous decisions has caused problems for top-down approaches and resulted in a wide variety of perspectives. For example, Dennis et al. [38] argue a different use of ethical theory. They claim that within a civilian context, stopping all actions that violate at least one ethical principal is not a "satisfactory method of assuring ethical behaviour" [38]. They instead implement an ethical theory stipulating that an unethical option may sometimes be chosen as long as it is "the least of all evils" [38]. Another view is that of Bendel, who proposes the the use of annotated decision trees for modeling moral machines [16]. Top-down approaches like these allow for a rigorous process of decision making [30]. However, the ambiguity over universal ethical theories means it is not clear as to whether top-down approaches dependent on specified ethical theory can be used to solve ethical problems in practice. Attention has, therefore, turned towards bottom-up approaches that do not require hard coded ethical theory or rules. [30]

5.2 Bottom Up Related Work

Bottom-up approaches in engineering require a description of a problem and then the development of a method to find a solution in terms of its parameters [30]. Machine learning approaches use a bottom-up perspective to artificial intelligence problems. In their general ethical analyzer, "GenEth", Anderson and Anderson use machine learning to learn an ethical theory [6]. GenEth was created alongside input from ethicists to help codify ethical principles in any given domain. In order to learn ethical principals GenEth uses inductive logic programing, a machine learning technique that learns relations based on First Order Horn Clauses. Contrastingly, Abel et al. use reinforcement learning to learn the most moral

decision [2]. Reinforcement learning is heavily influenced by behavioural psychology and teaches agents how to take actions in particular environments to maximise rewards. Abel et al. use Markov Decision Processes and Partially Observable Markov Decision Processes, typical decision problem formulations used in reinforcement learning, as mechanisms to frame a decision problem with an associated reward function. Contrastingly, D.Dewey in his work, "Learning What To Value" [40], argues that reinforcement learning can only learn preferences based on potential rewards. Dewey continues to present "expected observation utility maximization" as a mechanism that, unlike reinforcement learning , can be used to define agents with multiple final goals. Similarly, Bostrom in his recent work, "contemplating the feasibility of super-intelligence" [22], also questions the use of reinforcement learning in learning moral theory. Bostrom suggests that a sufficiently intelligent machine could realise that it could maximise its reward by exploiting or "wireheading" [22] its reward function. Although only applicable to some reinforcement learning scenarios, this vulnerability to cheating shifts attention to Bayesian approaches.

Both Bostrom [22] and Dewey [40] suggest utility functions as a preferential way of ensuring that an artificial intelligent machine learns about moral values instead of decision outcomes. Building on this, Bayesian learning is an alternative bottom-up technique that allows agents to make decisions that optimize a meta-utility function [2]. In his paper, "Motivated Value Selection for Autonomous Agents" [12], Armstrong experiments with an agent that uses Bayesian learning to update beliefs about a "true" ethical theory [2]. The agent works by performing the following action

$$argmax_{a \in A} \sum_{w \in W} P(w|e,a)\big(\sum_{u \in U} u(w)P(C(u)|w)\big). \tag{5.1}$$

In this maximisation, A is the set of possible actions, W the set of possible worlds and e denotes evidence seen by the agent. Armstrong applies the theory to toy experiments and concludes that this model is sufficient to define a value selecting agent.

This section has motivated the use of Bayesian approaches to learning a model of ethical behaviour. We shall now present the state of the art work by Kleiman et al., "Learning a Common Sense Moral Theory" [72], that uses both hierarchical Bayesian inference and utility calculus to learn a model of moral theory. This paper forms the theoretical basis for the remainder of this dissertation.

5.3 Moral Theory as presented by Kleiman-Weiner et al.

In their paper, "Learning a Common Sense Moral Theory" [72], Kleiman-Weiner et al. introduce a novel computational framework for understanding and learning moral theory. They first introduce a recursive utility calculus that captures welfare trade-offs in interactions between individuals. They then introduce hierarchical Bayesian inference as a mechanism to understand the actions and decisions of individuals.

In order to frame their definition of morality, Kleiman-Weiner et al. begin by defining commmon sense, in the words of Nagel [91], as a judgement that suggests we should live our lives with autonomy and consideration for specific others, the greater good and decency. Despite the potential for moral conflict, Nagel claims that common sense should have the final word. Kleiman-Weiner et al. expand this definition by suggesting that behind the notion of common sense there should be a set of values that determine the treatment of others. They conjecture that complex moral constructs such as altruism, loyalty and justice have their grounding in these values and their paper attempts to provide a framework to understand these values.

Chapter 3 discusses the psychological theory behind moral development and learning and establishes the ongoing debate in the field over the role learning has in moral behaviour. Kleiman-Weiner et al. believe that despite the possibility of innate morality, there must be a substantial role for learning in the way humans trade-off underlying moral values in interactions. Kleiman-Weiner et al. also discuss the way in which moral behaviour has changed as a result of society over time. They draw on the celebrated research of Blake et al. [20] who drew attention to the impact culture and age have on the idea of fairness. Kleiman-Weiner et al. continue to highlight extensive related work [56], [65], that collectively imply the extreme variability of morality. Kleiman-Weiner et al. recognise the challenge in learning moral values across cultures and groups with such variability.

Kleiman-Weiner et al. use moral learning during childhood to explain the existence of abstract moral principles that guide moral judgment: children cannot possibly learn how to trade off specific values in specific interactions but must be instead guided by a set of abstract moral principles, which Kleiman-Weiner et al. term, "moral theory". They present utility calculus and Bayesian hierarchical inference as tools to help learn their notion of moral theory.

5.4 Representing Moral Theory through Utility Calculus

The first component of the model presented by Kleiman-Weiner et al. is the expression of moral values as utility functions. They are able to capture the idea of altruism through recursive utility functions where an agent's utility is increased by other's utilities. Under this perspective, moral learning is the process of learning how to value the utilities of different groups of people [72].

Kleiman-Weiner identify the first challenge of learning moral theories under their recursive utility framework as representing moral theories under an "infinitude" of scenarios [72]. They begin by considering a simplified decision scenario where N represents the set of agents indexed by i, S is a set of states and A_s is the set of all possible actions in a particular state. The probability

$$P(s'|a, s),$$

denotes the probability of reaching state s' given action a and initial state s. $R_i(s)$ is a mapping of outcomes to a real number that represents the welfare agent i experiences in state s. Kleiman-Weiner et al. then define moral theories using $R(s)$, where the level 0 moral theory, referring to an agent who values only the welfare they receive themselves is represented as

$$U_i(s) = R_i(s).$$

Kleiman-Weiner et al. then use the work of Hurka [66] to expand the level 0 moral theory to include ideas of altruism. Hurka describes people's attitudes to good and bad through their attitudes to "base goods" and "base evils". Kleiman-Weiner et al. expand this idea to recursively value others in explaining moral preferences. Their level 1 moral theory therefore uses the moral 0 theory, such that

$$U_i^1(S) = (1 - \gamma_i) + \gamma_i \sum_{j \in N j \neq i} \alpha_{i,j} U_j^0(S), \tag{5.2}$$

where $\gamma \in [0, 1]$ represents how willing an agent is to trade off their own welfare with the welfare of others. Each $\alpha_{i,j}$ denotes the weight agent i places on the welfare of agent j. Since the proposed moral theories are grounded in the welfare of individual agents, Kleiman-Weiner et al. claim that under all possible parameterisations, these specify an expansive space of moral theories.

Having specified moral theories within a utility calculus framework, Kleiman-Weiner et al. focus on agents who act accordingly to moral theories by considering classical probabilistic decision making tools. Kleiman-Weiner et al. define the expected utility of an action as

$$EU(a, s) = \sum_{s'} U(s')P(s'|a, s).$$

Kleiman-Weiner then use the Luce-choice decision rule to define action selection probabilistically. The Luce-choice decision axiom was formulated by R. Luce in 1959 [83] and is used to model individual choice behaviour in economics. The axiom states that the probability of selecting one item from a set of possible items is not affected by the presence or absence of other items. Applied to Kleiman-Weiner's definition of expected utility, the probability of choosing action a from state s is

$$P(a|s) = \frac{\exp(\beta EU()a, s)}{\sum_{a' \in A_s} exp(\beta EU(a', s))}. \tag{5.3}$$

As the limit $\beta \to \infty$, the agent chooses the action with highest associated utility.

In the model so far, each $\alpha_{i,j}$ describes how a specific individual j should be treated by agent i. This is problematic, as without abstract principles an agent would have to specify a new $\alpha_{i,j}$ for each new individual it interacts with [72]. Kleiman-Weiner et al. therefore propose that opinions about individuals

should be instead determined by abstract relationships which are captured within abstract moral principles. Kleiman-Weiner et al. propose six different types of relationship that carry moral obligation: Self, Kin, In-group, All, Direct-reciprocity (moral obligation to specific individuals within a community) and Indirect-reciprocity (moral obligation to unknown individuals within a community).

Kleiman-Weiner et al. believe that these abstract principles capture simplified relationships between agents and explain why a particular individual may act towards another. Formalising these principals, Kleiman-Weiner et al. let $P = \{kin, group, all, drecip, irecip\}$ be the set of all possible principles. For each principle there is a function $f^p(i, j)$ for all possible pairings of agents that returns one if the relationship between agents holds. For example $f^{kin}(i, j) = 1$ if i and j belong to the same family. [72] Kleiman-Weiner et al. then propose a weighting to the above principles where each $w_i p \in W_i$ is the weight agent i places on the principle p. $\alpha_{i,j}$ can then be re-written using the weights over abstract principles as

$$\alpha_{i,j}(W_i) = \phi_{i,j} + \sum_{p \in P} w_i^p f^p(i, j), \tag{5.4}$$

where $\phi_{i,j}$ is introduced as a quantity that reflects factors, like empathy, outside the moral principles, that affect the way i feels about j. Kleiman-Weiner et al. state that the adjustments now reflect **what** agents value instead of **who** they value.

5.5 Inferring Moral Theories

The second component of Kleiman-Weiner et al.'s model is an approach to learning about abstract moral theories from observations about their behaviour in interactions. They identify from historical research [31] the existence of a "poverty of stimulus" [72] which encapsulates the gap between the data that is available to the learner (sparse and noisy interactions) and what is learned (abstract moral principles). This challenge motivates the use of Bayesian hierarchical models which have been used historically to learn over sparse and noisy data [72]. Kleiman-Weiner et al. continue to justify their decision to represent moral theory in terms of abstract principles by claiming that a learner needs only to infer the weightings over a small set of principles, instead of inferring weightings over every individual, which makes learning tractable [72]. Further, it is the abstractness of the principles that Kleiman-Weiner et al. claim overcomes the "poverty of stimulus" problem. If a learner observes that a particular agent values kin highly, when that agent is introduced to a new member of its kin, the learner will already have a pretty good idea of how the two agents will interact. Kleiman-Weiner et al. also claim that reducing moral theories to moral principles makes it easier to incorporate the idea of moral variation within cultures or groups. For example, people within a certain group may value kin more highly that those in a different group. This further motivates learning abstract principles at the group level and not the individual level [72].

Kleiman-Weiner make the assumption that $self, kin, group, all$ are principles observable to the learner. This is a phenomenon that is recognised in multiple pieces of biological research, including that of Lisa DeBruine [36] who observes that organisms are sensitive to cues of genetic relatedness when making decisions about social behaviour. However, for the abstract principles $drecip, irecip$ Kleiman-Weiner et al. state that these relationships are not observed by the learner and must be observed from behaviour [72]. Kleiman-Weiner et al. then present their approach to learning abstract moral preferences. Let T be the set of all observed interactions. Each observation (a_i, s) denotes the choice a_i made by agent i in state s. To infer the moral theories of others, a learner needs to infer the weights of the moral principals conditioned on the observations. This conditional inference utilises Bayes' Rule in the following problem:

$$P(W_i|(a_i^0, s^0), ..., (a_i^T, s^T)) \propto \sum_{f^{drecip}} \sum_{f^{irecip}} P(a_i^0, ..., a_i^T|s^0, ..., s^T, W_i, f^{drecip}, f^{irecip}) P(W_i) P(f^{drecip}) P(f^{irecip}). \tag{5.5}$$

The likelihood, $P(a_i^0, ..., a_i^T|s^0, ..., s^T, W_i, f^{drecip}, f^{irecip})$, is a probabilistic rational action as shown in Equation 5.3 with $\alpha_{i,j}$ derived from Equation 5.4 [72]. Kleiman-Weiner et al. continue to specify priors over the unobserved principles, direct reciprocity and indirect reciprocity as well as over the weights themselves. For the direct reciprocity principle, Kleiman-Weiner et al. decide to represent the sparsity of this relationship with the following exponential prior:

$$P(f^{drecip}) = \prod_{i \in N} \prod_{j \in N j \neq i} \gamma \exp(-\gamma f^{drecip}(i,j)), \tag{5.6}$$

where γ controls the likeliness of these relations [72]. Kleiman-Weiner et al. formulate indirect reciprocity relationships as inference over a group rather than an individual relation. Only if both interacting agents are in the "co-operating" group will they value each other under the indirect reciprocity relation. This principle is therefore given the following prior distribution:

$$P(f^{irecip}) = \prod_{i \in N} p^{i \in C}(1-p)^{i \notin C}, \tag{5.7}$$

where C represents the "co-operating" group [72].

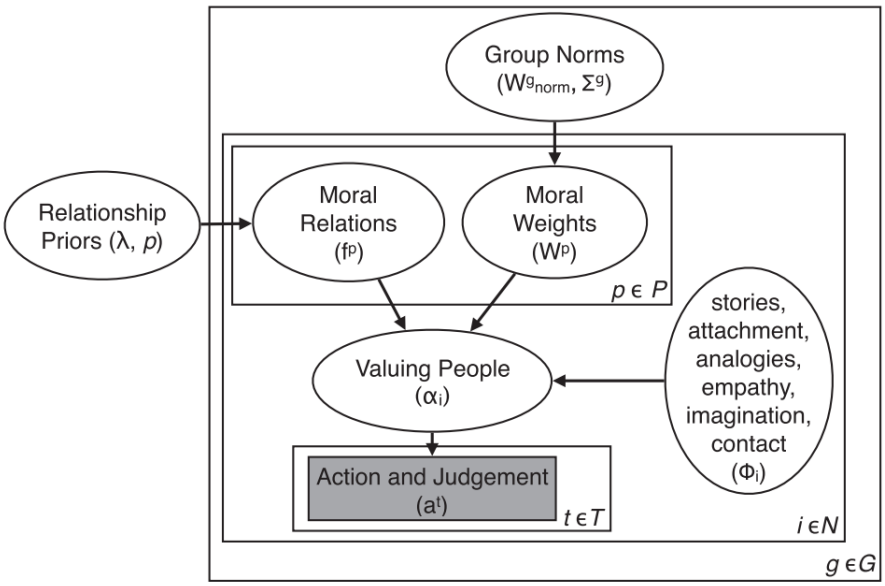
Next, Kleiman-Weiner et al. turn their attention to defining a generative model over the possible ways the abstract principles could be weighted. A simple model may treat each agent's weights as generated independently from a common prior. However, Kleiman-Weiner et al. propose a more structured model where each individual's principles are generated from a prior dependent on the group that individual belongs to [72]. This means the group norms should be inferred alongside the individual weights. Kleiman-Weiner et al. specify that each individual weight is drawn from a Gaussian distribution centred around the average weighting of that principle in a particular group, such that

$$W_i \sim \mathcal{N}(W^g_{norm}, \Sigma^g), \tag{5.8}$$

where Σ^g denotes the amount by which principles covary within a group. The full hierarchical model is shown in Figure 5.1.

Kleiman-Weiner et al. report the success of their model on fabricated data using the optimization technique, maximum a posteriori to infer moral theories (their results can be found in Appendix A). We recognise the need to test this theoretical model on real data to test its applicability to real scenarios and this process is detailed in the following chapters.

Figure 5.1: Kleiman-Weiner et al.'s proposed hierarchical probabilistic model for inferring latent moral theories from sparse behavior [72]



Chapter 6

Data

6.1 Big Data Explosion

In Chapters 2 and 4 we saw how machine learning is being used to solve artificial intelligence problems. One factor that is both motivating and facilitating the explosion of machine learning applications is the availability of large quantities of data. As professed by Kevin Murphy [108], we are living in the era of big data. Zhobre and Dhafa, for example, claim that the world produces over 2.5 quintillion bytes of data daily [43]. With this overwhelming amount of complicated, often unstructured, data being generated from any device, any time, anywhere, it is undeniable that "we are drowning in information, starving for knowledge" [92]. This "data deluge" [92] has caused data based artificial intelligence to explode over the past decade. The hype is inescapable: "Chinese robot, Xiaoyi passes its National Medical Licensing Examination"; "Google teaches AI to bake cookies" [24]. Despite all the exciting opportunities data and data mining bring to the world of artificial intelligence, there is also an ever increasing array of challenges and ethical issues that must be addressed if we are to continue using data sustainably [62]. This chapter therefore seeks to raise questions surrounding the use of of big data, focusing on the domain of data based ethical machines.

6.2 Problems with Big Data

Google Flu trends (GFT) was a web service operated by Google that provided flu influenza activity for over 25 countries. In February 2013, GFT made global headlines, but not for the reasons Google would have hoped. There were reports that GFT was predicting more than double the proportion of doctor visits for influenza like illnesses than the Centre for Diseases Control and Prevention (CDC), despite the fact that GFT was build to predict CDC reports [77].

Mistakes like these highlight the worrying assumptions the public and industry alike make about big data. In their cautionary work, "The Parable of Google Flu: Traps in Big Data Analysis" [77], Lazer et al. warn that we are still far from a place where big data can replace more traditional methods of data collection. Furthermore, they conjecture that big data hubris and algorithm dynamics are two of the biggest pitfalls for the use of big data in artificial intelligence [77].

Lazer et al. claim that the GFT error occurred due to what is commonly known as algorithm dynamics. Algorithm dynamics are the changes made by the developers and engineers to improve the commercial service offered by the algorithm [77]. Lazer et al. deem it likely that the GFT gave volatile predictions of the flu because of algorithm dynamics affecting Google's search algorithm. [77]. Due to the complexity and variability of the search algorithm, it is impossible to either reproduce GFT's results or to spot errors like GFT. This lack of algorithmic transparency is a rhetoric that is becoming increasingly poignant as black box neural networks drive progress in artificial intelligence. How are we to trust in results if both algorithm and data are too complex for human comprehension?

Lazer et al. present the idea of big data hubris as the assumption that big data is the substitute, rather than the supplement, for traditional data collection and analysis. They assert that the quantity and easy availability of the data should not allow researchers to ignore traditional methods of validation,

measurement techniques and reliability. Further, most of the recent big data results that have received media attention have not been handled reliably and validly. It is thus important to remember that conclusions are only ever as powerful as the data behind them. This is a rhetoric reflected in Noam Chomsky's address at the "Brains, Minds and Machines Symposium" [96], held at MIT in 2011, where he expressed that there exists a "notion of success" within machine learning, "which I think is novel in the history of science. It interprets success as approximating unanalyzed data" [96]. This dissertation continuously considers its results in light of this statement. We do not claim to solve the conundrum of "moral machines" but use both data and results to raise and address questions as well as suggest potential future work.

6.3 MIT Moral Machine Dataset

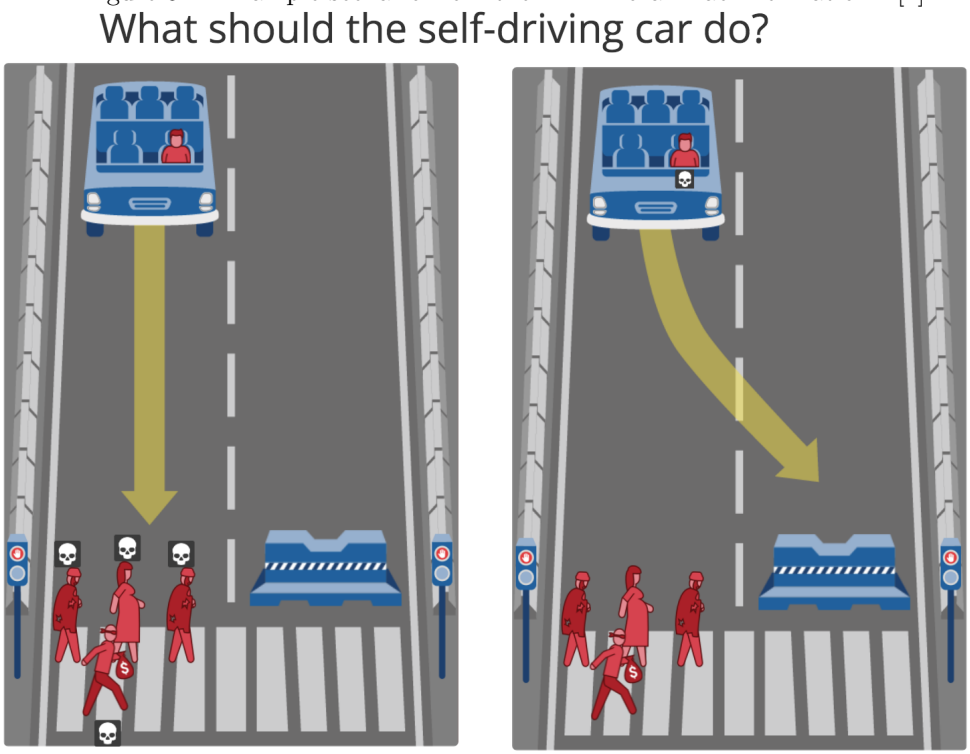
As addressed in the previous chapter, there is a multitude of literature surrounding the field of engineering moral machines. The majority of this work is concerned with building theoretic models that formulate moral theory or ethical actions. However, if we are to succeed in building machines capable of making ethical decisions then we will need to test these models in real world contexts. Therefore, there is an urgent need for data that encapsulates information about how humans make moral decisions. Francis et al. [48] identify moral dilemmas as key in helping researchers to understand and interpret moral decision making. Francis et al. therefore use their study to collect data about human decision making in perhaps the most famous of all moral dilemma thought experiments, the trolley problem. Francis et al. use virtual reality as an immersive technique to compare participants' responses to a variety of trolley problem scenarios. Contrastingly, researchers at MIT take a different approach to collecting trolley problem data. Their platform, The Moral Machine, is the first of its kind to "crowdsource morality" [116].

The Moral Machine platform was commissioned by Rahwan, Bonnefon, Shariff following their well cited article, "The social dilemma of autonomous vehicles" [21]. The article presents experimental results conducted by Bonnefon et al. using the Amazon Mechanical Turk Platform. Bonnefon et al. conduct a study with just under 2000 participants looking at responses to a variety of extended trolley dilemma scenarios, e.g. "Is it more moral to sacrifice one passenger for the lives of ten passengers?". Bonnefon et al. conclude that the discrepancies in their results show that ethical autonomous vehicles is a serious social dilemma. They discuss the need for autonomous vehicle algorithm design to take into consideration more "intricate decisions" than those presented in their study [21]. This conclusion prompted the creation of the Moral Machine platform, which aims to provide both a crowd-sourced image of public opinion on how machines should act in moral dilemmas and generate discussion over potential scenarios of moral consequence [1].

The platform presents 13 randomly generated scenarios to the participant. An example of a scenario is shown in Figure 6.1 [1]. When faced with this scenario, the participant is faced with a hypothetical moral dilemma and must choose which group of characters to kill and which group of characters to save. There are 20 possible characters that are randomly generated for every scenario including children, pregnant women, overweight men and criminals. In the example shown in Figure 6.1 the participant must choose between killing two homeless men, a pregnant woman and a criminal or one man who is also the driver of the car. Also notable is the fact that the first group are crossing the road on a red light thus making their actions illegal. As of October 2017 the platform had collected over 30 million responses from over three million respondents from over 180 countries from across the world [71].

The dataset compiled by Bonnefon et al. has motivated further research [71], [98], who have used the data to learn underlying moral preferences. These studies are generating a lot of interest, both having been accepted to the AAAI/ACM Conference on Artificial Intelligence, Ethics and Society of February, 2018. However, we have previously seen the dangers of using unqualified big data to over hype experimental results. As attention turns towards implementing real systems that make ethical decisions we need to ask the right questions concerning the data that these systems may be trained on. One particular concern with the data generated by the MIT Moral Machine platform is the question surrounding the gamification of data collection. Dergousoff et al. raise questions surrounding the quality of results collected through a gamified approach to data collection when compared to traditional methods [39]. Similarly, Versteeg establishes gamification as a potentially "manipulative construct" [122]. This opinion reinforces our own opinions about the Moral Machine's data quality: are participants taking it seriously? What happens if someone gets bored in the middle of playing? Do people understand that this data is being used in academic research? The data collected from the Moral Machine platform has been used in the study

Figure 6.1: Example scenario from the MIT Moral Machine Platform [1]



of Kim et al., "A Computational Model of Commonsense Moral Decision Making" [71]. Kim et al. use data to test an expanded version of Kleiman-Weiner et al.'s theoretical model presented in the previous chapter. Kim et al. achieve exciting results when attempting to learn the underlying moral preferences of Moral Machine respondents which are discussed in detail later in the dissertation. We also want to implement a computational model of Kleiman-Weiner' et al.'s proposed theory [72] using real world data. We opt to select a new dataset which allows us to question the repeatability of Kim et al.'s results [71]. Additionally, we select a dataset that has been collected in a traditional laboratory environment to avoid the ethical problems we have raised over the Moral Machine dataset. We now present our dataset.

6.4 Autonomous Vehicle Dataset: Wachter et al.

The data used for this project's implementation is from the German study "Human Decisions in Moral Dilemmas are Largely Described by Utilitarianism: Virtual Car Driving Study Provides Guidelines for Autonomous Driving Vehicles" [46] and is referred to as the "German Autonomous Vehicle" dataset for the remainder of the dissertation. Faulhaber et al. conduct a set of experiments in which participants experience modified trolley problems as the driver of a car in a virtual reality environment. Participants are forced to make one of two decisions, choosing between the left or right lane when faced with obstacles. The obstacles consist of a variety of human-like avatars of different ages and numbers.

Each participant is presented with one training track and five different experiment tracks. The car being driven by the participant is travelling at 36 mph and the tracks range in length between 180m to 200m. The experiment tracks consist of five different environments: two mountain, a suburban and two city levels. In the city scenarios, the participant has to decide whether to swerve and mount the pavement or continue on the road. Example scenarios are shown in Figure 6.2. The presented avatars are: middle aged man, old man, young boy, kneeling man, self (the participant who is driving). The avatars are all male, range in number and presented in random combinations.

Prior to starting the experiment Faulhaber et al. ensure each participant is aware of the nature of the experiment and make sure they have signed a consent form clarifying they are able to terminate the experiment at any time. Each participant is then presented with a sequence of training scenarios followed by 24 test scenarios across a combination of the five different test environments. Finally, the participants

Figure 6.2: Example scenario from the German Autonomous Vehicle Study [46]

are asked a series of questions to ascertain age, gender and driving experience alongside a questionnaire containing high level philosophical questions. 216 unpaid participants take part in the study and 201 participants complete the study.

Chapter 7

Model Design and Implementation

This chapter details a computational model that learns moral preferences by drawing on the theoretical model of Kleiman-Weiner et al. [72]. We follow the approach of Kim et al. [71] but make some adjustments such that the model can be evaluated on the German Autonomous Vehicle dataset as presented in the previous chapter.

7.1 Introduction to Kim et al. Computational Model

The paper, "A Computational Model of Commonsense Moral Decision Making" [71] introduces a novel computational model of moral decision making using the theory presented by Kleiman-Weiner et al. [72] as a basis. Kim et al.'s model describes moral dilemmas as utility functions that use abstract moral principles to compute trade-offs in decision making. Kim et al. use a Bayesian hierarchical model to categorise social structures of individuals and groups and to show that individual moral preferences can be inferred as interpretable parameters from limited data. A further contribution of Kim et al.'s proposed model is its ability to rapidly infer group norm moral theories, which is critical in designing an artificially intelligent agent that makes socially optimal choices [71]. Kim et al. evaluate their model using a sub sample from the Moral Dilemma dataset, presented in the previous chapter [1].

7.2 Decomposing Character Vectors to Abstract Feature Space

The primary challenge faced by Kim et al. is the devision of a mechanism that maps observable objects onto the abstract feature space. The model presented by Kleiman-Weiner et al. is concerned only with fabricated interactions between hypothetical agents (See Appendix A for more details). Kim et al.'s model, however, uses data from the Moral Machine platform which represent binary choice tasks in trolley problems. As discussed in Chapter 6, the Moral Machine dataset contains randomly generated scenarios which force the participant to choose which set of characters to save and which to kill. Every scenario has two choices, which Kim et al. represent with the random variable $Y \in [0, 1]$. A respondent's choice to swerve (change lanes) is represented as $Y = 1$ and their choice to stay in their initial lane is represented as $Y = 0$. The respondent's choice produces a resultant state of a set of saved characters, represented as $\Theta_y \in \mathbb{N}^K$ where y is the resultant state of choice y and K denotes the set of all possible characters where $|K| = 24$. Figure 7.1 shows the 24 possible characters.

In their model, Kleiman-Weiner et al. derive abstract moral principles over kinmanship, self, group membership, all, direct reciprosity and indirect reciprosity [72]. Kim et al. extend this model to derive abstract moral principles that can be combined to describe each of the 24 characters in the Moral Machine dataset. Kim et al. claim that the original character vector Θ_y can be decomposed into a new vector in the abstract feature space $\Lambda_y \in \mathbb{N}^{\mathbb{D}}$ where D represents the set of all possible abstract moral principles and $D \leq K$ by the mapping $F : \Theta \rightarrow \Lambda$. Specific to their model and dataset, Kim et al. use the linear mapping $F(\Theta) = A\Theta$ where A is a 16x24 matrix shown in Figure 7.1.

For our implementation, the mapping between observable objects onto the abstract feature space is altered slightly due to the differences between the Moral Machine dataset and The German Autonomous Vehicle dataset. The characters represented by the German Autonomous Vehicle dataset are: the driver of the

Figure 7.1: Binary matrix, A, that decomposes the features from the character space to features from the abstract principle space [71]

car (the respondent), man, old-man, young boy, kneeling man and pedestrians. Our character vector is therefore represented as $\Theta_y \in \mathbb{N}^K$ where $|K| = 6$. Following Kim et al.'s approach, we believe that each of the six characters can be described by one or more of the following abstract principles: self-preservation, elderly, infancy, middle-aged, kneeling-down, pedestrian. Our implementation therefore uses the linear mapping $F(\Theta) = A'\Theta$ where A' is a 6x6 matrix.

7.3 Composing Utility Functions

Kim et al. choose to model the utility value of a resultant state as a linear combination of the features in the abstract dimension:

$$\mu(\Theta_i) = w^\top F(\Theta_i), \tag{7.1}$$

where Θ_0 represents the state achieved from choosing not to switch lanes and Θ_1 represents the state achieved by choosing to switch lanes. A respondent's decision to switch lanes, $Y = 1$, is represented by the sigmoid function of net utility of the two choices [71]:

$$P(Y = 1|\Theta) = \frac{1}{1 + \exp^{-U(\Theta)}}, \tag{7.2}$$

where the net utility $U(\Theta)$ is given by

$$U(\Theta) = \mu(\Theta_1) - \mu(\Theta_0). \tag{7.3}$$

7.4 Hierarchical Model

Kim et al. begin by considering N respondents from the dataset belonging to a group $g \in G$ [71]. Similarly to Kleiman-Weiner et al., Kim et al. state that this group could represent a country, culture or group within which customs and moral norms are shared. The moral principles of an individual i within group g are then drawn from a multivariate Gaussian distribution:

$$w_i \sim \mathcal{N}_D(w^g, \Sigma^g) \tag{7.4}$$

Where w^g denotes the mean values of the group g over D dimensions and the diagonal of the covariance matrix Σ^g represents the difference between members of group g over the abstract principles. The covariance matrix Σ^g also holds information about the strength of relationships individuals place between

abstract principles. For example, if an individual values infancy highly they may also value the elderly highly. The covariance matrix allows the learner to quickly infer moral principles of one dimension after inferring those of a highly correlated dimension [71].

Let $\boldsymbol{w} = \{w_1, ..., w_i, ...w_n\}$ be the set of moral principles of N respondents and the vector $\boldsymbol{\Theta} = \{\Theta_1^1, ..., \Theta_i^t, ..., \Theta_N^T\}$ represent the resultant states of i respondents over T scenarios. The decision made by respondent i is represented by a random variable Y_i^t [71]. From this foregrounding, Kim et al. are then able to define the posterior distribution as

$$P(\boldsymbol{w}, w^g, \Sigma^g | \boldsymbol{\Theta}, \boldsymbol{Y}) \propto P(\boldsymbol{\Theta}, \boldsymbol{Y}) | \boldsymbol{w}) P((w) | w^g, \Sigma^g) P(w^g) P(\Sigma^g), \tag{7.5}$$

where the likelihood is given by

$$P(\boldsymbol{\Theta}, \boldsymbol{Y}) | \boldsymbol{w}) = \prod_{i=1}^{N} \prod_{t=1}^{T} p_{ti}^{y_i^t} (1 - p_{ti})^{(1-y_i^t)}, \tag{7.6}$$

and p_{ti} is the probability respondent i chooses to swerve in scenario T, which is given by equation 7.2. Kim et al. define a prior over the covariance matrix Σ^g as a LKJ covariance matrix with parameter $\eta = 2$, such that

$$\Sigma^g \sim LKJ(\eta). \tag{7.7}$$

They specify a prior over the group weights as

$$w^g \sim \mathcal{N}_D(0, \Sigma^g). \tag{7.8}$$

7.5 LKJ Covariance Matrix

Selecting a prior for a covariance matrix has been long established as a difficult challenge. The poor selection of a prior could result in an improper posterior distribution [4]. In 2000, Barnard et al. attempted to simplify the complexity of estimating a covariance matrix by decomposing the covariance matrix into its correlation matrix and variances [14]. However, whilst their decomposition was appealing from a theoretical perspective it turned out to be difficult to use and resulted in computational problems [4]. The LKJ covariance matrix was presented by Lewandawski, Kurowicka and Joe (LKJ) in their paper, "Generating random correlation matrices based on vines and extended onion method" [80]. Lewandawski et al. propose an alternative way of generating random correlation matrices and therefore an alternative way of achieving the separation method first proposed by Barnard et al..

Lewandawski et al. propose two methods of generating random correlation matrices. The first extends the preceding work of Joe [69]. Joe presents a method of generating random correlations based on D-vines. A vine is a special type of nested connected tree. A regular vine is a vine in which an edge in tree T_i is connected by an edge from tree T_{i+1} only if these edges share a connected node. C-vines and D-vines are specific types of regular vines. Lewandawski et al. extend the work of Joe to use both C-vines and D-vines to randomly generate correlation matrices.

Lewandawski et al. also propose an alternative method for generating correlation matrices using the oninon method, originally introduced by Ghosh et al. [55]. The onion method is derived from the fact that any matrix of size $(k+1) \times (k+1)$ can be partitioned as

$$\boldsymbol{r}_{k+1} = \begin{bmatrix} \boldsymbol{r}_k & \boldsymbol{z} \\ \boldsymbol{z}^T & 1 \end{bmatrix}, \tag{7.9}$$

where $\boldsymbol{r}_k$ is an $k \times k$ correlation matrix and z is a k-vector of correlations [80]. Lewandawski et al. then use the onion method to generate random correlation matrices of d dimensions with one free parameter, η, and density proportional to $[det(\boldsymbol{r})]^{\eta\text{-}1}$, where $\eta > 1$ [80].

7.6 Implementation of Model in PyMC3

Kim et al. continue to infer the model's individual weights, group norms and covariance matrices. Kim et al. do not specify what technique they use to infer these parameters but we assume that they use a maximum a posteriori estimate. Due to the reasons given in Chapter 4.5 we decide to use Markov Chain Monte Carlo to infer the posterior distributions over the model parameters.

Probabilistic programming allows for automatic Bayesian inference on user defined Bayesian models. PyMC3 is an open source, probabilistic programming framework that features the most recent Markov Chain Monte Carlo algorithms such as the No U-Turn Sampler. Probabilistic programming in Python offers many advantages including compatibility with many statistical and mathematical libraries. Despite most of PyMC3's user facing functionality being written in Python, it also leverages the Theano framework for computation optimisation. We therefore use PyMC3 for our model implementation [111].

We begin the implementation by performing pre-processing on the data which involves codifying the mapping $F : \Theta \rightarrow \Lambda$ transforming character vectors into abstract feature vectors. We continue to specify the model. Specifying the model in PyMC3 is relatively easy as the notation closely resembles mathematical notation.

We begin by specifying the hyperparameters and hyperpriors. To define the covariance matrix Σ^g we use the PyMC3 inbuilt function for specifying LKJ priors, *LKJCholeskyCov*. This distribution is determined by its parameters, shape and standard deviations. We set shape to six as $|D| = 6$. The standard deviation parameter controls the diagonals of the covariance matrix. We model the standard deviation parameters with normally distributed priors, centered around 1 with narrow variance of 0.1^2. Each LKJ correlation matrix, Ω, has a Cholesky factorization, $\Omega = LL'$, where L is a lower triangular matrix (a matrix where values are all 0 above the main diagonal). For efficiency reasons the PyMC3 function *LKJCholeskyCov* only stores the vector $\boldsymbol{v}$ which contains diagonal and sub-diagonal entries. We therefore expand $\boldsymbol{v}$ into the lower diagonal matrix L using the PyMC3 function *expand_packed_triangular*. From here, we could compute the full correlation matrix by computing $\Omega = LL'$ however, it is often more efficient to specify a multivariate Gaussian distribution using the Cholesky decomposition instead of the full correlation matrix [111].

We can then specify the group-norm hyperparameter which we do using a the PyMC3 inbuilt distribution *MvNormal* which defines a multivariate Guassian distribution over the group norm vector. The *MvNormal* distribution has mean and covariance parameters . The mean parameter μ is set to 0 and covariance parameter is the Cholesky decomposition of the LKJ correlation matrix. We then model the individual weights parameters as another multivariate Guassian distribution with *MvNormal*. We set the mean parameter to be the group norm hyperparameter and model the covariance as the Cholesky decomposition of the LKJ correlation matrix.

Finally, we can model the likelihood as the Bernoulli distribution using the inbuilt PyMC3 distribution *Bernoulli* which takes parameters p, the probability of success, and a vector of observed data. The probability of success is defined as $P(Y = 1|\Theta)$ as given in Equation 7.2. The observed data is given by the vector $\boldsymbol{Y}$ representing the decisions of all respondents over all scenarios.

After specifying the model, we can sample from it using one of PyMC3's featured Markov Chain Monte Carlo algorithms. We select the No U-Turn Sampler (NUTS) algorithm for reasons described in Chapter 4.6. The NUTS algorithm has multiple parameters to specify:

- Draws: This parameter determines the number of samples to be taken from the posterior distribution.
- Tune: This parameter specifies how many extra samples should be taken from the posterior and be discarded which allows the sampler to adjust its parameters.
- Step: This parameter determines the function that controls the step size and method. The NUTS algorithm tunes these things automatically.
- Init: This parameter determines how the sampling algorithm should be initialised. The default is *jitter+adapt_diag* which makes the NUTS algorithm change the mass matrix during tuning to

match the samples already drawn from the posterior.

- Chains: This parameter specifies the number of chains to sample, multiple chains are often required to determine whether the sampler has performed optimally.

The NUTS sampler returns a trace object which contains all generated samples. We now run the NUTS sampler algorithm on our specified model with samples = 500, chains=2, init=jitter+adpat_diag and tunes = 500.

7.7 Evaluation of Model

One of the advantages of using Markov Chain Monte Carlo is the multitude of associated evaluation heuristics which we can use to ascertain biased inference.

7.7.1 Gelman-Rubin Diagnostic

The Gelman Rubin diagnostic is one of the most effective methods for ascertaining whether the sampling algorithm has converged. Convergence determines whether the simulated Markov chain has fully explored the target posterior distribution. A common approach to establish convergence is to compare multiple chains. The Gelman Rubin statistic was proposed in the instructive work, "Inference from Iterative Simulation Using Multiple Sequences" [52], to analyze the difference between two Markov chains. Gelman and Rubin approach the problem by letting $m \geq 2$ be the number of Markov chains sampled from the target posterior distribution, where each chain has length $2n$. They assert that to ignore the effects of the starting distribution, the first n iterations should be discarded [52]. Then, for each model parameter the following scalars are calculated: $\frac{B}{n}$, which represents the variance between the m sequence means, and W, the average of m within chain variances. Then, the estimated marginal posterior variance of a particular parameter is equal to.

$$V = \frac{n-1}{n}W + \frac{m+1}{nm}B. \tag{7.10}$$

This heuristic is unbiased assuming that the target distribution is equal to the posterior distribution [52]. The convergence of the simulation can then be ascertained by calculating how much the current distribution for the parameter might be reduced if the chains continued infinitely [52]. This potential scale reduction factor (psrf) is given by:

$$\sqrt{R} = \sqrt{\frac{V}{W}\frac{df}{df-2}} \tag{7.11}$$

.

R declines to 1 as $n \to \infty$. Brooks and Gelman expand on this theorem [25] by normalising R to take sample variance into account, thus deriving R_c. They suggest that if $R_c \leq 1.2$ then it is fairly certain that the sampler has converged where $R_c \leq 1.1$ is optimal and $R_c \geq 1.4$ indicates the sampler has not converged at all [25]. Figure 7.2 shows a histogram of Gelman Rubin statistics for all parameters. All values are below 1.05 showing promising convergence of the sampling algorithm. However, it would be preferential to have more of the statistics concentrated between 1 and 1.01 to be more certain of convergence for all parameters.

7.7.2 Poorly Chosen Kinetic Energies

One particular behaviour that makes approximation via Markov Chain Monte Carlo methods challenging is a heavy tailed target distribution [19]. Further, if the kinetic energy is poorly chosen for Hamiltonian Monte Carlo algorithms then the marginal energy distribution will itself become heavy tailed and Markov chain exploration will become so slow that the chain will not explore the full space. Fortunately, these badly chosen kinetic energies are easy to spot by visualising the marginal energy density and transition energy density together. When these energies are well matched and similar in shape, the Hamiltonian Markov chain should perform robustly. However if the shape of the transition energy density is significantly narrower than that of the marginal energy density then we can assume that the chain has not explored the target distribution entirely [19]. Figure 7.4 shows the energies explored by the Hamiltonian Markov chain under the NUTS algorithms. The graph shows that the transition energy density is much narrower than the marginal energy density which indicates that the Markov chain has not explored the target distribution effectively.

Figure 7.2: Histogram showing Gelman Rubin statistic for all model parameters under a centred parametrization

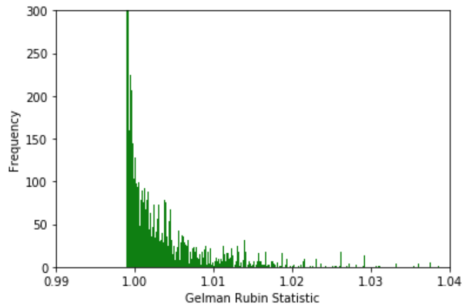


Figure 7.3: Histogram showing Gelman Rubin statistic for all model parameters under a non-centred parametrization

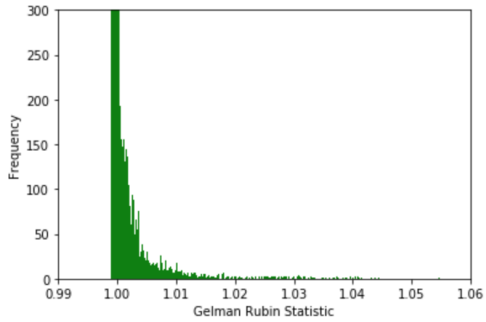


Figure 7.4: Energies explored by Hamiltonian Markov Chain under the NUTS algorithm for a centred parametrization.

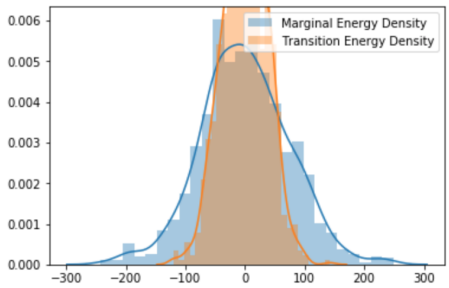
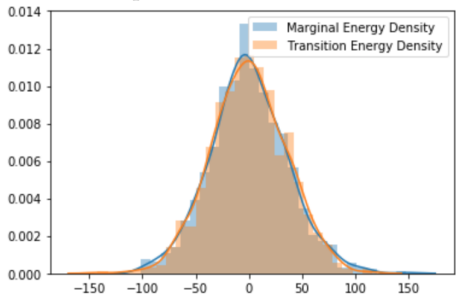


Figure 7.5: Energies explored by Hamiltonian Markov Chain under the NUTS algorithm for a non-centred parametrization.



7.7.3 Centred vs Non Centred Model

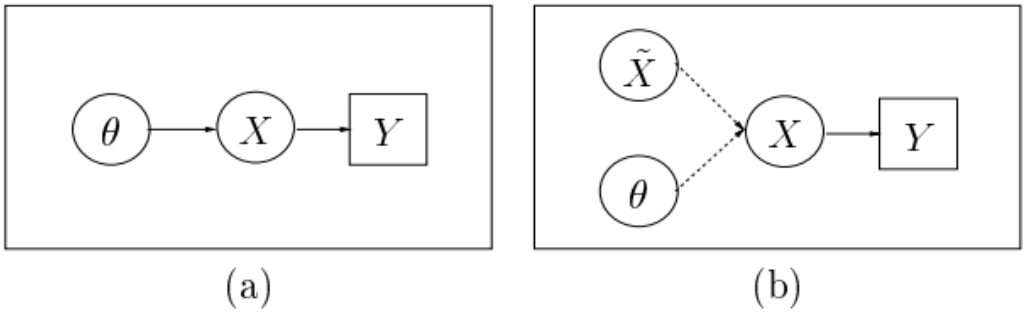
To address the convergence and exploration issues we consider using a non-centred parametrisation. The differences between a centred and non-centred parametrization are neatly explained by Figure 7.6 [18]. In non-centred parametrisations, the aim is not to fit the model parameters X directly but to use dummy variables $\tilde{X}$ that are then used to recover the model parameters under transformations. This often simplifies the shape of the target distribution and makes it easier for the sampler to explore. The benefits of non-centred parametrization over centred parametrization are summarised by Bernardo et al. in "Non-centered parameterisations for hierarchical models and data augmentation" [17]. Bernardo et al. argue that despite the computational advantages of a centred approach, non-centred parametrisations improve convergence of Markov Chain Monte Carlo techniques in cases where the latent process is weakly identified by the data [18]. In fact, PyMC3 directly suggests to use a non parametrized model when defining the covariance prior of a multivariate distribution with a LKJ covariance matrix [111].

To specify a non-centred parametrisation of the multivariate Guassian priors over the group-norms and individual weight parameters, we first define a normally distributed prior over dummy group norms and dummy individual weights centred around zero with standard deviation of one. We then compute the dot product of these dummy parameters with the Cholesky decomposition of the LKJ correlation matrix.
Figure 7.4 shows how the non-centred parametrisation results in a larger proportion of parameter Gelman-Rubin statistics are concentrated between 1 and 1.01, indicating better convergence than the centred parametrisation. Figure 7.5 shows how the non-centred parametrisation results in less difference in shape between the marginal energy density and transition energy density indicating that the Markov Chain has explored the target distribution much better than the centred parametrisation.

7.8 Results

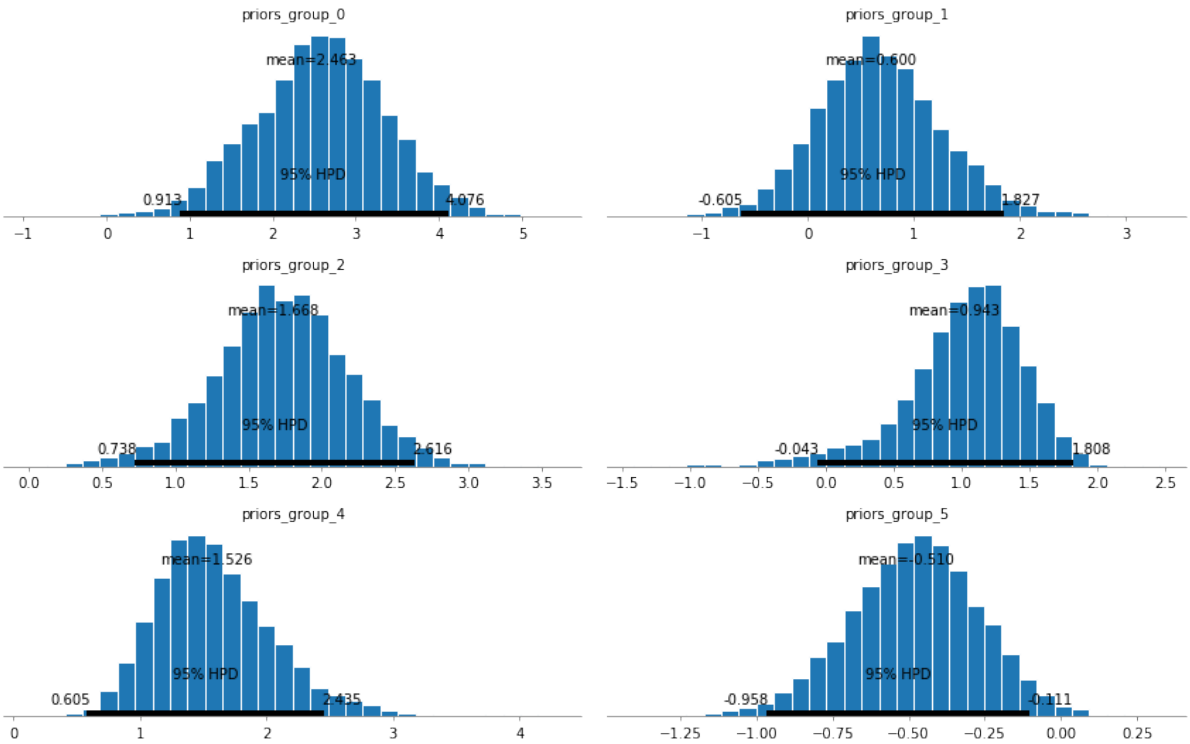
One of the main advantages of using Markov Chain Monte Carlo is its ability to approximate posterior distributions for model parameters. Solonen expresses the importance of drawing enough samples from

Figure 7.6: Example of a) A Centred Parametrization b) A Non-Centred Parametrization



the posterior distribution to allow the sampler the greatest chance of converging [117]. The posterior distributions for the model's group norm parameters over 5000 samples, with 1000 tuning samples, is shown in Figure 7.7. As these posterior distributions seem to closely resemble normal distributions, Markov Chain Monte Carlo techniques can be used to derive parameter point estimates by taking the average over the posterior distribution. Kim et al. use point estimates to discuss the success of their model in inferring moral principles. Therefore, below we discuss the accuracy of the model in terms of point estimates that have been obtained from Markov Chain Monte Carlo.

Figure 7.7: Inferred posterior distributions for group norm model parameters. Top Left: self preservation, Top Right: elderly, Middle Left: infancy, Middle Right: middle aged, Bottom Left: kneeling down, Bottom Right: pedestrian



When collecting data to evaluate their model , Kim et al. randomly select 10,000 respondents from the MIT Moral Machine website and filter each of the respondent's first 13 answers to randomly generated dilemmas. Kim et al. continue to split each respondent's answer set into a training set of the first eight answers and a testing set consisting of the remaining five answers. In order to evaluate the effectiveness of their model in inferring the moral preferences of individuals Kim et al. compare the accuracy of their model in predicting individual responses against a benchmark model, Benchmark 1, which models respondent values along the abstract moral principles Λ. It models weights as

$$w^f \sim Normal_D(\mu, \sigma^2 I), \tag{7.12}$$

which ignores hierarchical structure and assumes that inferring the moral preferences of an individual does nothing to inform the inference of others. We also introduce a new benchmark, Benchmark 2, that models individual weights as

$$w_i^l \sim [1, 1, 1, 1, 1, 1], \tag{7.13}$$

representing the assumption that an individual assigns equal weight to each abstract principle.
The results of Kim et al. can be found in Appendix B, where Kim et al. note the success of their model in inferring individual weights. Whilst we recognise that comparing the predictive accuracy of our implementation with that of Kim et al. is not significant due to different datasets, we still want to compare the results to ascertain model generality. Further, we recognise the fragility of Kim et al.'s evaluation due to their lack of cross validation. Cross validation is critically important to generalize results. Picard et al. identify that without performing cross validation, bias and variance could mislead the naive into thinking results are better than they actually are [104]. We therefore apply cross validation to our data set, using four separate partitions. The variance across partitions is represented by the vertical bars in our graphs. Figure 7.8 shows the comparative predictive accuracies of the implemented model when compared to the two proposed benchmarks. We can see that the hierarchical model outperforms both benchmarks in predicting out of sample individual responses. Kim et al. note during their evaluation that the flat model, Benchmark 1, as well as having a lower predictive accuracy than the hierarchical model rapidly decreased in accuracy as the number of respondents used in training decreased which we can also see reflected in our own results as the predictive accuracy for Benchmark 1 dramatically decreases as the sample size drops below 20. This confirms the hypothesis of Kim et al. that a Hierarchical model is an effective mechanism for learning moral preferences from sparse data.

The comparative success of both Benchmark 1 and the Hierarchical model, when compared to Benchmark 2, seems to provide experimental evidence to support the existence of moral principles that are unique to individuals and govern the judgements we make in moral dilemmas. Interestingly, the variance between partitions is greatest within Benchmark 3, which suggests that whilst most humans operate in a utilitarian way there is great variability amongst individuals in their beliefs that all humans should be treated equally. This conclusion was also made by Faulhaber et al. upon conducting the original study [46]. The proposed hierarchical model therefore has offered a solution that deals with inter-person moral variability.
One of our earlier observations was that expectations computed via Markov Chain Monte Carlo techniques explored the posterior distribution more effectively than optimization techniques which could result in more accurate and reliable inferences. Whilst this is true theoretically, often Markov Chain Monte Carlo algorithms can often get stuck in local optima and generate inaccurate expectations. We therefore wanted to compare the differences between obtaining parameter expectations from averaging the inferred posterior distribution and point estimates achieved through maximum a prosteriori. Figure 7.9 shows that, for larger sample sizes, Markov Chain Monte Carlo estimates outperform maximum a posteriori (MAP) which seems to confirm the theory outlined in Chapter 4, that point estimates obtained by approximation techniques are better than those obtained by optimization alone.

This chapter has seen a model successfully infer the individual moral preferences of our dataset population. The hierarchical model achieved a predictive accuracy of 82.17% with a variance of 0.92 over four partitions when evaluated using a sample size of 201 respondents. These results suggest that individuals within the dataset use a unique set of moral principles to calculate an optimum decisions. However, many assumptions were made about human behaviour throughout the creation of the model. When attempting to model an abstract characteristic such as morality it is inevitable that certain assumptions must be made. However, the proceeding chapters emphasize the sensitivity of Bayesian models to these assumptions.

Figure 7.8: Pointplot showing the differences in predictive accuracies between three different models. The Hierarchical model is the implemented model detailed above, Benchmark 1 shows results from the flat model, Benchmark 2 shows results from a model that assumes all individuals have equal weights over all moral principles. The Y axis accuracy shows predictive accuracy over individual decisions. The X axis represents the number of respondents in sample. The vertical bars show variance from cross validation.

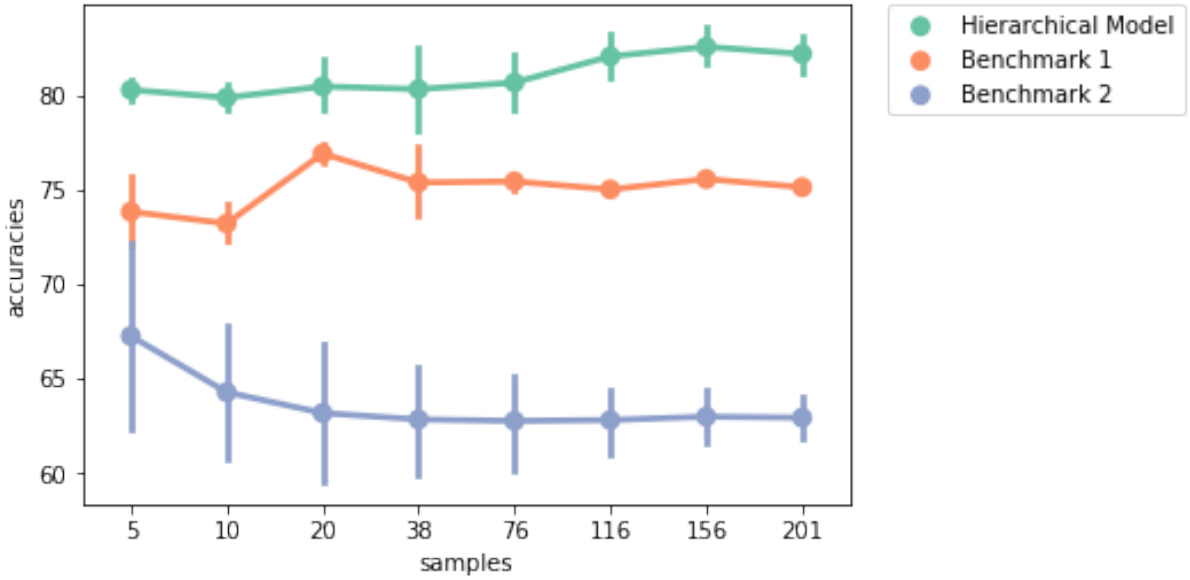
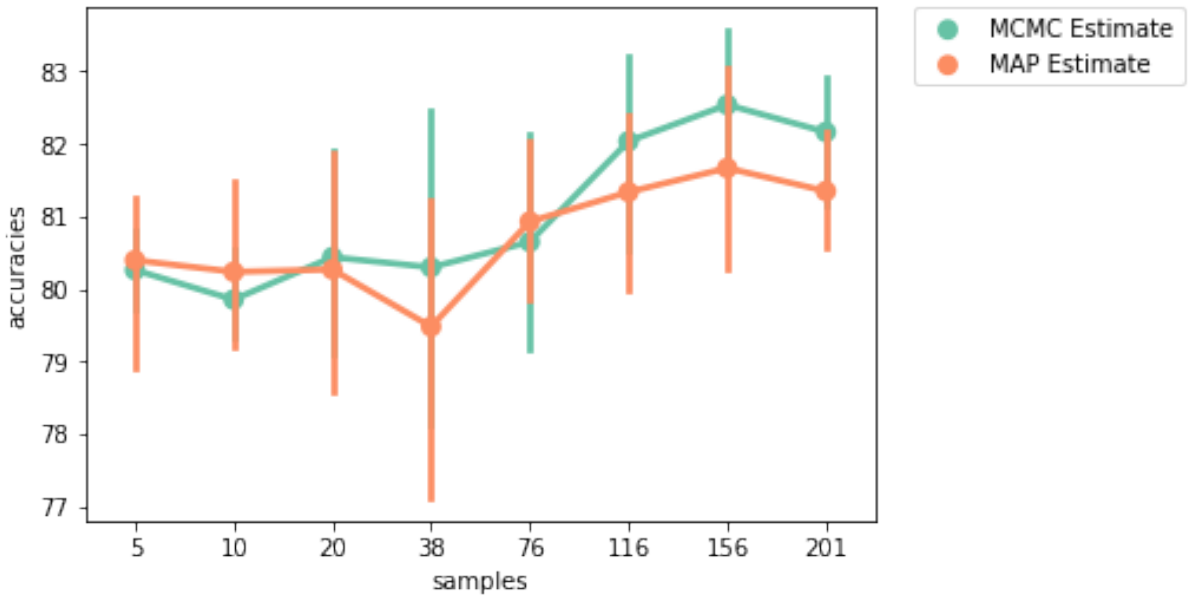


Figure 7.9: Pointplot showing the differences in predictive accuracies between point estimates of individual moral weights. The MCMC Estimate shows point estimates achieved by taking the mean of inferred posterior distributions obtained from the MCMC, the MAP estimate shows accuracy achieved through point estimates obtained through maximum a posteriori.



Chapter 8

Questioning Prior Assumptions

This chapter starts to raise questions surrounding the assumptions made by Kim et al. and Kleiman-Weiner et al. when deriving their models. We begin by considering the prior assumptions made by both models. Historically, prior assumptions have been identified as highly influential on statistical models. Further, Kleimen-Weiner et al. identify the insufficient exploration of optimal prior distributions for their model and highlight this as an area for future work. We therefore make the hypothesis that the priors chosen by Kim et al. were chosen primarily for practicality and that, with further consideration, prior distributions could be found that more accurately reflect prior beliefs in moral theories.

8.1 Importance of Prior Assumptions

One of the biggest differences between Bayesian reasoning and traditional methods is the inclusion of subjective beliefs about a probability in calculations. Bayesian inference is therefore composed of both current and prior knowledge which act together to make up the posterior [120]. Van Dongen voices his opinion that, despite the multitude of advantages of Bayesian methods, there remain two central challenges today [120]. The first being the convergence of Markov Chain Monte Carlo techniques which have been discussed in the previous chapter. The second, Van Dongen claims, is prior specification. The prior distribution plays a central role in Bayesian inference, particularly in circumstances where the likelihood does not dominate the posterior, i.e. the volume of data is limited. Prior specification becomes particularly challenging when using hierarchical models for Bayesian inference, as these models require hyperparameters which require their own prior distributions [49].

Gelman presents some considerations that should influence decisions on prior distributions [49]. The first are "conditionally conjugate prior families". He defines these as the class of prior distributions whereby the conditional posterior $p(\phi|y)$ is in the same class [49]. For a computational perspective, conditional conjugacy means that if it is posssible to draw ϕ from the special class prior distributions, then it is possible for a Markov Chain Monte Carlo sampler to draw ϕ from the posterior distribution. This property is useful in hierarchical models as it is preserved when models are expanded hierarchically whereas conjugacy is not. Gelman continues to introduce the idea of improper priors which are priors that integrate to a non-finite value. These priors sometimes lead to improper posterior distributions [49]. Gelman characterizes weakly informative priors as distributions that are proper yet set up to reveal less information than the prior knowledge that is actually available, whereas non-informative priors are used to "let the data speak" [51] when informing the posterior. In models where little information is known about certain parameters, non-informative priors are usually used. However, many researchers may use informative priors based on their own subjective beliefs. These assumptions have been subject to rigorous debate [120]. Indeed, the entire subject of non-informative priors remains an open discussion in academia, with many researchers using trial and error to ascertain the optimum distributions for their models in cases where prior knowledge is scarce. We propose a similar form of experimentation for our implemented model.

We conjecture that as a society we still know little about human morality, where it originates, and how it varies from individual to individual or group to group. The argument presented in Chapter 3 is testament to this belief. We argue that the priors chosen in the model of Kim et al. make assumptions about the moral preferences of the underlying data that do not reflect theory. The decision of Kim et al. to model

individual weights and group norms as precisely normally distributed is particularly questionable. Firstly, considering group norms: do we expect these values to be closely concentrated around zero? How do we expect these values to be correlated? Secondly, considering individual weights: do we consider these values to be closely concentrated around the group norm with low mass in distribution tails? Is this an accurate reflection of society? We now present alternative prior distributions that attempt to address these issues.

8.2 The Strength of Non-informative Priors

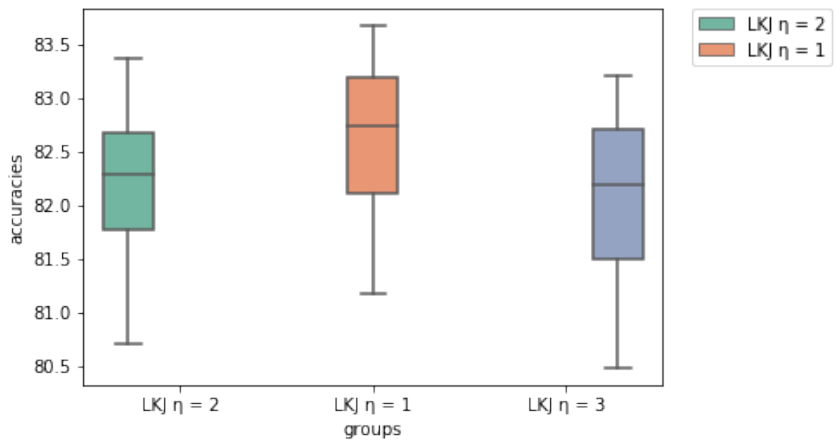
8.2.1 LKJ Covariance Prior

The Separation Strategy is widely considered an optimal approach to specifying a prior on covariance matrices as it ensures independence between correlations and standard deviations [4]. Furthermore, the LKJ method is a computationally efficient implementation of the Separation Strategy and is thus recommended as the default for various probabilistic programming platforms. Alvarez et al. however, express the difficulty in specifying the value of the free parameter η. Setting $\eta > 1$ favours diagonal correlation matrices and is thus suited to models where there is a prior assumption of independence. Setting $\eta = 1$ implies a jointly uniform prior on all correlations. The model of Kim et al. model the covariance matrix as,

$$\Sigma^g \sim LKJ(\eta = 2) \tag{8.1}$$

However, there is little moral theory or evidence that accounts for the extent as to which moral principles are related to each other [33], if they are even related at all! We therefore propose experimentation with the LKJ distribution under different values to η to ascertain the prior over the covariance matrix that best suits the structure of the data. Figure 8.1 shows the results of our experimentation varying η. The boxplot shows that the model achieves greatest predictive accuracy when modeled using the LKJ covariance prior with $\eta = 1$. This implies our earlier hypothesis, that weakening prior assumptions more accurately reflects the structure of the data. However, it is important to consider the variances across subsamples. The model where $\eta = 1$ has the lowest variance of all three parameterisations, which suggests a more robust model. However, the inter-quartile range is lowest in the model where $\eta = 2$, which again could suggest optimised robustness. Further comparisons could be performed to confirm a preferential hyperparameter for the covariance matrix, however, our results begin to show that weakening assumptions could be beneficial.

Figure 8.1: Box plot showing results of experimentation with the parametrisation of the LKJ covariance prior.



8.2.2 The Half Cauchy Distribution

A main concern over the model of Kim et al. is the normally distributed priors with narrow variance. The standard deviations for the mutivariate Gaussian distributions are currently normally distributed themselves as:

$$\sigma \sim \mathcal{N}(1, 0.1) \tag{8.2}$$

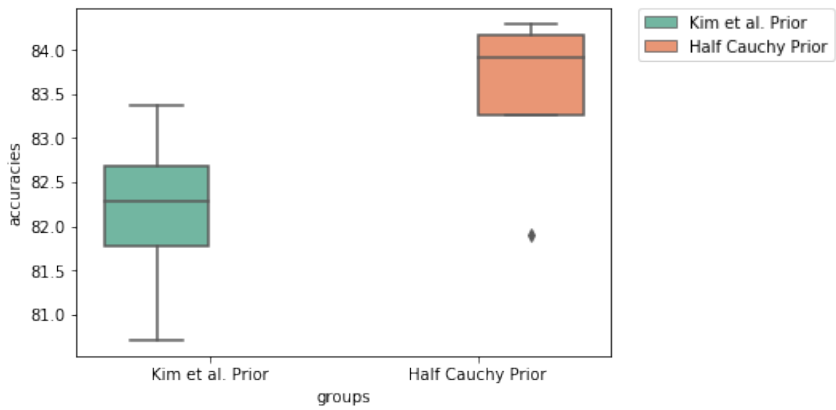
We propose the use of a weaker prior over the standard deviations of the model. This will challenge the idea that individual moral principles are tightly clustered around a group norm and the idea that group norms are constrained to lie between a narrow range.

Gelman identified that many of the distributions that have been historically used for non-informative priors introduce levels of subjectivity into results [49]. Furthermore, Gelman proposes the half Cauchy density as a reference non-informative prior that should be used for the standard deviation term in a hierarchical model. The half Cauchy is a special case of the conditionally-conjugate folded-non central-t family of prior distributions for variance [50]. It has a wide peak at 0 and also has a scale parameter A. As A tends towards infinity, the half Cauchy distribution becomes a uniform prior on the standard deviation. Further, large and finite values of A represent a weak prior as the distribution has a gentle slope in the tail. We propose modeling the standard deviations of the group-norm parameters and individual-weights as,

$$\sigma \sim halfCauchy(10) \tag{8.3}$$

Figure 8.2 shows the comparative predictive accuracies between the standard deviation hyperprior specified by Kim et al. and the half Cauchy hyperprior. From the position of the median of the half Cauchy distribution we can see that samples are concentrated in the upper quartile and the short boxplot emphasizes low variability between partitions. As the lowest accuracy sample is statistically classified as an outlier, we can see that the half Cauchy prior significantly improves the predictive accuracy of the model. This re-enforces our hypothesis that Kim et al.'s prior assumptions are too specific to represent the data. Following this conclusion, we propose a new line of questioning over the use of normal distributions to model moral weights. There is no concrete evidence to suggest that human moral beliefs are concentrated around a central group value. We therefore propose to model individual weights with a more general distribution that has more mass in its tails.

Figure 8.2: Box plot showing predictive accuracy of the hierarchical model, comparing modeling the standard deviation hyperprior as a narrow normal distribution and a half Cauchy distribution.



8.2.3 The Student T Distribution

In their work "Student-t Processes as Alternatives to Gaussian Processes" [112], Shah et al. highlight the popularity of the normal distribution due to its interpretability, large support base and the success of its empirical results. They continue to present Student-t processes as a family of elliptical processes that generalise the normal distribution, derive a definition for the Student-t-process by placing an inverse Wishart process on the kernel of a Normal distribution and give a definition of the Student-t-distribution as:

$$(f(x_1), ..., f(x_n))^T \sim MVT_n(v, \phi, K) \tag{8.4}$$

Where, $K \in \Pi(n)$ is the covariance matrix and $\phi \in \mathbb{R}^n$ is the mean vector [112]. Shah et al. continue to stress the importance of the v parameter which controls how heavy tailed the Student-t distribution

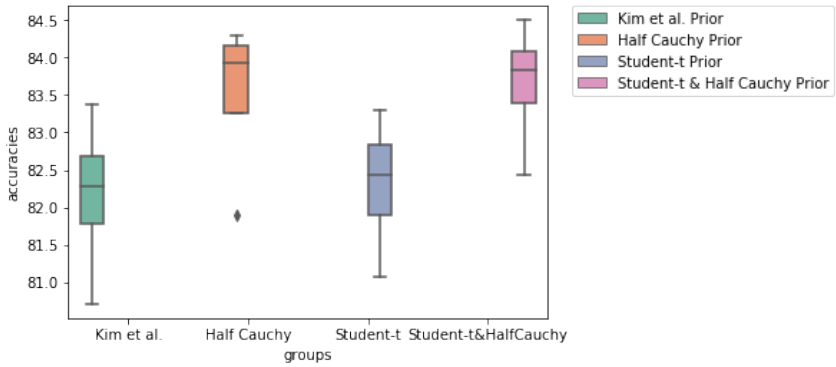
is. As v increases, the Student-t distribution converges to a normal distribution. We hypothesise that this property of the Student-t distribution may reflect a more accurate prior assumption of the group norms and individual weights. By using a distribution with more mass in the tails we are weakening the assumption on the model parameters, releasing them from the pressure to be concentrated around a central value. Shah et al. finish by expressing the advantages of using a Student-t distribution over a normal distribution, naming its robustness to model mis-specifications, improved predictive covariances and to be especially useful for Bayesian inference where the covariances are important. We therefore propose to model the group norms and individual weights as:

$$w^g \sim MVT(5, 0, \Sigma^g) \tag{8.5}$$

$$w^i \sim MVT(5, w^g, \Sigma^g) \tag{8.6}$$

Figure 8.3 is a box plot showing the predictive accuracy of the hierarchical model comparing modeling the individual weight parameters and standard deviation hyperprior as normally distributed; modeling the individual weight parameters as normally distributed and hyperpriors as half Cauchy; modeling the individual weight parameters with a Student-t distribution and hyperpriors with a normal distribution; modelling the individual weight parameters with a Student-t distribution and modeling the standard deviation hyperpriors with a half Cauchy distribution. Figure 8.3 shows that whilst the combined model achieves the greatest predictive accuracy over a single partition, the box plots reveal that the half Cauchy model remains the most robust with the lowest variance value across partition. These results show the difficulty in selecting a prior distribution for a model. Is it more important to have a consistently performing model or achieve the greatest predictive accuracy? However, through demonstrating the sensitivity of the model to parameterisations, we have motivated the importance of carefully considering prior distributions, particularly in cases where prior knowledge is scarce. Furthermore, in the context of morality we must be careful to view the results in light of our dataset and stress the importance of further study in this area to generalise results.

Figure 8.3: Box plot showing predictive accuracy of Hierarchical model when modeled with different hyperprior distributions



Chapter 9

Questioning Utility Functions

One of the central components of the model presented by Kim et al. is the use of utility calculus as a mechanism to compute welfare-trade offs that determine individual moral decisions. The idea of quantifying weights over moral principles that can be summed to determine an overall utility is synonymous with the ethical perspective of utilitarianism and places a big assumption over the ways in which humans make moral decisions. This chapter therefore raises questions over the use of utility functions to infer individual moral principles and attempts to ground the experimental results found by our model in ethical theory.

9.1 Moral Principles

The first question we wish to raise is whether humans use moral principles to make decisions and indeed, whether these principles even exist. This debate has been ongoing in the field of ethics and is encapsulated by the Particularism vs. Generalism argument, which refers to the families of attitudes surrounding moral principles [59]. The difference between both schools of logic is nicely detailed by J.Horty in his formative work, "Reasons as Defaults" [64]. Horty presents Generalism as the attitude that our ability to reach decisions is governed by, at some level, general principles. In contrast, Horty presents Particularism as the attitude that downplays the importance of general principles, instead emphasizing the role of receptivity to features of particular situations as the mechanism behind decsion making. In his work, Horty presents a counter-argument to the popular opinion given by J Dancy in support of Particularism [35]. Dancy argues that "a principle based approach to ethics is inconsistent with the holism of reasons" [35], where the "holism of reasons" refers to the idea that what counts as a reason for an action in a particular scenario may not apply to another scenario.

From the German Autonomous Vehicle dataset, we can clearly see evidence for the "holism of reasons". An individual who in one context chooses to save an elderly person over a child may do the opposite when given an alternate scenario. Contrastingly, Horty uses default logic to present a counter argument to Dancy [64]. Horty develops a framework that supports the "holism of reasons", whilst being built on a system of principles, which Horty terms defaults. Horty's central argument is based on the idea that reasons be thought of as being provided by defaults, where the default $A \rightarrow B$ means that A provides the reason for B. Horty continues to present the idea of priority relations between defaults, such that if default $\delta_1 : A \rightarrow B$ default $\delta_2 : C \rightarrow\!\!\!/ B$ and default δ_3 specifies the priority relation $T : \delta_1 \succ \delta_2$. This ordering means δ_2 has a higher priority than δ_1 and therefore, if there was a scenario where A and C both hold, δ_1 would be defeated. Horty continues to expand this argument to derive a framework that uses the idea of defaults and priority relations to support the notion of "reason holism" [35]. We argue that our model implementation shows some experimental evidence for Horty's argument. From our model's inference of individual moral weightings we can define a priority default for each individual that ranks the defaults δ_1 : Child $\rightarrow$ Swerve; δ_2 : Elderly $\rightarrow$ Swerve; δ_3 : Self $\rightarrow$ Swerve; δ_4 : Kneeling $\rightarrow$ Swerve; δ_5 : Man $\rightarrow$ Swerve; δ_6 : Pedestrian $\rightarrow$ Swerve.

If an individual therefore has the priority default that gives δ_1 higher priority over δ_2, given a moral dilemma that presents a child and elderly man in each lane of the scenario, the priority relation would determine the outcome. However, Horty also presents the idea that priority relations are variable which supports the evidence we see in the data that decisions are not consistent over every scenario. Similarly,

Guarini et al. present neural networks as an artificial intelligence tool that could, in the future, help to resolve the Particularism vs. Generalism debate [59]. We have seen how experimental evidence gathered by our model can be applied to the context of ethical theory. We do not pretend that this contribution has brought a resolution to the Particularism vs. Generalism argument, however, we have highlighted how exploration into artificial morality could contribute to historical ethical conundrums and help to illuminate ways forwards.

9.2 Do Humans Use Utility Functions to Make Decisions?

The second question we want to raise is the applicability of utility theory to moral decision making. Utility theory has been used successfully in economics since its introduction by Von Neuman et al. in 1945 [123], who defined utility theory in terms of the attempts of the individual to obtain a maximum of utility or a maximum of profit. However, the differences between moral theory and behavioural decision theory is nuanced and difficult to define, thus questioning the use of utility theory in moral decision making. Further, this approach to moral decision making falls under the utilitarianism approach to ethics. Historically, researchers have identified utilitarianism and deontology as the two central competing modes of moral decision making [57]. Ditto and Liu discuss the differences between both perspectives and attempt to explain how they both contribute to moral decision making [42]. The findings of Ditto and Liu can be summarised in the conclusion that humans tend to balance utilitarian outcomes with deontological principles [42]. For example, if an individual believes it is wrong to kill children and they are presented with a scenario that forces them to choose between saving five men or one child, this individual would increase their weighting of the child to be in line with their deontological standing. Ditto and Liu therefore propose that deontology and utilitarianism are inextricably linked [42]. We now incorporate the perspective of Ditto and Liu into the model proposed by Kim et al..

9.3 Incorporation of Deontological Statements Into Model

The previous chapter showed how weakening prior assumptions by generalising could lead to better results when prior beliefs in model parameters are limited. However, Gelman refers to these non informative prior distributions as "reference models" [49] with the view that these should be replaced with more informative prior distributions once more knowledge was available [49]. We therefore propose to use individual questionnaire results to inform the prior of our model.

As part of the German Autonomous Vehicle study, Faulhaber et al. [46] ask participants a series of questions following their completion of the moral dilemma trials. These questions are intended to gauge the respondent's perspective on general ethical questions, i.e. "Protecting oneself should have priority over protecting others" and "It is sometimes acceptable to sacrifice elderly people in order to prevent the death of another person". Participants are asked to give each statement a score ranging from one to seven. A score of seven indicates that the participant strongly agrees with the statement whereas a score of one indicates complete disagreement. These questions obtain a sense of the individual's view of the rightness or wrongness of the action, removed from a specific context, which can be equated to their deontological perspective. We therefore propose to take the quantification of a participant's deontological beliefs and combine this into the model such that an individual's weights are modeled by:

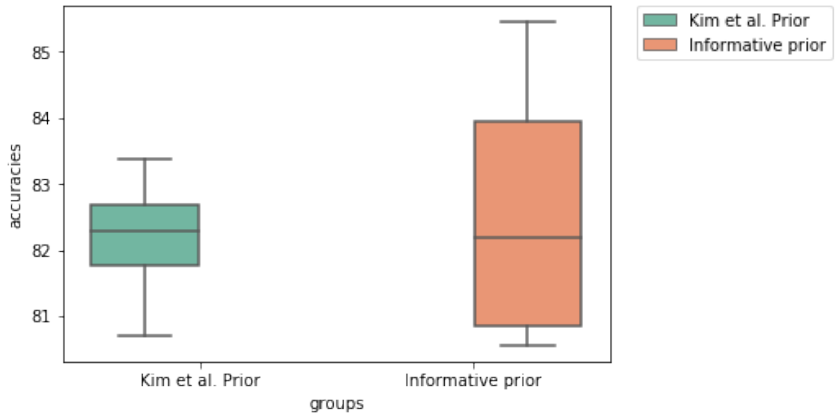
$$w_i \sim \mathcal{N}((w^g + \phi)/2, \Sigma^g), \tag{9.1}$$

where ϕ represents the vector of individual i deontological beliefs.

Figure 9.1 shows that by incorporating deontological statements into the model's prior distributions the model achieves its highest recorded predictive accuracy of 85.5% which seems to support our hypothesis that deontological statements are intrinsically linked to how we weight alternatives in moral dilemmas. However, the high variability across the partitions for this prior specification suggests that our hypothesis is not true over all individuals and that there is high variability amongst our data as to the extent in which individuals use deontological rules to shape their actions. These results, however, motivate further study to provide more conclusive results as to whether deontological and utilitarian perspectives are linked. Further, if these two ethical perspectives are proven to be linked, this result could be invaluable

in resolving the debate over optimum ethical theories, as discussed in Chapter 3, which we have seen as being a substantial hurdle to ethical artificial intelligence.

Figure 9.1: Box plot showing predictive accuracy of the hierarchical model with priors specified by Kim et al. [71] when compared to the model initialised with informative, deontological priors.



Chapter 10

Challenging the Hierarchical Model

10.1 Decomposing a Population into Sub-Groups

The theoretical framework proposed by Kleiman-Weiner et al. [72] heavily emphasizes the influence of culture in the formation of moral preferences and uses this idea to motivate the use of hierarchical models. Inter-cultural moral difference has been strongly confirmed throughout existing literature [20], [56], [65]. Futhermore, the decomposition of a population into its composite cultures generates some interesting questions when we start to consider the possibility of universal ethical machines. Namely, what is the optimum way of combining moral preferences from multiple cultures into an autonomous ethical decision maker? Contrastingly, the existence of inter-cultural moral variance could negate the possibility, and perhaps the need, for an universal ethical machine. From the perspective of an autonomous vehicle manufacturing company, would the company make more profit by marketing different versions of the same car with different ethical values to different groups of people. There are obviously huge ethical and legal problems associated with this approach. However, inferring moral principles over different cultural groups is a key area for future study. Unfortunately, the model of Kim et al. assumes all respondents from the Moral Machine dataset belong to the same cultural group [71]. Similarly, the study conducted by Faulhaber et al. [46] uses participants from the same region of Germany and provides no further cultural insight. This dissertation, does, however question whether the idea of decomposing a population into subgroups can be extended to differentiate over age and gender. Not only does this experimental perspective appeal to car manufacturing companies (a car for him, a car for her!), but, more importantly, it could help moral theorists to better understand moral origins and development.

In the German Autonomous Vehicle study, participants were required to give both age and gender as responses in the questionnaire [46]. We therefore propose to expand our implemented model to infer group norms over the categories: men, women, old and young. We then propose to predict group membership of out of sample responses from these posterior. To perform this evaluation, we first divide a subsample of the population into male and female and old and young subgroups. The old and young groups are determined by the boundary: young < 23, old ≥ 23. We then infer the group norms and covariance matrices for these populations.

Next, we infer the individual moral weights for the remainder of the data. For each parameter distribution for each individual we wish to ascertain whether it is more likely to have been drawn from the male or female, old or young posterior. We first use the two sided Kolmgorov-Smirnov test, a statistical measure that determines the difference between two samples from two distributions [114] and operates under the null hypothesis that both samples are drawn from the same distribution . Using the Kolmogorov-Smirnov test to predict individual membership to the male and female posterior distributions results in 50.51% accuracy whereas predicting membership of age subsgroups results in 48.68% accuracy.

These results seem to indicate that there is no difference in the way in which men, women or old and young make moral decisions within our dataset. We can therefore perform further inspection on the inferred group norm values between different sub samples to confirm this hypothesis. Tables 10.1 and 10.2 show the inferred group-norm weights over gender and age differentiated groups. The tables also show the results of the double sided Kolmogorov-Smirnov test used to evaluate the differences between inferred group norm weights.

Tables 10.1 and 10.2 show how different group norm weights represent bigger differences between men and women and old and young. In the Kolmogorov-Smirnov test, the larger the value of the KS statistic and the smaller the p-value mean that the two distributions are less likely to belong to the same distribution. We can therefore reattempt to predict group membership using only the moral features with most differentiable Kolmogorov-Smirnov results.

For the gender differentiated group, using the kneeling down weight and self preservation weight results in a group membership predictive accuracy of 63.64% whilst for the age differentiated group, using only the middle aged and the elderly weights result in a predictive accuracy of 51.52%. Whilst the results suggest that there is no way of predicting age of an individual based on their moral preferences, the results for gender are more promising and could suggest that there exists a link between gender and moral beliefs.

There is an infinite number of characteristics that can be used to differentiate populations: culture, age, gender, class, profession, political views... This chapter has therefore served to question the limit of population decomposition. We have raised questions over its ethicality and conclude with an argument over the assumptions decomposition places over populations. Further, with every division, we make increasingly more specific assumptions over the nature of moral preferences. Indeed, our results mirror the conclusions made in Chapter 8. They have shown how certain characteristics cannot be used to differentiate moral preferences, how making stronger assumptions does not necessarily strengthen results and does not accurately reflect our knowledge of human morality.

Table 10.1: Table showing the difference in inferred moral norms between gender groups

	Man Mean	Woman Mean	K-S Statistic	p-value
Elderly Weight	0.4534	0.1478	0.184	2.7090e-15
Infancy Weight	1.3499	1.5279	0.148	4.8046e-10
Middle Aged Weight	1.0976	1.0287	0.148	4.8046e-10
Kneeling Down Weight	1.8822	1.0038	0.603	4.1527e-160
Pedestrian Weight	-0.4231	-0.5594	0.193	8.7638e-17
Self Preservation Weight	2.3371	1.6801	0.408	1.5762e-73

Table 10.2: Table showing the difference in inferred moral norms between age groups

	Old Mean	Young Mean	K-S Statistic	p-value
Elderly Weight	0.9490	-0.2571	0.629	3.5575e-174
Infancy Weight	1.3436	1.6401	0.273	3.7225e-33
Middle Aged Weight	1.1983	1.5821	0.3710	47.1539e-61
Kneeling Down Weight	1.2599	1.6880	0.405	1.8524e-72
Pedestrian Weight	-0.4387	-0.4996	0.171	2.8809e-13
Self Preservation Weight	2.0244	2.1908	0.140	4.9372e-9

Chapter 11

Aggregating Preferences

The previous chapters have focused on building a computational model of human morality. Now we turn our attention to ethical machines. Can models of human moral decision making be used to automate ethical decisions. Kim et al. identify that an expansion of their model could be to investigate the design of an ethical agent that, when faced with a specific moral dilemma, is capable of making a decision that maximises the overall social utility of a population. Noothigatttu et al. [98] use recent developments in computational social choice to suggest a way of aggregating individual preferences to identify an overall desirable choice for a given ethical dilemma. We now present the theory of Noothigattu et al. and apply it to our model.

11.1 Ethics without Rules

Noothigattu et al. identify the central challenge that has so far prevented progress in automating ethical decision making as the lack of ground-truth principles. In most successful artificial intelligence applications, there exists a bonified right answer that is provable from data. However, in the world of ethics, there is no correct definition of ethical behaviour. Dwork et al. [45] claim that in the absence of ground truth rules, the optimum decision should be agreed upon by society. Kim et al. expand this idea to suggest aggregation of individual preferences to automate ethical decision making. Further, when faced with the dilemma of best incorporating human behaviour into an automated ethical system, Greene et al. consider a solution in the longstanding field of preference aggregation [58]. Connitzer at al. extend this idea by suggesting that we let models of human moral decision making vote over the best possible outcome [34]. Noothigattu et al. use these studies to propose a framework of ethical decision making, utilising computational social choice [98].

11.2 Voting Based Ethical Decision Making

Noothigattu et al. summarise their framework into three key stages: 1. Learning 2. Summarisation 3. Aggregation. The first stage learns individual models of preferences using pairwise comparisons over all possible scenarios. The second stage combines all individual models into a combined model of preferences. The third stage uses the summarised model such that, at run time, when presented with an ethical dilemma, a voting rule is applied to aggregate preferences into a collective decision [98]. One of the biggest challenges faced by Noothigattu et al. is the selection of a voting process that aggregates individual preferences into one overall winner. Traditional methods consist of techniques such as the Borda count, which determines an outcome by giving each alternative, for each individual, a score that is then summed over all individuals such that the alternative with the highest score wins. Noothingattu et al. note the potential computational cost of voting techniques such as the Borda count when used on large datasets [98]. They therefore propose a more efficient summarised model [98].

The learning stage learns a Thurstone-Mosteller (TM) process that represents an individual's preferences. A Thurstone-Mosteller process is a consistent permutation process whose utility process U is modeled over a set of all possible alternatives $(x_1, x_2, ..., x_n)$ such that the utilities $(U_{x_1}, U_{x_2}, ..., U_{x_n})$ are independent and $U_{x_i} \sim \mathcal{N}(\mu_{x_i}, \frac{1}{2})$. Noothigattu et al. specify that the mode utility μ_{x_i} of an individual depends linearly on its underlying features such that $u_x = \beta^T x$. Noothigattu et al. continue to use maximum

likelihoood estimation to learn the feature vector, β, which represents the preferences of individual i [98]. To summarise, Noothigattu et al. suggest bundling all inferred individual TM processes into one TM process, Π [98]. They stress the importance of this offline summarisation step as decisions need to be made quickly in the aggregation step. This necessity for speed can be exemplified by an autonomous car that has to make decisions in a fraction of a second. Noothigattu et al. prove mathematically that defining the summarised model Π in terms of feature vectors β such that $\beta' = \frac{1}{N}\sum_{i=1}^{N}\beta_i$ maximises overall social utility. Finally, for the aggregation stage, Noothigattu et al. show that the alternative that is selected is the one that maximises mode utility over $\beta_1' 1^T x_1, \beta_2' 1^T x_2, ..., \beta_m' 1^T x_m$.

Noothigatu et al. evaluate their model on a subset of data from the Moral Machine dataset [1]. They compare the performance of their summarised model to an aggregation achieved through the Borda count. Noothigatu et al. are then able to conclude that not only is their summarised model more computationally efficient, it also matches the performance of traditional voting mechanisms [98].

We now test the theory of Noothigattu et al. on our implemented model. We note that the feature vectors β_i of Noothigattu et al. are equivalent to the moral weight vectors, w_i, of Kim et al. Furthermore, Noothigattu et al. achieve their summarised model, β', by averaging all individual feature vectors. To achieve a summarised model we must average over all individual weight vectors such that $w' = \frac{1}{N}\sum_{i=1}^{N} w_i$. Additionally, we note that the group norm vector w^g is intended to represent the mean values of the group. We can therefore evaluate the success of Noothigattu et al.'s framework on the model by comparing the decisions made by the aggregation of the averaged individual weights, an aggregation via inferred group norms and an aggregation achieved through the application of the Borda count to individual preferences.

Figure 11.1 shows that the aggregated model achieved by averaging individual moral weights as specified by Noothigattu et al. matches more closely the aggregation achieved via the Borda count, than that of the aggregation achieved by using the ranking of group norms. This immediately raises interesting questions over the discrepancy between the mean of inferred individual responses and the inferred mean of the distribution from which the individual weightings are drawn from. Investigating this inconsistency to determine whether it is caused by a property of the Borda count or a misspecification of the model would be an interesting direction for future work. Our results show that the summarised model of Noothigattu et al. encapsulates the properties of the Borda count voting rule whilst avoiding computational inefficiency.

Finally we wish to raise the question of the ethicality of this framework. Can ethical behaviour actually be defined as the average behaviour of humans in moral dilemmas? Furthermore, how representative is this framework? How many individuals would agree with the decisions made by the aggregate model? We evaluate this question by using the aggregation of the summarised model to predict individual out of sample decisions. Figure 11.2 shows how the Borda count is the most representative aggregation process but is nearly matched by the summarised model. Furthermore, for an aggregated model to match the decisions of 78% of the population seems to suggest that this is a good starting point to begin engineering ethical machines that behave most desirably according to the majority.

Figure 11.1: Boxplot comparing model accuracy of the aggregation via the summarised method of Noothigattu et al. with group norm aggregation. Accuracy here refers to the number of individual decisions predicted by each aggregated model that are equivalent to applying the Borda count to individual preferences.

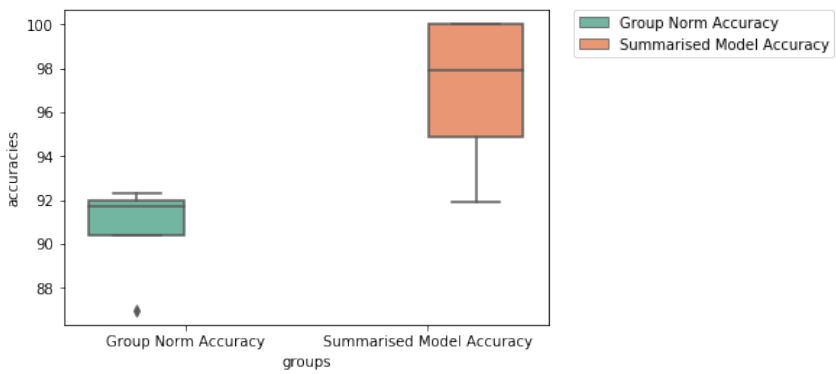
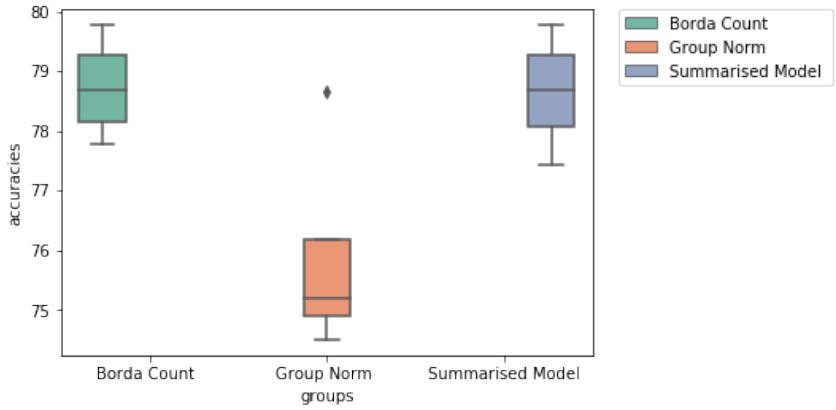


Figure 11.2: Boxplot comparing the out of sample predictive accuracy of model under different aggregation methods when using different methods of aggregation to generate universal moral weightings.



Chapter 12

Cross Domain Model

One of our main motivations for implementing the model of Kim et al. was to test results on an alternate dataset. We have seen how the model proposed by Kim et al. has provably inferred moral principles over two alternate datasets. So far, however, we have only considered moral dilemmas in the context of trolley problems. In his PhD thesis, Awad highlights some of the limitations of using the trolley problem to frame moral theory [13]. Awad claims that trolley dilemmas are too simple, too rare, and unrealistic [13]. We do not, therefore, pretend to have implemented a model capable of extracting the full spectrum of human morality, however, we have used the trolley problem as a framing device to show the feasibility of this goal. We can therefore identify one critical area for future work as the applicability of the model of Kim et al. to other domains of moral theory [71]. We now present the application of Kim el.'s model to more general moral dilemmas.

12.1 Transferring Model to a New Domain

The study conducted by Haykawa et al. [61] investigates the contribution of foreign language in responses to moral dilemmas. Hayakawa et al. present a wide range of scenarios to participants, which extend the idea of moral dilemmas beyond trolley problems. An example of these scenarios is shown in Figure 12.1. 20 variations of these scenarios were shown to respondents gathering their opinions on moral dilemmas ranging from coma patients to bear attacks. Every scenario poses a concluding statement and the respondent gives a binary response, which represents the extent to which they agree with the statement. From Figure 12.1 we can see that the presented moral dilemmas from Hayakawa et al. are more complex and realistic than the original dataset. Hayakawa et al. present six experiments, each containing a slight variation of dilemmas, and a sample size of 224. We propose to apply the model of Kim et al. to the data generated by one of the experiments of Hayakawa et al.

One of the biggest challenges in transferring the model of Kim et al. to the new data domain is redesigning the mapping of character vectors to the abstract feature space. Figure 12.1 shows the complexity of the moral dilemmas as presented by Hayawaka et al. However we believe that, similarly to trolley problems, these moral dilemmas can be entirely described by a set of abstract features. The example from Figure 12.1 could, for example, be entirely described by the abstract features, criminal and torture, for one decision and property damage for the other. This, we believe, encapsulates the spirit of the moral dilemma. Due to the high variability between the 20 moral dilemmas presented in the experiments of Hayakawa et al. the resulting abstract feature vectors are much more difficult to design. We therefore experiment with different abstract feature vectors, encapsulating different numbers of moral feature parameters that can be used to describe individual moral preferences of the dataset.

After defining the abstract feature vectors, we use the model parametrisation of Kim et al. as presented in Chapter 7. We are then able to infer individual moral preferences over the 224 respondents and use these to predict out of sample decisions. Figure 12.2 shows the predictive accuracy of our implemented model using abstract feature vectors of varying length to represent the dataset from Hayakawa et al. Figure 12.1 shows how the predictive accuracy of the model increases with the number of abstract moral features used to describe underlying moral dilemmas, where error bars represent the variance in accuracy over different permutations of chosen abstract features. Our results show that the accuracy decreases as the number of parameters exceeds seven. Furthermore, when we attempt to define eleven abstract moral

Figure 12.1: An example moral dilemmma posed to respondents in the Hayakawa et al. study [61]

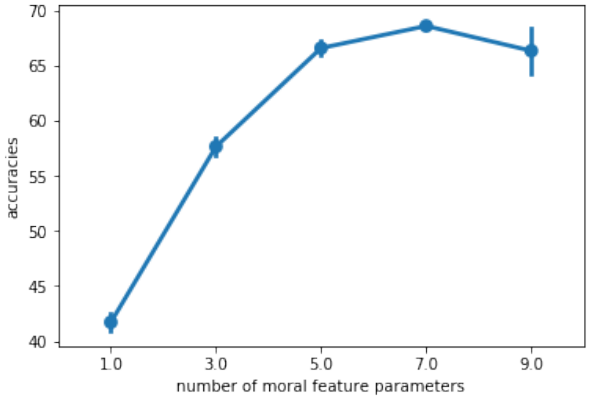
You are a police officer, and have recently caught a criminal you have been hunting for some time. He has made many bombs: some that have already exploded and some that have not exploded yet.

He places bombs outside cafes and they explode when no one is around. His bombs are inside paint cans so that they spray nearby objects with paint. He has sprayed many cafes with paint and might spray many more.

Now that the criminal is caught, you want to know where the unexploded bombs are so you can stop them. The only way to make him talk is to use torture. For example, holding his head under water and beating him.

Is it morally correct for you to use torture in order to prevent further property damage, even though the criminal will suffer?

Figure 12.2: Pointplot showing the relationship between model predictive accuracy and the number of abstract moral features used in parametrisation on the Hayakawa et al. dataset [61]



features the model encounters sampling issues and does not converge. This behaviour can be explained by the incapability of the model in dealing with sparse abstract feature vectors. Furthermore, we cannot conclude that seven abstract moral features are sufficient to describe individual moral preferences. In fact, these results show the impact of placing different assumptions over how many, or which, abstract features are used to describe moral dilemmas and represents a significant challenge to the method of Kim et al. if the model is to be expanded to learn moral preferences over more complex moral dilemmas.

Our results have shown how significant model selection is in statistical analysis which is a rhetoric explored in Chapter 4. A commmon approach of data scientists when tackling this problem is to fit multiple models to their data and then select one depending on model comparison heuristics [53]. Gershman et al. introduce Bayesian Non-parametric models (BNP) that offer an alternative approach to this problem [53], where non-parametric models allow model complexity to grow as more data is observed. Gershman et al. continue to discuss latent feature models, which describe observations composed of a number of hidden factors. The model as proposed by Kim et al. is an example of a finite latent fixture model, whereby the number of abstract moral features is specified prior to inference. However, our results have shown that it is not always straightforward to select the number of latent factors that perfectly describe data. Griffiths et al. [54] introduce a BNP model, the Indian Buffet Process (IBP), that allows for an unbounded number of latent features.

The IBP is based on an analogy of an Indian buffet restaurant, whereby N customers walk into the restaurant and are presented with an infinite number of dishes. If the first customer chooses to load their plate with the first k dishes, the ith customer chooses a dish to try with probability proportional to its popularity. Griffiths et al. explain that the selection of dishes chosen by each customer can therefore be represented by a binary matriz $\mathbf{Z}$ with N finite rows (customers) and an infinite number of columns (dishes). Griffiths et al. continue to derive the probability $P(\mathbf{Z})$ for a particular matrix to be chosen by the IBP and show how the IBP can be used as a prior in non parametric models for learning in contexts where the number of latent features for observable data is unknown.

This chapter has found that a serious limitation in the model proposed by Kim et al. is the ambiguity of selecting the number of latent features that are capable of completely describing the moral data. Furthermore, we have discussed how the Indian Buffet Process could offer a non-parametric solution. However, we must return now to the "No Free Lunch Problem" and remind the reader that there is no optimal solution to this problem. One significant disadvantage of non-parametric models is their intractability. Furthermore, future work in this area must work out how to balance representation of complex data with computational power, a problem that has plagued data science since its origin. We therefore conclude with the argument that the future success of engineering ethical machines is intrinsically dependent on the resolution of classical mathematical problems.

Chapter 13

Critical Evaluation and Future Work

Within this dissertation, we have implemented a model of human moral decision making and used mathematical and philosophical theory to suggest adaptations to the model. These adaptations are not necessarily intended to improve the model but are aimed to raise awareness about the sensitivity of these models to assumptions made throughout development. Our main results and conclusions can be summarised below:

- A model that infers individual weights over abstract moral principles has been proven to perform significantly better at predicting future decisions than a model that assumes equal weighting over abstract moral weights.
- We have replicated the conclusion of Kim et al. that a Bayesian hierarchical model outperforms a flat model in inferring individual moral weights, particularly over small data samples.
- We have shown that by using a non-informative prior, not only does the model produce more accurate predictions but this assumption captures the ambiguity of moral theory and is, therefore, arguably a more accurate reflection of prior knowledge.
- By incorporating deontological statements into prior distributions we have shown that there appears to be a tenuous link between deontological and utilitarian approaches to moral decision making.
- We have argued that there could be a link between gender and certain moral beliefs. However, this conclusion highlights the difficulty in dividing populations into subgroups.
- We have found that aggregating individual moral preferences, using a summarised voting rule matches the performance of traditional voting rules. By questioning the ethicality of using a voting rule to aggregate individual models of morality, we have discovered that the decisions predicted by the summarised model is representative of 78% of our data population.
- We have uncovered the challenges faced by the model of Kim et al. when applied to complex and realistic moral dilemmas and have shown that non-parametric techniques could offer a solution. However, this conclusion has shown the mathematical difficulties future models of morality could face.

This dissertation has established the enormity of the ethical artificial intelligence challenge and throughout, we have emphasized the importance of questions and issues over quantitative results. To motivate the importance of our conclusions, we draw your attention to a recent report published by the House of Lords, Select Committee on Artificial Intelligence on 16th April 2018 [101]. The report recommends that the UK "take an active role as a moral and ethical leader on the global stage in this area" and highlights the importance of industry ensuring that artificial intelligence is developed for the common benefit of society whilst calling for a "shared ethical artificial intelligence framework". This directive emphasizes the timeliness of our dissertation and encapsulates our own motivation for studying this topic. We hope that the arguments presented in this dissertation could inform future direction and realisation of the goals outlined by the report.

To conclude, we present an extended critical evaluation that raises further questions and suggests directions for future work. This section is critical in continuing the discursive quality of the dissertation and leaving the reader with some ideas as to how this field may develop.

13.1 Oversimplification of Decision Making

13.1.1 Modeling Deliberation

One significant criticism of the implemented model is the decision to model decision making as a linear combination of moral preferences that remains invariant over time. However, we question whether a richer model of moral decision making would be achieved by modeling, not just the decision outcome, but also, the cognitive process of deliberation. Busemeyer et al. identify the shortcomings of historical methods of modeling decision making using expected utility theory [28]. They draw on the psychology theory of James [67] to highlight the importance of cognitive deliberation processes in making decisions under uncertainty. Busemeyer et al. build on historic work of static models of decision making to propose a dynamic model that shows how preference relations change as a function of deliberation time.

Busemeyer et al. begin by explaining the difference between probabilistic and deterministic models, voicing how deterministic approaches use expected utility maximisation to model choices as preference relations that are either true or false at any time, whereas probabilistic models use random utility theory to model preference relations under probability distributions. Busemeyer et al. argue that it is unavoidable fact that human choices are inconsistent and therefore probabilistic approaches are the only kind that make sense [28]. Busemeyer et al. continue by comparing dynamic and static approaches to decision making, concluding that human decisions take time and the final decision made is heavily impacted by the amount of time taken, thus ruling out static methods of modeling decision making.

Busemeyer et al. continue by presenting decision field theory as providing a mathematical foundation for modeling dynamic, stochastic decision making under uncertainty. It provides, amongst other economic uses, a method for predicting choice probabilities and response time distributions. Busemeyer et al. continue to present the different mathematic mechanisms that formalise complex cognitive deliberation. Importantly, these formalisms show how the "preference state" changes as a function of time. Busemeyer et al. argue how decision field theory is an effective tool for explaining the correlation between response time and choice accuracy. That is, how decision confidence is affected by decision time? Contained in the dataset of Hayakawa et al. is information concerning the length of time it takes respondents to select an answer to a moral dilemma. This would allow for the incorporation of a drift diffusion model, such that moral weights were themselves weighted by the time it took the respondent to answer. Alternatively, the dataset of Hayakawa et al. also contains information as to the number of times a respondent changes their mind before selecting a final answer for a given moral dilemma. This data could be related to decision field theory to show how moral preferences change over time.

13.1.2 Modeling Empathy

An alternative oversimplification of decision making assumed by the Kim et al. model is the decision to model decision making processes as utility functions representing underlying moral preferences. Furthermore, there has been huge debate in the ethics sphere over "Principilism", the practice of using principles to replace moral theories. Clouser et al. argue that misguided moral principles obscure moral reasoning by misrepresenting and over simplifying moral theories [33]. We can see this oversimplification of moral theories within our implementation. When defining their utility calculus, Kleiman-Weiner et al. include both a linear representation of moral preferences **and** another term representing other abstract individual qualities that affect moral decision making such as empathy. Furthermore, when defining their model, Kim et al. ignore this additional term and focus on inferring weights over moral principles. An extension of this model would therefore be incorporating a method of modeling the additional source of moral valuation.

Considering empathy as a factor that shapes moral decision making in the trolley problem has been approached in the literature by Wilson et al. [128] who begin by modeling decision making as utility scores, a linear combination of propositions that have an associated weight. To account for empathy in their model, Wilson et al. introduce a new weight on the utility of a proposition that determines the empathetic response of an action. Their model for empathy draws on three well known characteristics of neurological and psychological empathy: cognitive empathy, which models the extent of emotional appraisal an individual should feel after a given action; emotional empathy, which refers to how connected an individual is to the target; and pro-social motivation, which refers to the extent an individual feels concern for a target. Wilson et al.'s empathy weight is therefore a linear combination of all three char-

acteristics. This model fits the idea of external factors influencing moral decision making mentioned by Kleiman-Weiner et al. [72]. Furthermore, this empathy model could be used to enrich the model of Kim et al. by accounting for individuals whose behaviour could not be explained by moral principles alone: a naturally empathetic person, a mother who had recently given birth or a husband who had recently had an argument with his wife.

13.2 Towards a Hybrid Model

Throughout this dissertation, we have discussed the difference between bottom-up and top-down approaches to engineering ethical machines. We have presented a bottom-up approach to learning human morality and a method for combining individual models into an aggregated model of morality. The next stage in engineering an ethical machine would therefore be to incorporate ethical or legislative rules into the system presented in this dissertation to produce a hybrid ethical model. Noothigattu et al. recommend that ethical rules could control behaviour of the machine. However, they claim it impossible to specify ethical rulings that cover every possible scenario, in which case the aggregated model of human moral principles would control behaviour [98]. Loreggia et al. also propose an alternative method for encoding ethics within models of human moral behaviour [82]. Their proposed system uses CP-nets to evaluate whether human preferences are compatible with specified ethical principles. CP-nets were presented by Boutiller et al. in 2004 as a graphical representation of preference data and allow for the qualitative instead of quantitative assessment of preferences using preference relations [23]. Loreggia et al. continue to construct two alternative CP-nets, with the first over human moral preferences and the second over ethical principles. This formulation allows Loreggia et al. to calculate the distance between both CP-nets and a way of measuring whether moral principles are "close enough" to ethical rules. This would be a really interesting extension of this dissertation. However, the question remains, what set of ethical rules should be used? We propose the use of the recently defined German ethical guidelines for autonomous vehicles commissioned by the German federal government and the first of its kind to standardise the behaviour of autonomous vehicles in ethical dilemmas [100]. The incorporation of these guidelines as ethical rules would show the difference between human behaviour and legal requirements defined by lawyers which could make for a very interesting comparison.

13.3 Limits of Data

In Chapter 6 we go to great lengths to discuss the constraints data place on results and conclusions. We re-iterate that the results and arguments presented in this dissertation are contextualised to our datasets and are by no means a resolution to the challenge of designing ethical machines. Furthermore, future work must continue to gather quality data. One specific related ethical issue this dissertation has raised is the lack of fully representative data. Both the Moral Machine dataset [116] and German Autonomous Vehicle dataset [46] represent the moral beliefs of Western cultures. If future studies wish to use models of morality to automate ethical decision-making, consideration must be made to ethnic and socio-economic groups that are traditionally under represented.

13.4 Bias in Aggregation

Chapter 11 presents a discussion on the ethicality of utilising a voting rule to aggregate individual moral decisions. Whilst we have seen that the aggregated model is representative of the majority of our data population, there are other ethical arguments that should be considered. Two of these central arguments are presented in the work of Risse, in his informative work "Arrows Theorem, Indeterminacy, and Multiplicity Reconsidered" [107]. Risse presents the indeterminate problem as a historical negative result in social choice theory. The theorem highlights inconsistencies in preference aggregations such that a sequence of pairwise votes may not lead to a majority vote but to a cycle. Consider, for example, in the context of our moral data, a population who prefers the elderly to infancy and infancy to self preservation but prefers self preservation to elderly. Risse argues that if cycles occur, voting rules do not produce a result and leaves the decision indeterminate [107]. The second ethical challenge of voting rules presented by Risse is that of the multiplicity method, which refers to the problems incurred due to the existence of a wide variety of mutually inconsistent voting rules. Upon the selection of a particular voting rule there are guaranteed "winners" and "losers" whose identity is dependent on the specific voting rule [107]. Future work that attempts to aggregate moral preferences must therefore consider the underlying

properties of their selected voting rule. An interesting extension of Chapter 11 would therefore be to compare the decisions generated by the summarised model [98] to other traditional voting rules.

13.5 Algorithmic Fairness

Currently, one controversial area within ethical artificial intelligence is the concept of algorithmic fairness. A 2016 study found that the algorithm used in criminal sentencing, to assess risk of recidivism, was displaying signs of racial bias [10]. The fact that artificial intelligence has already spread to areas in which it has a big impact on human lives means that developers must start thinking beyond the predictive accuracy of their systems and consider the social implications [76]. Up to this point, we have considered aggregating models of individual moral preferences to automate decision making. We have considered representation as an ethical issue. However, we now present the ethical issue of fairness. Learning a model of human morality is inherently biased. We have seen how our data population are likely to place higher preference on humans with certain characteristics. Is it therefore ethical to use models of human morality in automating ethical decision making? Many critics would argue no, that humans are not an example of ethicality and that ethical intelligent systems should be designed without individual moral influence. Advocates, however, would argue that ethics is a human construct and therefore can only be considered within human conduct. If we are to use human moral behaviour to automate decision making, extensions to models must be made to prevent the use of protected attributed in decision making [76]. In their paper, "Counterfactual Fairness", Kusner et al. describe protected attributes as variables that must not be discriminated against, relative to a particular system. Kusner et al. present counterfactual fairness as the idea that a decision is only fair if it is fair in both the real world and a counterfactual world, where the target individual belongs to an alternative demographic group. An interesting extension of Chapter 11 would incorporate this idea into the summarisation process, such that only individual preferences that were counterfactually fair could be included in the summarisation process.

The results of the implemented model show that, within our data population, individuals place a higher weight on the infancy moral principle when compared to the elderly moral principle. This result shows that individuals are likely to discriminate based on potentially protected attributes, such as age. The ability of the model to infer quantified rankings over moral principles could, potentially, be expanded to recognize certain biases in certain populations, or software. An interesting extension of this project would be to apply the model of Kim et al. to the COMPAS open source dataset used for the recidivism study cite [10]. By assigning a weighting over race moral principles, we hypothesise that the model would be able to extract racial bias. A model that was capable of verifying the neutrality of a piece of software or a person would be invaluable in industries such as law, banking and recruitment.

13.6 Transparent Artificial Intelligence

Another key component of ethical artificial intelligence that has not been mentioned in this dissertation is the issue of transparency in intelligent systems. In their cautionary work "Standardizing Ethical Design for Artificial Intelligence and Autonomous Systems" [26] Bryson et al. sepcify some motivations for artificial intelligence transparency as: building user trust in a system; certification against safety standards; traceability following malfunctions; legal evidence following an accident; and public trust in disruptive technologies. These motivations are particularly prominent for artificial intelligence systems that are required to make ethical systems. The recent Arizona driverless Uber tragedy is c a particularly stimulating example for transparency in decision making [78]. Kroll et al. have recently presented a selection of computational techniques that work towards algorithmic transparency and accountability including, software verification; cryptographic commitments; zero knowledge proofs and fair random choices. Kroll et al. argue the effectiveness of their proposal for systems "to create cryptographic commitments as digital evidence of their actions" [75]. Furthermore, with the increasing volume of legislation in this area, any future work that attempts to build a system of automated ethical decision making must consider methods like those expressed by Kroll et al., such that all decisions are traceable and accountable.

Bibliography

[1] *Moral machine*. http://moralmachine.mit.edu/. Accessed: 2018-04-18.

[2] D. Abel, J. MacGlashan, and M. L. Littman, *Reinforcement learning as a framework for ethical decision making.*, in AAAI Workshop: AI, Ethics, and Society, vol. 16, 2016, p. 02.

[3] C. Allen, I. Smit, and W. Wallach, *Artificial morality: Top-down, bottom-up, and hybrid approaches*, Ethics and information technology, 7 (2005), pp. 149–155.

[4] I. Alvarez, J. Niemi, and M. Simpson, *Bayesian inference for a covariance matrix*, arXiv preprint arXiv:1408.4050, (2014).

[5] M. Anderson and S. L. Anderson, *Machine ethics*, Cambridge University Press, 2011.

[6] ——, *Geneth: A general ethical dilemma analyzer.*, in AAAI, 2014, pp. 253–261.

[7] M. Anderson, S. L. Anderson, and C. Armen, *Towards machine ethics*, in AAAI-04 workshop on agent organizations: theory and practice, San Jose, CA, 2004.

[8] S. L. Andresen, *John mccarthy: father of ai*, IEEE Intelligent Systems, 17 (2002), pp. 84–85.

[9] C. Andrieu, N. De Freitas, A. Doucet, and M. I. Jordan, *An introduction to mcmc for machine learning*, Machine learning, 50 (2003), pp. 5–43.

[10] J. Angwin, J. Larson, S. Mattu, and L. Kirchner, *Machine bias theres software used across the country to predict future criminals. and its biased against blacks.*, May 2016.

[11] R. C. Arkin, P. Ulam, and A. R. Wagner, *Moral decision making in autonomous systems: Enforcement, moral emotions, dignity, trust, and deception*, Proceedings of the IEEE, 100 (2012), pp. 571–589.

[12] S. Armstrong, *Motivated value selection for artificial agents.*, in AAAI Workshop: AI and Ethics, 2015.

[13] E. Awad, *Moral machines: perception of moral judgment made by machines*, PhD thesis, Massachusetts Institute of Technology, 2017.

[14] J. Barnard, R. McCulloch, and X.-L. Meng, *Modeling covariance matrices in terms of standard deviations and correlations, with application to shrinkage*, Statistica Sinica, (2000), pp. 1281–1311.

[15] I. Beichl and F. Sullivan, *The metropolis algorithm*, Computing in Science & Engineering, 2 (2000), pp. 65–69.

[16] O. Bendel, *Annotated decision trees for simple moral machines*, in The 2016 AAAI Spring Symposium Series, AAAI Press, 2016, pp. 195–201.

[17] J. Bernardo, M. Bayarri, J. Berger, A. Dawid, D. Heckerman, A. Smith, and M. West, *Non-centered parameterisations for hierarchical models and data augmentation*, in Bayesian Statistics 7: Proceedings of the Seventh Valencia International Meeting, vol. 307, Oxford University Press, USA, 2003.

[18] J. M. Bernardo and A. F. Smith, *Bayesian theory*, 2001.

[19] M. Betancourt, *A conceptual introduction to hamiltonian monte carlo*, arXiv preprint arXiv:1701.02434, (2017).

[20] P. Blake, K. McAuliffe, J. Corbit, T. Callaghan, O. Barry, A. Bowie, L. Kleutsch, K. Kramer, E. Ross, H. Vongsachang, et al., *The ontogeny of fairness in seven societies*, Nature, 528 (2015), p. 258.

[21] J.-F. Bonnefon, A. Shariff, and I. Rahwan, *The social dilemma of autonomous vehicles*, Science, 352 (2016), pp. 1573–1576.

[22] N. Boström, *Superintelligence: Paths, dangers, strategies*, 2014.

[23] C. Boutilier, R. I. Brafman, C. Domshlak, H. H. Hoos, and D. Poole, *Cp-nets: A tool for representing and reasoning with conditional ceteris paribus preference statements*, J. Artif. Intell. Res.(JAIR), 21 (2004), pp. 135–191.

[24] S. Bradley, *All the creepy, crazy and amazing things that happened in ai in 2017*. http://www.wired.co.uk/article/what-happened-in-ai-in-2017, December 2017. Accessed: 2018-05-01.

[25] S. P. Brooks and A. Gelman, *General methods for monitoring convergence of iterative simulations*, Journal of computational and graphical statistics, 7 (1998), pp. 434–455.

[26] J. Bryson and A. Winfield, *Standardizing ethical design for artificial intelligence and autonomous systems*, Computer, 50 (2017), pp. 116–119.

[27] B. G. Buchanan, *A (very) brief history of artificial intelligence*, Ai Magazine, 26 (2005), p. 53.

[28] J. R. Busemeyer and J. T. Townsend, *Decision field theory: a dynamic-cognitive approach to decision making in an uncertain environment.*, Psychological review, 100 (1993), p. 432.

[29] J. G. Carbonell, R. S. Michalski, and T. M. Mitchell, *An overview of machine learning*, in Machine Learning, Volume I, Elsevier, 1983, pp. 3–23.

[30] V. Charisi, L. Dennis, M. F. R. Lieck, A. Matthias, M. S. J. Sombetzki, A. F. Winfield, and R. Yampolskiy, *Towards moral autonomous systems*, arXiv preprint arXiv:1703.04741, (2017).

[31] N. Chomsky, *Rules and representations*, Behavioral and brain sciences, 3 (1980), pp. 1–15.

[32] R. Chrisley and S. Begeer, *Artificial intelligence: critical concepts*, vol. 1, Taylor & Francis, 2000.

[33] K. D. Clouser and B. Gert, *A critique of principlism*, The journal of medicine and philosophy, 15 (1990), pp. 219–236.

[34] V. Conitzer, W. Sinnott-Armstrong, J. S. Borg, Y. Deng, and M. Kramer, *Moral decision making frameworks for artificial intelligence.*, in AAAI, 2017, pp. 4831–4835.

[35] J. Dancy, *Ethics without principles*, Oxford University Press on Demand, 2004.

[36] L. M. DeBruine, *Facial resemblance enhances trust*, Proceedings of the Royal Society of London B: Biological Sciences, 269 (2002), pp. 1307–1312.

[37] J. Decety and T. Wheatley, *The moral brain: A multidisciplinary perspective*, MIT Press, 2015.

[38] L. Dennis, M. Fisher, M. Slavkovik, and M. Webster, *Formal verification of ethical choices in autonomous systems*, Robotics and Autonomous Systems, 77 (2016), pp. 1–14.

[39] K. Dergousoff and R. L. Mandryk, *Mobile gamification for crowdsourcing data collection: Leveraging the freemium model*, in Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems, ACM, 2015, pp. 1065–1074.

[40] D. Dewey, *Learning what to value*, in International Conference on Artificial General Intelligence, Springer, 2011, pp. 309–314.

[41] O. E. Dictionary, *Oxford english dictionary*, Simpson, JA & Weiner, ESC, (1989).

[42] P. Ditto and B. Liu, *Deontological dissonance and the consequentialist crutch*, in Artificial Intelligence: Its Scope and Limits, Psychology Press, 2016, pp. 51–70.

[43] C. Dobre and F. Xhafa, *Intelligent services for big data science*, Future Generation Computer Systems, 37 (2014), pp. 267–281.

[44] M. A. Dummett, *The logical basis of metaphysics*, Harvard university press, 1991.

[45] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, *Fairness through awareness*, in Proceedings of the 3rd innovations in theoretical computer science conference, ACM, 2012, pp. 214–226.

[46] A. K. Faulhaber, A. Dittmer, F. Blind, M. A. Wächter, S. Timm, L. R. Sütfeld, A. Stephan, G. Pipa, and P. König, *Human decisions in moral dilemmas are largely described by utilitarianism: Virtual car driving study provides guidelines for autonomous driving vehicles*, Science and Engineering Ethics, pp. 1–20.

[47] J. H. Fetzer, *What is artificial intelligence?*, in Artificial Intelligence: Its Scope and Limits, Springer, 1990, pp. 3–27.

[48] K. B. Francis, S. Terbeck, R. A. Briazu, A. Haines, M. Gummerum, G. Ganis, and I. S. Howard, *Simulating moral actions: An investigation of personal force in virtual moral dilemmas*, Scientific reports, 7 (2017), p. 13954.

[49] A. Gelman et al., *Prior distributions for variance parameters in hierarchical models (comment on article by browne and draper)*, Bayesian analysis, 1 (2006), pp. 515–534.

[50] A. Gelman and J. Hill, *Data analysis using regression and multilevel/hierarchical models*, Cambridge university press, 2006.

[51] A. Gelman and A. Jakulin, *Weakly informative priors*, (2007).

[52] A. Gelman and D. B. Rubin, *Inference from iterative simulation using multiple sequences*, Statistical science, (1992), pp. 457–472.

[53] S. J. Gershman and D. M. Blei, *A tutorial on bayesian nonparametric models*, Journal of Mathematical Psychology, 56 (2012), pp. 1–12.

[54] Z. Ghahramani and T. L. Griffiths, *Infinite latent feature models and the indian buffet process*, in Advances in neural information processing systems, 2006, pp. 475–482.

[55] S. Ghosh and S. G. Henderson, *Behavior of the norta method for correlated random vector generation as the dimension increases*, ACM Transactions on Modeling and Computer Simulation (TOMACS), 13 (2003), pp. 276–294.

[56] J. Graham, P. Meindl, E. Beall, K. M. Johnson, and L. Zhang, *Cultural differences in moral judgment and behavior, across and within societies*, Current Opinion in Psychology, 8 (2016), pp. 125–130.

[57] K. Gray and C. Schein, *Two minds vs. two philosophies: Mind perception defines morality and dissolves the debate between deontology and utilitarianism*, Review of Philosophy and Psychology, 3 (2012), pp. 405–423.

[58] J. Greene, F. Rossi, J. Tasioulas, K. B. Venable, and B. C. Williams, *Embedding ethical principles in collective decision support systems.*, in AAAI, 2016, pp. 4147–4151.

[59] M. Guarini, *Particularism and generalism: how ai can help us to better understand moral cognition*, in Machine ethics: Papers from the 2005 AAAI fall symposium, 2005.

[60] J. Hamlin, *The infantile origins of our moral brains*, in The moral brain: A multidisciplinary perspective, MIT Press, 2015.

[61] S. Hayakawa, D. Tannenbaum, A. Costa, J. D. Corey, and B. Keysar, *Thinking more or feeling less? explaining the foreign-language effect on moral judgment*, Psychological science, 28 (2017), pp. 1387–1397.

[62] R. Herschel and V. M. Miori, *Ethics & big data*, Technology in Society, 49 (2017), pp. 31–36.

[63] M. D. Hoffman and A. Gelman, *The no-u-turn sampler: adaptively setting path lengths in hamiltonian monte carlo.*, Journal of Machine Learning Research, 15 (2014), pp. 1593–1623.

[64] J. F. Horty, *Reasons as defaults*, Oxford University Press, 2012.

[65] B. R. House, J. B. Silk, J. Henrich, H. C. Barrett, B. A. Scelza, A. H. Boyette, B. S. Hewlett, R. McElreath, and S. Laurence, *Ontogeny of prosocial behavior across diverse societies*, Proceedings of the National Academy of Sciences, 110 (2013), pp. 14586–14591.

[66] T. Hurka, *Virtue, vice, and value*, Oxford University Press, 2000.

[67] W. James, *The principles of psychology*, Read Books Ltd, 2013.

[68] E. T. Jaynes, *Probability theory: the logic of science*, Cambridge university press, 2003.

[69] H. Joe, *Generating random correlation matrices based on partial correlations*, Journal of Multivariate Analysis, 97 (2006), pp. 2177–2189.

[70] J. Katz, *Ethics without morality*, The Journal of Philosophy, 52 (1955), pp. 287–291.

[71] R. Kim, M. Kleiman-Weiner, A. Abeliuk, E. Awad, S. Dsouza, J. Tenenbaum, and I. Rahwan, *A computational model of commonsense moral decision making*, AAAI 2018: Proc. 32nd AAAI Conference on Artificial Intelligence, February 2018 (forthcoming), (2018).

[72] M. Kleiman-Weiner, R. Saxe, and J. B. Tenenbaum, *Learning a commonsense moral theory*, cognition, 167 (2017), pp. 107–123.

[73] A. Kolmogorov, *Foundations of the Theory of Probability: Second English Edition.*

[74] K. B. Korb and A. E. Nicholson, *Bayesian artificial intelligence*, CRC press, 2010.

[75] J. A. Kroll, S. Barocas, E. W. Felten, J. R. Reidenberg, D. G. Robinson, and H. Yu, *Accountable algorithms*, U. Pa. L. Rev., 165 (2016), p. 633.

[76] M. J. Kusner, J. Loftus, C. Russell, and R. Silva, *Counterfactual fairness*, in Advances in Neural Information Processing Systems 30, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, eds., Curran Associates, Inc., 2017, pp. 4066–4076.

[77] D. Lazer, R. Kennedy, G. King, and A. Vespignani, *The parable of google flu: traps in big data analysis*, Science, 343 (2014), pp. 1203–1205.

[78] S. Levin, *Self-driving uber kills arizona woman in first fatal crash involving pedestrian.* https://www.theguardian.com/technology/2018/mar/19/uber-self-driving-car-kills-woman-arizona-tempe, March 2018. Accessed: 2018-04-24.

[79] R. Levy, *Probabilistic models in the study of language*, Online Draft, Nov, (2012).

[80] D. Lewandowski, D. Kurowicka, and H. Joe, *Generating random correlation matrices based on vines and extended onion method*, Journal of multivariate analysis, 100 (2009), pp. 1989–2001.

[81] D. Li and Y. Du, *Artificial intelligence with uncertainty*, CRC press, 2007.

[82] A. Loreggia, N. Mattei, F. Rossi, and K. B. Venable, *Preferences and ethical principles in decision making*, (2018).

[83] R. D. Luce, *Individual choice behavior: A theoretical analysis*, Courier Corporation, 2005.

[84] D. J. MacKay, *Introduction to monte carlo methods*, in Learning in graphical models, Springer, 1998, pp. 175–204.

[85] R. S. Masters, *Knowledge, knerves and know-how: The role of explicit versus implicit knowledge in the breakdown of a complex motor skill under pressure*, British journal of psychology, 83 (1992), pp. 343–358.

[86] T. May, *Theresa may's speech at the world economic forum annual meeting.* https://www.weforum.org/agenda/2018/01/theresa-may-davos-address, January 2018. Accessed: 2018-04-24.

[87] J. McCarthy, *What is artificial intelligence?*, (1998).

[88] J. McCarthy, M. L. Minsky, N. Rochester, and C. E. Shannon, *A proposal for the dartmouth summer research project on artificial intelligence, august 31, 1955*, AI magazine, 27 (2006), p. 12.

[89] M. Minsky, *Semantic information processing*, (1982).

[90] J. H. Moor, *The nature, importance, and difficulty of machine ethics*, IEEE intelligent systems, 21 (2006), pp. 18–21.

[91] T. Nagel, *The view from nowhere*, oxford university press, 1989.

[92] J. Naisbitt and J. Cracknell, *Megatrends: Ten new directions transforming our lives*, tech. rep., Warner Books New York, 1984.

[93] R. M. Neal, *Bayesian training of backpropagation networks by the hybrid monte carlo method*, tech. rep., Citeseer, 1992.

[94] M. Negnevitsky, *Artificial intelligence: a guide to intelligent systems*, Pearson Education, 2005.

[95] A. Newell, J. C. Shaw, and H. A. Simon, *Report on a general problem solving program*, in IFIP congress, vol. 256, 1959, p. 64.

[96] C. Noam, *The golden age a look at the original roots of artificial intelligence, cognitive science, and neuroscience.* MIT150 Symposia: Brains, Minds and Machines Symposium, 2011.

[97] I. Nonaka and G. von Krogh, *Tacit knowledge and knowledge conversion: Controversy and advancement in organizational knowledge creation theory*, Organization Science, 20 (2009), pp. 635–652.

[98] R. Noothigattu, S. Gaikwad, E. Awad, S. Dsouza, I. Rahwan, P. Ravikumar, and A. D. Procaccia, *A voting-based system for ethical decision making*, AAAI 2018: Proc. 32nd AAAI Conference on Artificial Intelligence, February 2018 (forthcoming), (2017).

[99] D. Nualart, *Kolmogorov and probability theory*, Arbor, 178 (2004), pp. 607–619.

[100] F. M. of Transport and D. Infrastructure, *Ethics commission automated and connected driving*, 2017.

[101] S. C. on Artificial Intelligence, *Ai in the uk: ready, willing and able?*, 2018.

[102] I. M. on behalf of The Royal Society, *Public views of Machine Learning April 2017 Findings from public research and engagement conducted on behalf of the Royal Society*, 2017.

[103] C. Pennachin and B. Goertzel, *Contemporary approaches to artificial general intelligence*, in Artificial general intelligence, Springer, 2007, pp. 1–30.

[104] R. R. Picard and R. D. Cook, *Cross-validation of regression models*, Journal of the American Statistical Association, 79 (1984), pp. 575–583.

[105] L. Pretot and S. Brosnan, *The evolution of morality: A comparative approach*, in The moral brain: A multidisciplinary perspective, MIT Press, 2015.

[106] B. Puza, *Bayesian Methods for Statistical Analysis*, ANU Press, 2016.

[107] M. Risse, *Arrow's theorem, indeterminacy, and multiplicity reconsidered*, Ethics, 111 (2001), pp. 706–734.

[108] C. Robert, *Machine learning, a probabilistic perspective*, 2014.

[109] J. N. Rouder, R. D. Morey, and M. S. Pratte, *Bayesian hierarchical models*, New handbook of mathematical psychology, 1 (2014).

[110] S. J. Russell and P. Norvig, *Artificial intelligence: a modern approach*, Malaysia; Pearson Education Limited,, 2016.

[111] J. Salvatier, T. V. Wiecki, and C. Fonnesbeck, *Probabilistic programming in python using pymc3*, PeerJ Computer Science, 2 (2016), p. e55.

[112] A. Shah, A. Wilson, and Z. Ghahramani, *Student-t processes as alternatives to gaussian processes*, in Artificial Intelligence and Statistics, 2014, pp. 877–885.

[113] S. Shead, *Microsoft exec: 'ai is the most important technology that anybody on the planet is working on today'*, May 2016. Accessed: 2018-03-28.

[114] N. Smirnov, *Table for estimating the goodness of fit of empirical distributions*, The annals of mathematical statistics, 19 (1948), pp. 279–281.

[115] A. F. Smith and G. O. Roberts, *Bayesian computation via the gibbs sampler and related markov chain monte carlo methods*, Journal of the Royal Statistical Society. Series B (Methodological), (1993), pp. 3–23.

[116] C. Snyder, *This mit website makes you decide who a self-driving car should kill in an accident.* http://uk.businessinsider.com/mit-research-crowdsource-moral-decision-making-self-driving-cars-2016-9, September 2016. Accessed: 2018-04-18.

[117] A. Solonen et al., *Monte carlo methods in parameter estimation of nonlinear models*, (2006).

[118] A. Steinbauer, *Ethics versus morality*, Philosophy Now, 26 (2000), pp. 9–9.

[119] A. Talwar and Y. Kumar, *Machine learning: An artificial intelligence methodology*, Int. J. Eng. Comput. Sci, 2 (2013), pp. 3400–3405.

[120] S. Van Dongen, *Prior specification in bayesian statistics: three cautionary tales*, Journal of Theoretical Biology, 242 (2006), pp. 90–100.

[121] D. Vanderelst and A. Winfield, *The dark side of ethical robots*, arXiv preprint arXiv:1606.02583, (2016).

[122] M. Versteeg, *Ethics & gamification design: a moral framework for taking responsibility*, Master's thesis, 2013.

[123] J. Von Neumann and O. Morgenstern, *Theory of games and economic behavior*, Bull. Amer. Math. Soc, 51 (1945), pp. 498–504.

[124] R. K. Wagner and R. J. Sternberg, *Practical intelligence in real-world pursuits: The role of tacit knowledge.*, Journal of personality and social psychology, 49 (1985), p. 436.

[125] W. Wallach and C. Allen, *Moral machines: Teaching robots right from wrong*, Oxford University Press, 2008.

[126] P. Weiss, *Morality and ethics*, The Journal of Philosophy, 39 (1942), pp. 381–385.

[127] B. Whitby, *Reflections on artificial intelligence*, Intellect Books, 1996.

[128] J. R. Wilson and M. Scheutz, *A model of empathy to shape trolley problem moral judgements*, in Affective Computing and Intelligent Interaction (ACII), 2015 International Conference on, IEEE, 2015, pp. 112–118.

[129] D. Wolpert, *No free lunch theorem for optimization*, IEEE transactions on evolutionary computation, (1997), pp. 467–482.

Appendix A

Results from "Learning a commonsense moral theory"

Figure A.1: Kleiman-Weiner et al. maximum a posteriori (MAP) estimate of direct reciprocity relations, indirect reciprocity relations and individual weights from a learner observing behavior of the society shown in Figure A.2 [72]

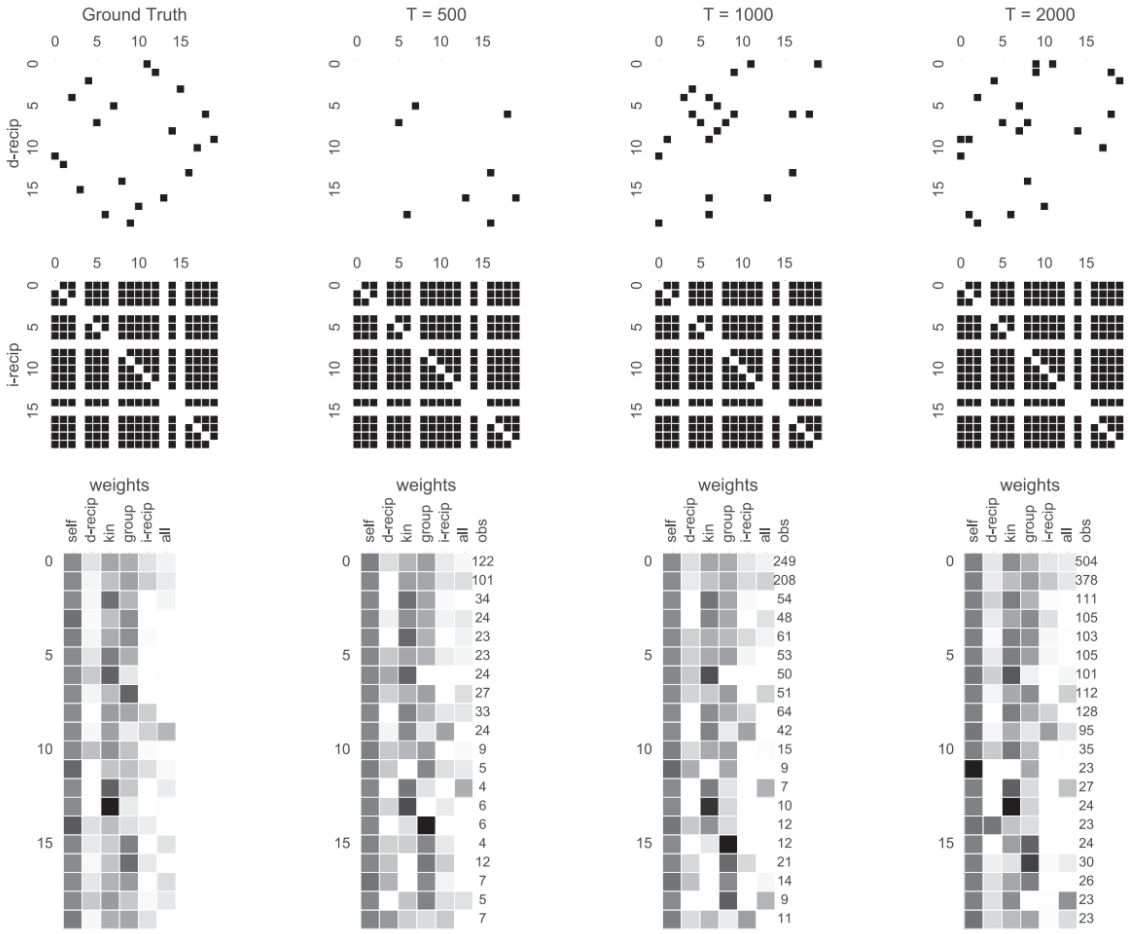
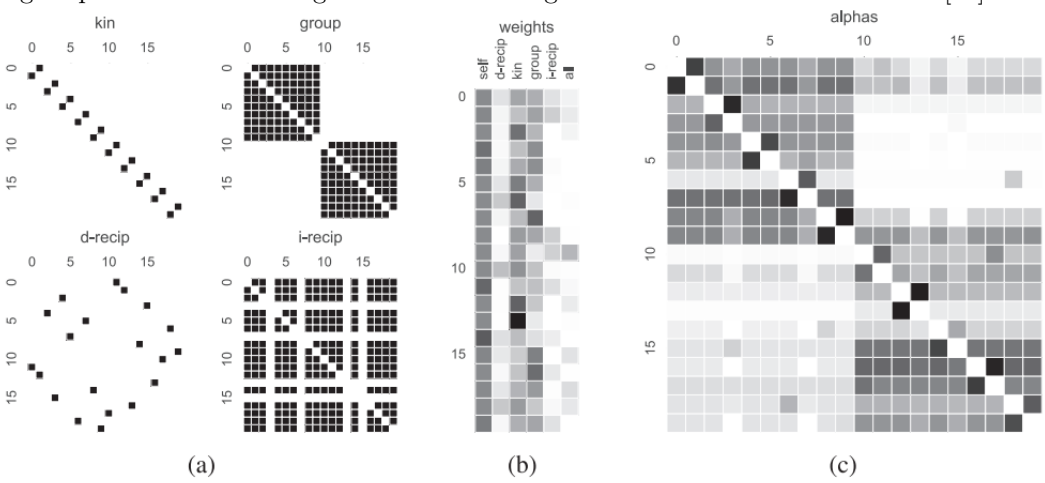


Figure A.2: A population of 20 agents used throughout Kleiman-Weiner et al. work showing a) their relationship to other agents b) the weights they place over abstract moral principles and c) the weighting each agent places on all other agents which is a weighted sum over abstract features. [72]



Appendix B

Results from "A Computational Model of Commonsense Moral Decision Making"

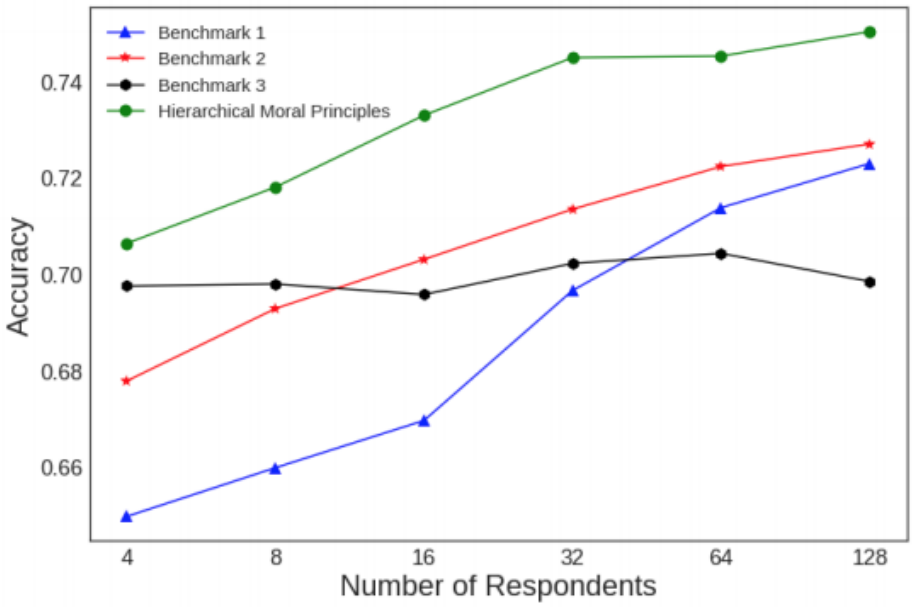



Figure B.1: Kim et al. reported individual predictive accuracy of their proposed hierarchical model when compared to three benchmarks. Benchmark 2 is the equivalent of Benchmark 1 used in this dissertation's results [72].